Hydraulic Turbines

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Final Project

Design and Simulation of the Turbine for the Siah-Bisheh Hydropower Plant

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Abstract

This project presents the design, meshing, and numerical performance evaluation of a high-head Francis turbine runner based on the selected operating conditions of the Siah Bisheh hydropower case study. The main design parameters used in this work were a net/design head of 490m, flow rate of 59 m^3/s, rotational speed of 254 rpm, water density of 998 kg/m^3, and design turbine efficiency of 0.94. Based on these parameters, the available hydraulic power was estimated to be approximately 283 MW, while the expected shaft power was about 266 MW.

The runner geometry was first developed in CFturbo. The design process included the definition of the main dimensions, meridional contour, blade mean lines, blade loading distribution, blade angle variation, blade thickness profile, and leading and trailing edge geometry. A NACA 0015 profile was selected for the blade thickness distribution because of its symmetric shape, suitable thickness ratio, geometric simplicity, and good meshing stability. The final runner geometry was checked in three dimensions and prepared for CFD analysis.

The meshing process was performed in ANSYS TurboGrid using a structured single-passage turbomachinery mesh. Since the runner contained 14 blades, one periodic blade passage was modeled instead of the full runner. The final mesh consisted of approximately 169,047 nodes and 156,910 elements. Mesh quality assessment showed zero bad elements, positive minimum element volume, and acceptable skewness and orthogonality for the scope of the CFD simulation.

The CFD simulation was carried out in ANSYS CFX using a rotating domain, periodic interfaces, $SST\ k-\omega$ turbulence model, mass-flow inlet, and pressure outlet. Several numerical setups were tested during the simulation process. The initial simulations using a normal inlet direction did not reproduce the expected runner performance. Therefore, the inlet condition was corrected using cylindrical velocity components to represent the swirl entering the Francis runner. A two-step solution strategy was then used: first, a stabilization run with first-order schemes, followed by a final high-resolution run. The final simulation successfully converged to the RMS residual target of 10^{-5}.

The final CFD results showed a mass imbalance of only 0.073%, indicating good mass conservation. The mass-flow-averaged total pressure in the stationary frame decreased from 5.50467×10^6 Pa at the inlet to 5.32743×10^5 Pa at the outlet, giving a total pressure drop of 4.972 MPa. The corresponding CFD runner head was calculated as 507.8 m. The extracted torque for one periodic passage was -760708 N.m, resulting in a total runner torque of approximately -10.65 MN.m. The calculated shaft power was 283.3 MW, and the runner-passage efficiency was estimated as 96.6%.

Keywords: Francis turbine, runner design, CFturbo, TurboGrid, ANSYS CFX, CFD, swirl inlet, pressure drop, shaft power, runner efficiency.

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Introduction

Hydropower is one of the most important renewable energy sources for large-scale electricity generation. In a hydropower plant, the potential energy of water stored at a high elevation is converted into mechanical energy by a hydraulic turbine, and then into electrical energy by a generator. The amount of energy that can be extracted from water mainly depends on the available head, flow rate, turbine efficiency, and hydraulic losses in the system.

Among different types of hydraulic turbines, the Francis turbine is one of the most widely used machines for medium- and high-head hydropower applications. A Francis turbine is a reaction-type mixed-flow turbine, in which water enters the runner with both radial and tangential velocity components and leaves the runner in a more axial direction. Energy extraction occurs because the runner blades change the angular momentum of the flow. As a result, a torque is produced on the rotating runner, which is transferred to the shaft.

In the present project, a Francis turbine runner was designed and numerically evaluated for a high-head hydropower condition inspired by the Siah Bisheh hydropower plant. The main objective of the project was to design the runner geometry, generate a suitable turbomachinery mesh, and perform CFD simulation to evaluate the hydraulic performance of the runner passage.

1-1 General Description of Francis Turbines

A Francis turbine consists of several main hydraulic and mechanical components, including the spiral case, stay vanes, guide vanes, runner, shaft, head cover, and draft tube. The general arrangement of these components is shown in Figure 1.

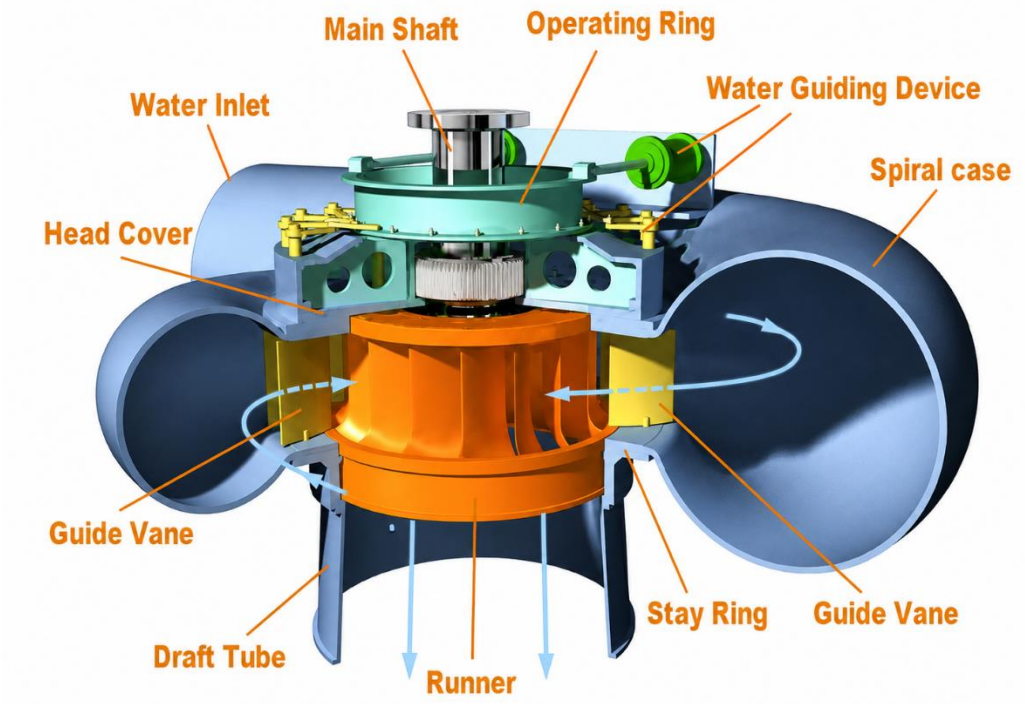


Figure 1: Main components and flow path of a Francis turbine

The water first enters the turbine through the spiral case. The spiral case distributes the incoming water around the circumference of the turbine. After that, the flow passes through the stay vanes and guide vanes.

The guide vanes control the flow rate and direct the water toward the runner with the required inlet angle. This is a very important part of Francis turbine operation because the runner inlet flow is not simply normal to the inlet section. Instead, the flow has a strong circumferential component, also known as swirl.

The runner is the main component responsible for energy conversion. When water enters the runner passage, it interacts with the curved runner blades. The blade geometry forces the flow to change direction, and this change in angular momentum generates torque on the runner. The runner torque is then transferred to the main shaft and converted into useful shaft power.

Finally, the water leaves the runner and enters the draft tube. The draft tube helps recover part of the kinetic energy remaining at the runner outlet and converts it into pressure energy. Although the draft tube is important for overall turbine efficiency, the present project focuses only on the runner and its flow passage.

1-2 Siah Bisheh Hydropower Plant as the Reference Case

The Siah Bisheh hydropower plant is located in a mountainous region and operates under high-head hydraulic conditions. A general view of the dam and reservoir area is shown in Figure 2.



Figure 2: General view of the Siah Bisheh hydropower plant area

In this project, the Siah Bisheh power plant was used as the reference case for defining a high-head Francis turbine design problem. However, the numerical model developed in this project does not represent the full hydropower plant. Instead, the project focuses on the design and CFD evaluation of the Francis runner passage under the selected design conditions.

The main design parameters used in this project are summarized in Table 1.

Table 1: Main design parameters used for the Francis runner design

Parameter	Symbol	Value	Unit
Net/design head	H_n	490	m
Flow rate	Q	59	m^3/s
Rotational speed	N	254	rpm
Turbine efficiency	η	0.94	–
Water density	ρ	998	kg/m^3
Gravitational acceleration	g	9.81	m/s^2

These parameters were used as the basis for the manual calculations, CFturbo runner design, mesh generation, and final CFD simulation.

1-3 Initial Hydraulic Power Estimation

Before starting the three-dimensional design process, the approximate hydraulic power available from the flow was estimated using:

$$P_h = \rho g Q H_n$$

Substituting the design values:

$$P_h = 998 \times 9.81 \times 59 \times 490 \approx 283.0 \times 10^6 \text{ W} = 283\ MW$$

Considering the design turbine efficiency of $\eta = 0.94$ the expected mechanical shaft power can be estimated as:

$$P_{shaft} = \eta P_h$$

$$P_{shaft} = 0.94 \times 283.0 \approx 266\ MW$$

This value was used as a reference for evaluating the final CFD results. The final CFD simulation predicted a shaft power in the same order of magnitude, which confirmed that the designed runner was operating close to the intended hydraulic condition.

1-4 Runner Design Objective

The main purpose of the project was to design a Francis turbine runner suitable for the selected high-head operating condition. The runner must be able to receive water at the design flow rate, extract hydraulic energy from the flow, and generate the required torque and shaft power.

The runner design process included the following main stages:

1. defining the basic hydraulic design parameters;
2. estimating the main dimensions of the runner;
3. designing the meridional contour;
4. defining the blade angle distribution;
5. adjusting the blade loading distribution;
6. selecting a suitable blade thickness profile;
7. generating the final three-dimensional runner geometry;
8. preparing the geometry for meshing and CFD simulation.

A technical runner drawing is shown in Figure 3 as an example of the level of geometrical detail required in Francis runner design. This type of drawing illustrates that the runner geometry is not a simple blade shape, but a complex three-dimensional structure involving blade profiles, hub and shroud contours, leading and trailing edge details, and manufacturing-related dimensions.

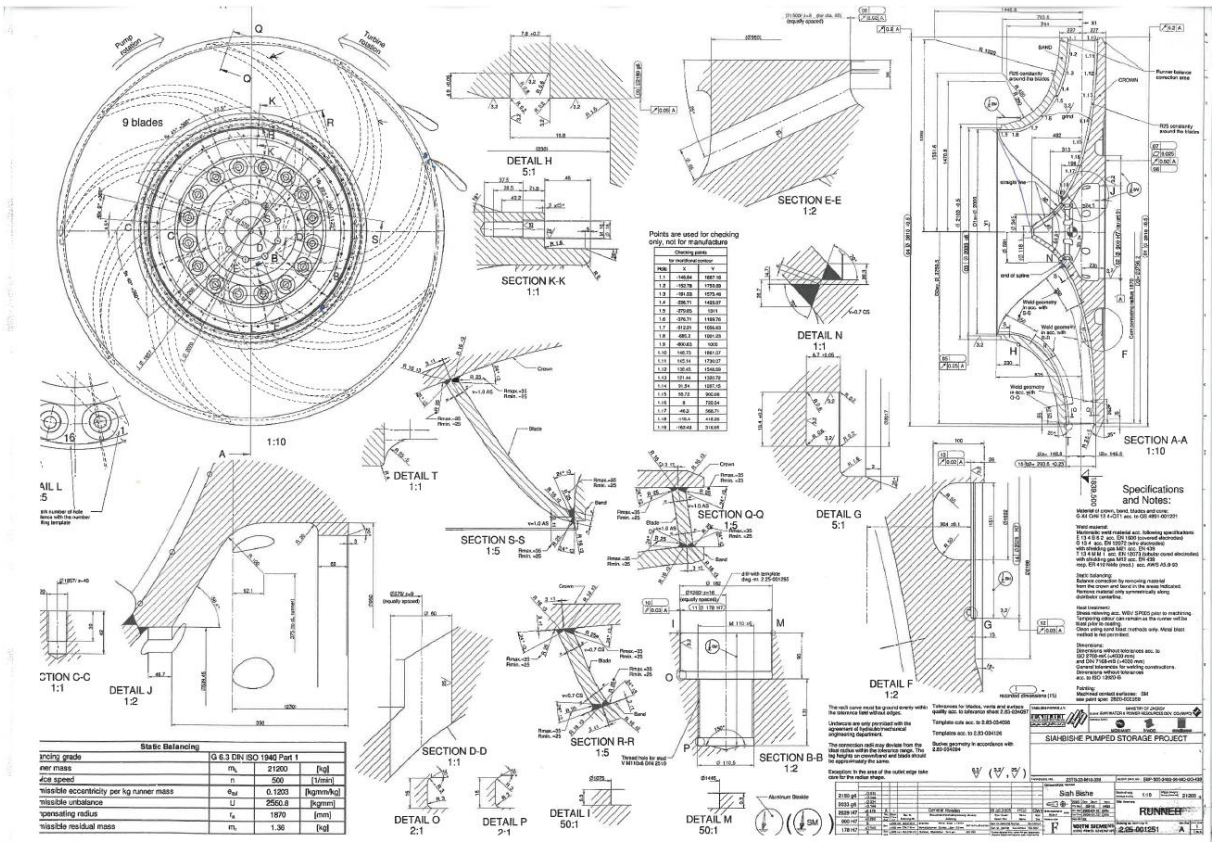


Figure 3: Technical drawing of a Francis turbine runner showing the complexity of runner geometry

In this project, the runner geometry was developed using CFturbo, where the main dimensions, meridional profile, blade mean lines, blade angle distribution, and thickness profile were defined and corrected step by step. The final runner geometry was then prepared for turbomachinery meshing and CFD analysis.

1-5 Importance of Runner Inlet Swirl

One of the most important points in the simulation of a Francis runner is the correct definition of the inlet flow direction. In a real Francis turbine, the flow entering the runner is generated by the guide vanes. Therefore, the inlet flow has both meridional and tangential velocity components.

At the beginning of the CFD work, a simple normal inlet direction was tested. However, the resulting pressure drop and torque were not consistent with the expected design performance. This showed that a normal inlet condition was not sufficient for a runner-only model. For this reason, the inlet boundary condition was later corrected by applying cylindrical velocity components, including a radial component and a tangential component.

This correction was essential because the runner extracts energy mainly through the change in angular momentum of the flow. If the tangential component of the inlet velocity is not defined correctly, the runner cannot reproduce the expected turbine behavior. After applying the swirl inlet condition, the CFD results became much closer to the design point.

1-6 Numerical Simulation Strategy

The CFD simulation was performed using a single periodic runner passage rather than the full 360-degree runner. Since the runner has repeated blade passages around the circumference, one passage can represent the complete runner if periodic boundary conditions are applied properly. This approach

significantly reduces the computational cost while preserving the main flow physics inside the runner passage.

The numerical workflow used in the project was:

$$CFturbo \rightarrow TurboGrid \rightarrow CFX \rightarrow CFD - Post$$

The runner geometry was first designed in CFturbo. Then, the blade passage was meshed in ANSYS TurboGrid using a structured turbomachinery mesh. The mesh was transferred to ANSYS CFX, where the rotating-domain CFD simulation was performed. Finally, the results were analyzed in CFD-Post.

The final CFD setup used:

Table 2: Final CFD setup and boundary conditions used for the runner passage simulation.

Setting	Description
CFD domain	Single periodic runner passage
Flow model	Steady – state turbulent flow
Turbulence model	SST k – ω
Domain motion	Rotating domain
Boundary condition at inlet	Mass flow with cylindrical velocity components
Boundary condition at outlet	Static pressure outlet
Periodicity	Rotational periodic interface
Wall condition	No – slip rotating wall

This setup was selected because it represents the runner passage more realistically than a simple stationary-domain or normal-inlet simulation.

1-7 Scope of the Present Study

The present study focuses on the runner-level hydraulic performance of the Francis turbine. The model includes the runner passage, blade walls, hub, shroud, inlet, outlet, and periodic interfaces. However, the model does not include the complete turbine system.

The following components were not included in the final CFD model:

- spiral case;
- stay vanes;
- guide vanes;
- draft tube;
- leakage paths;
- mechanical losses;
- generator losses.

Therefore, the calculated efficiency should be interpreted as a runner-passage efficiency, not the total plant or full turbine efficiency. This distinction is important because a complete turbine simulation would include additional hydraulic losses before and after the runner.

Even with this limitation, the runner-passage CFD model is highly useful. It allows the pressure distribution, velocity field, streamlines, total pressure drop, torque, shaft power, and runner efficiency to be evaluated in detail. These outputs are sufficient for assessing whether the designed runner geometry is hydraulically reasonable.

1-8 Project Aim and Structure

The aim of this project is to design and evaluate a Francis turbine runner for the selected high-head operating condition. The design and analysis are carried out through a complete engineering workflow, including theoretical calculation, geometric design, mesh generation, CFD simulation, and post-processing.

The main objectives are:

- to calculate the initial hydraulic and mechanical power of the turbine;
- to design the main runner geometry using CFturbo;
- to define the blade profile, blade angle distribution, blade loading, and thickness distribution;
- to generate a structured runner-passage mesh using TurboGrid;
- to set up a rotating-domain CFD model in ANSYS CFX;
- to apply a realistic swirl inlet condition;
- to evaluate the CFD convergence behavior;
- to extract pressure drop, head, torque, shaft power, and efficiency;
- to visualize the pressure field, velocity distribution, streamlines, and wall y^+.

The final CFD results showed that the corrected runner-passage model produced a physically meaningful pressure drop and shaft power. The calculated CFD runner head was close to the selected design head, and the predicted shaft power was in the expected range for the designed runner. Therefore, the final numerical model was considered suitable for evaluating the hydraulic behavior of the designed Francis turbine runner

Basic Design

2-1 Introduction

The objective of this section is to perform the basic or preliminary design of a Francis turbine for the Siah Bisheh hydropower plant. The design is based on the given operating conditions of head, discharge, rotational speed, and efficiency. At this stage, empirical Francis turbine design correlations are used to estimate the main runner dimensions, including the reference runner diameter, inlet and outlet diameters, and characteristic heights.

This stage is not the final hydraulic design of the runner blades. Instead, it provides the first estimation of the turbine size and geometry, which will later be used for velocity triangle analysis, blade angle calculation, cavitation checking, and more detailed hydraulic design.

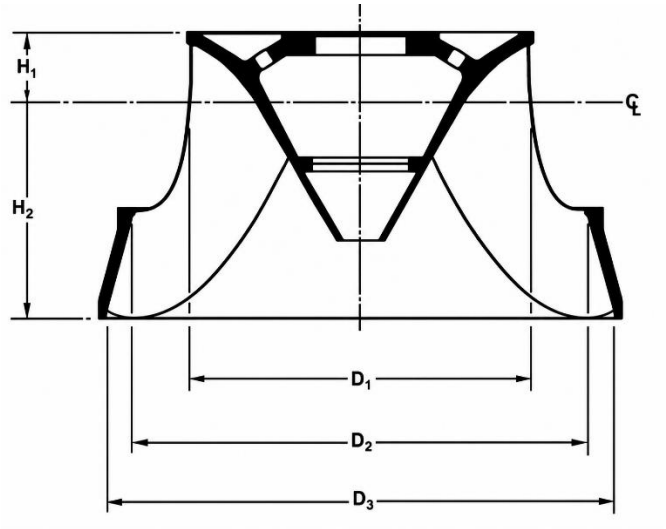


Figure 4: Initial meridional contour of the Francis runner used for the basic design stage.

2-2 Given Design Data

The design data for the Francis turbine are:

Table 3: Main design parameters of the Siah Bisheh Francis turbine.

Parameter	Symbol	Value	Unit
Net/design head	H_n	490	m
Flow rate	Q	59	m^3/s
Rotational speed	N	254	rpm
Turbine efficiency	η	0.94	–
Water density	ρ	998	kg/m^3
Gravitational acceleration	g	9.81	m/s^2

The density of water is taken as:

$$\rho = 998\ \mathrm{kg/m^3}$$

This value is consistent with the previous example used for the Masjed Soleyman Francis turbine design.

2-3 Design Assumptions

The following assumptions are considered in the basic design stage:

1. The turbine is a Francis turbine operating under steady design conditions.
2. The given head $H_n = 490$ m is assumed to be the net design head.
3. The given discharge $Q = 59\ \mathrm{m^3/s}$ is considered for one turbine unit.
4. The turbine efficiency is assumed to be the design-point efficiency.
5. Empirical correlations from the Francis turbine design handout are used for preliminary sizing.
6. The metric specific speed relation based on power is used because H_n, Q, N, and η are all known.

2-4 Turbine Power Calculation

The hydraulic power available from the water flow is:

[2.1] $$P_h = \rho g Q H_n$$

Substituting the design values:

$$P_h = 998 \times 9.81 \times 59 \times 490$$
$$P_h = 283039885.8\ \mathrm{W}$$
$$\boxed{P_h = 283.04\ \mathrm{MW}}$$

The mechanical output power of the turbine is calculated by applying the turbine efficiency:

$$P = \eta P_h = 0.94 \times 283.04 = 266.06\ MW$$

Since the specific speed equation requires power in kilowatts:

$$P = 266057.49\ kW$$

2-5 Specific Speed Calculation

For the actual selected turbine, the metric specific speed is calculated using:

[2.2] $$N_s = \frac{N\sqrt{P}}{H_n^{5/4}}$$

where:

- N is in rpm,
- P is in kW,
- Hn is in m.

Substituting the values:

$$N_s = \frac{254\sqrt{266057.49}}{490^{5/4}}$$
$$\boxed{N_s = 56.83}$$

This value places the turbine in the low-specific-speed Francis turbine range. This is reasonable because the Siah Bisheh turbine operates under a very high head and relatively moderate flow rate.

A second empirical relation can also be used only as an initial estimate:

[2.3] $$N_s = 3470 H_n^{-0.625}$$

For $H_n = 490$ m:

$$N_s = 3470(490)^{-0.625} = 72.27$$

However, because the actual rotational speed, discharge, head, and efficiency are known, the power-based value

$$\boxed{N_s = 56.83}$$

is selected for the basic design calculations.

2-6 Peripheral Velocity Coefficient

The peripheral velocity coefficient is obtained from the empirical relation:

[2.4] $$k_u = 0.31 + 0.0025N_s$$

Substituting $N_s = 56.83$:

$$k_u = 0.31 + 0.0025(56.83)$$

$$\boxed{k_u = 0.4521}$$

This coefficient relates the runner peripheral velocity to the theoretical jet velocity based on the available head.

The theoretical velocity corresponding to the design head is:

$$\sqrt{2gH_n} = \sqrt{2 \times 9.81 \times 490} = 98.05\ m/s$$

Therefore, the peripheral speed is:

$$u = k_{u\sqrt{2gH_n}} u = 0.4521 \times 98.05$$

$$\boxed{u = 44.33\ \text{m/s}}$$

2-7 Calculation of Runner Reference Diameter D_3

The runner reference diameter D_3 is calculated from:

[2.5] $$D_3 = \frac{84.5 k_u \sqrt{H_n}}{N}$$

Substituting the values:

$$D_3 = \frac{84.5(0.4521)\sqrt{490}}{254}$$

$$\boxed{D_3 = 3.329\ \text{m}}$$

The value can be checked using the peripheral speed relation:

$$u = \frac{\pi D_3 N}{60} = \frac{\pi(3.329)(254)}{60} = 44.28\ m/s$$

This value is very close to the value obtained from $k_{u\sqrt{2gH_n}}$, so the calculated diameter is consistent.

2-8 Calculation of Runner Diameter D_2

The second runner diameter is calculated from:

[2.6] $$\frac{D_2}{D_3} = \frac{1}{0.96 + 0.00038N_s}$$

Substituting $N_s = 56.83$:

$$\frac{D_2}{D_3} = \frac{1}{0.96 + 0.00038(56.83)} = 1.01875$$

Therefore:

$$D_2 = 1.01875(3.329)$$

$$\boxed{D_2 = 3.392\ \text{m}}$$

2-9 Calculation of Runner Diameter D_1

The diameter D_1 is obtained from:

[2.7]
$$\frac{D_1}{D_3} = 0.4 + \frac{94.5}{N_s}$$

Substituting the specific speed:

$$\frac{D_1}{D_3} = 0.4 + \frac{94.5}{56.83}$$

Therefore:

$$D_1 = 2.063(3.329)$$

$$\boxed{D_1 = 6.868\ \text{m}}$$

The relatively large value of D_1 is due to the low specific speed of the turbine. Low-specific-speed Francis turbines are generally associated with high-head operation and require a more elongated runner geometry.

2-10 Calculation of Height H_1

The first runner height ratio is:

[2.8]
$$\frac{H_1}{D_3} = 0.094 + 0.00025N_s$$

Substituting $N_s = 56.83$:

$$\frac{H_1}{D_3} = 0.094 + 0.00025(56.83) = 0.1082$$

Thus:

$$\boxed{H_1 = 0.360\ \text{m}}$$

2-11 Calculation of Height H_2

For the second height H_2, the correct empirical relation must be selected based on the value of N_s. Since:

$$50 < N_s = 56.83 < 110$$

the following relation is used:

[2.9]
$$\frac{H_2}{D_3} = -0.05 + \frac{42}{N_s}$$

Substituting the value of N_s:

$$\frac{H_2}{D_3} = -0.05 + \frac{42}{56.83} = 0.6891$$

Therefore:

$$H_2 = 0.6891(3.329)$$

$$\boxed{H_2 = 2.294\ \text{m}}$$

It is important to note that the alternative relation

$$\frac{H_2}{D_3} = \frac{1}{3.16 - 0.0013N_s}$$

is only valid for:

$$110 < N_s < 350$$

Therefore, it must not be used for the present turbine because the calculated specific speed is only 56.83.

2-12 Shaft Torque Calculation

The angular velocity of the turbine shaft is:

$$\omega = \frac{2\pi(254)}{60}$$

$$\boxed{\omega = 26.60 \text{ rad/s}}$$

The shaft torque is:

$$T = \frac{P}{\omega} = \frac{266.06 \times 10^6}{26.60}$$

$$\boxed{T = 10.00 \text{ MN.m}}$$

This torque value shows that the turbine shaft must be designed for very high mechanical loading.

2-13 Summary of Basic Design Results

Table 4: Summary of basic design results for the Siah Bisheh Francis turbine.

Parameter	Symbol	Value	Unit
Net head	H_n	490	m
Flow rate	Q	59	m^3/s
Rotational speed	N	254	Rpm
Efficiency	η	0.94	–
Hydraulic power	P_h	283.04	MW
Mechanical output power	P	266.06	MW
Specific speed	N_s	56.83	–
Peripheral velocity coefficient	k_u	0.4521	–
Peripheral speed	u	44.33	m/s
Reference runner diameter	D_3	3.329	M
Runner diameter	D_2	3.392	M
Runner diameter	D_1	6.868	M
Height ratio	H_1/D_3	0.1082	–
Runner height	H_1	0.360	M
Height ratio	H_2/D_3	0.6891	–
Runner height	H_2	2.294	m
Shaft angular velocity	ω	26.60	rad/s
Shaft torque	T	10.00	MN. m

Geometry Design Using CFturbo

3-1 Impeller Design

3-1-1 Design Point Definition

The runner design process was initiated in CFturbo by defining the global operating point of the Siah Bisheh Francis turbine. As shown in **Figure 5**, the turbine was modeled as a **Francis high-head turbine**, which is consistent with the high net head of the project. The main design inputs included the volume flow rate, head, rotational speed, working fluid, inlet pressure condition, and additional hydraulic efficiency.

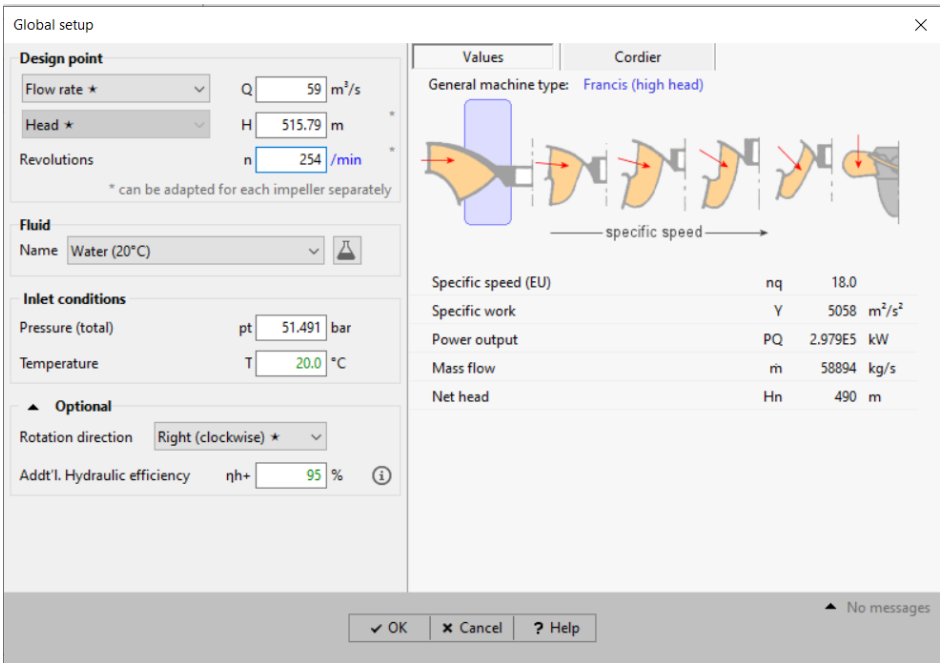


Figure 5: Global operating point definition in CFturbo for the Siah Bisheh Francis high-head turbine, including volume flow rate, rotational speed, head, and additional hydraulic efficiency settings.

The design flow rate was specified as:

$$Q = 59\ \mathrm{m}^3/\mathrm{s}$$

and the rotational speed was entered as:

$$\mathrm{n} = 254\ \mathrm{rpm}$$

In the global setup window, the entered head was:

$$H = 515.79\ m$$

The additional hydraulic efficiency was set to:

$$\eta_{h+} = 95\%$$

Therefore, CFturbo calculated the effective net head as:

$$H_n = 490\ \text{m}$$

This value matches the design net head of the Siah Bisheh turbine. The working fluid was selected as water at $20°C$, and the total inlet pressure was defined as approximately 51.491 bar, as shown in **Figure 6**. The rotation direction was set to right-hand clockwise rotation.

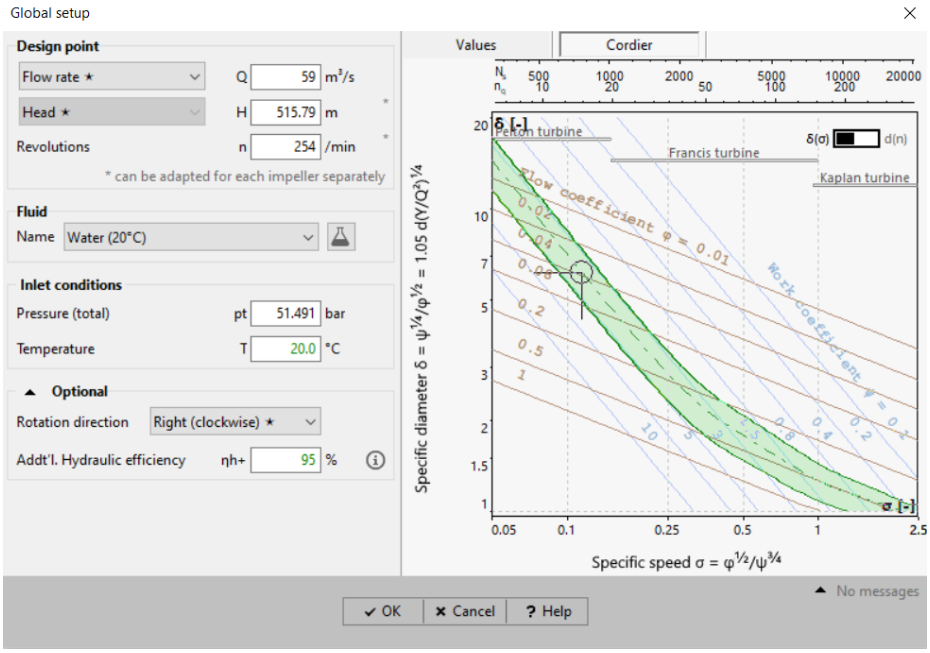


Figure 6: Working fluid selection (water at 20°C) and total inlet pressure definition (≈ 51.491 bar) in CFturbo.

Based on these inputs, CFturbo calculated a mass flow rate of:

$$\dot{m} = 58894\ \text{kg/s}$$

which is consistent with the volume flow rate of $59\ \text{m}^3/\text{s}$ for water. The calculated specific work was:

$$Y = 5058\ \text{m}^2/\text{s}^2$$

and the corresponding hydraulic power output was:

$$P_Q = 2.979 \times 10^5\ \text{kW}$$

or approximately:

$$P_Q = 297.9\ \text{MW}$$

This confirms that the selected design point represents a high-power Francis turbine operating under a high-head condition.

3-1-2 Turbine Classification Using the Cordier Diagram

The Cordier diagram was used as an initial verification tool to confirm whether the selected operating point is compatible with the Francis turbine type. As shown in **Figure 3**, the operating point lies within the Francis turbine region and close to the high-head side of the chart.

CFturbo calculated the European specific speed as:

$$n_q = 18.0$$

This relatively low specific speed is typical for high-head Francis turbines. A low specific speed indicates that the runner is expected to have a comparatively compact outlet region and a strong radial-to-axial flow transition. Therefore, the Cordier diagram confirms that the selected machine type and design point are hydraulically consistent.

3-1-3 Initial Runner Setup

After defining the global design point, the runner component was created. The setup parameters used for the runner are shown in **Figure 7**. The runner was defined as a standard impeller with a material density of:

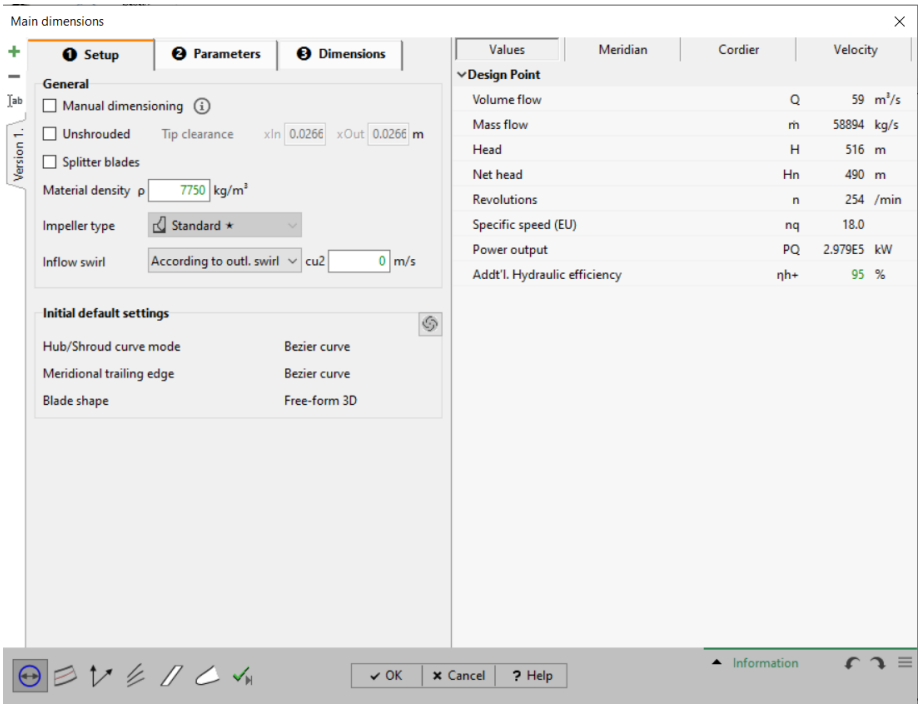


Figure 7: Runner component setup in CFturbo, showing the impeller type, material density, hub/shroud curve modes, blade shape, and inflow swirl definition.

$$\rho_m = 7750\ \mathrm{kg/m^3}$$

This value is representative of steel-based runner construction.

The initial runner geometry settings were selected as follows:

Table 5: Initial runner geometry settings in CFturbo.

Design item	Selected option
Impeller type	Standard
Hub/Shroud curve mode	Bezier curve
Meridional trailing edge	Bezier curve
Blade shape	Free – form 3D
Inflow swirl	According to outlet swirl

Bezier curves were selected for the hub and shroud contours because they provide smooth geometric control of the meridional flow path. This is important in Francis turbine runner design, since abrupt changes in curvature may lead to non-uniform flow acceleration, local separation, or unfavorable pressure gradients.

The blade shape was defined as **Free-form 3D**, as shown in **Figure 4**. This selection is suitable for a Francis turbine runner because the blade geometry is highly three-dimensional. The blade must simultaneously satisfy the meridional contour, inlet and outlet blade angles, blade wrapping, and spanwise variation from hub to shroud.

3-1-4 Runner Main Dimensions

The main runner dimensions were manually adjusted based on the preliminary hand calculations. The final input values used in CFturbo are shown in **Figure 4**.

The main dimensions were defined as:

Table 6: Main runner dimensions defined in CFturbo based on preliminary hand calculations.

Parameter	Symbol	Value	Unit
Hub diameter at inlet	dH1	6.686	m
Shroud diameter at inlet	dS1	6.686	m
Hub diameter at outlet	dH2	0	m
Shroud diameter at outlet	dS2	3.329	m
Horizontal inlet width	b1z	0.36	m

The inlet hub and shroud diameters were set equal, which means that the inlet section of the runner begins at the same radial position for both the hub and the shroud. The outlet shroud diameter was defined as:

$$dS2 = 3.329\ \text{m}$$

which gives an outlet shroud radius of:

$$r_2 = \frac{3.329}{2} = 1.6645 \text{ m}$$

The inlet width was entered as:

$$b_{1z} = 0.36 \text{ m}$$

This parameter controls the axial opening at the runner inlet and directly affects the inlet flow area and the meridional velocity.

The hub outlet diameter was set to:

$$dH2 = 0$$

This means that the hub contour reaches the rotational axis at the outlet. However, during later geometry refinement, this value may require adjustment if CFturbo produces geometric warnings near the outlet axis or if blade solid generation becomes unstable.

3-1-5 Calculated Global Values

After the main dimensions were entered, CFturbo calculated several global hydraulic parameters, shown on the right side of **Figure 8**. The most important calculated values are summarized below.

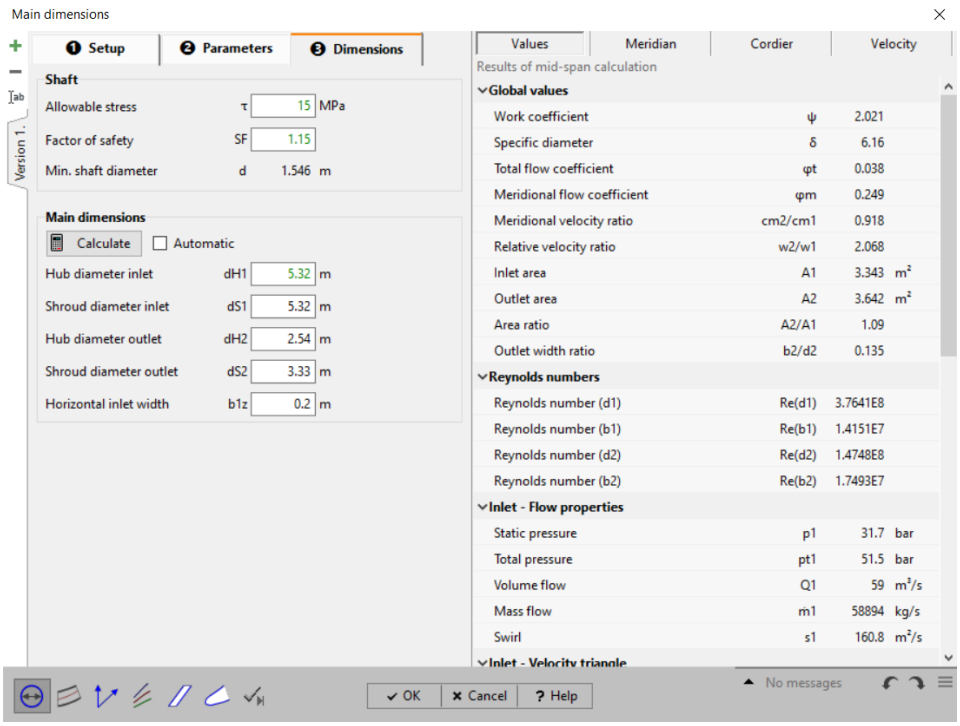


Figure 8: CFturbo interface showing the calculated global hydraulic parameters (work coefficient, specific diameter, flow coefficients, meridional velocity ratio, relative velocity ratio, and passage areas) after the main runner dimensions were defined.

Table 7: Calculated global hydraulic parameters of the runner in CFturbo.

Parameter	Symbol	Value

Work coefficient	ψ	2.253
Specific diameter	δ	5.831
Total flow coefficient	φ_t	0.044
Meridional flow coefficient	φ_m	0.033
Meridional velocity ratio	c_{m2}/c_{m1}	3.066
Relative velocity ratio	w_2/w_1	5.953
Inletarea	A_1	$26.688m2m^2m2$
Outletarea	A_2	$8.704m2m^2m2$
Area ratio	A_2/A_1	0.326
Outlet width ratio	b_2/d_2	1

The calculated inlet area is:

$$A_1 = 26.688\ \mathrm{m}^2$$

while the outlet area is:

$$A_2 = 8.704\ \mathrm{m}^2$$

Therefore, the area ratio is:

$$\frac{A_2}{A_1} = 0.326$$

This indicates a considerable reduction in flow area from the runner inlet to the outlet. Such a reduction is expected in a compact high-head Francis runner, where the water enters the runner at a larger radius and exits at a smaller radius after transferring energy to the runner.

The meridional velocity ratio was calculated as:

$$\frac{c_{m2}}{c_{m1}} = 3.066$$

This means that the meridional velocity increases significantly from inlet to outlet. This behavior is reasonable for a high-head Francis runner, but it should be checked carefully in later design stages because excessive acceleration may increase hydraulic losses or create unfavorable pressure gradients.

The relative velocity ratio was obtained as:

$$\frac{w_2}{w_1} = 5.953$$

This relatively high value indicates strong acceleration of the relative flow through the runner passage. Therefore, the blade angle distribution, blade passage area, and meridional contour must be refined carefully in the next steps.

3-1-6 Reynolds Number Evaluation

The Reynolds numbers calculated by CFturbo are also shown in **Figure 5**. The reported values are:

Table 8: Reynolds numbers calculated by CFturbo for the runner passage, confirming fully turbulent flow conditions.

Parameter	Value
Re(d1)	5.9452×10^8
Re(b1)	1.4991×10^8
Re(d2)	1.4739×10^8
Re(b2)	7.3694×10^7

These Reynolds numbers are very high, confirming that the flow inside the runner is fully turbulent. This is expected for a large-scale hydropower turbine. Although viscous effects are not dominant compared with inertial effects, wall friction, secondary flows, and blade-surface losses remain important and should be evaluated in the final performance assessment.

3-1-7 Inlet Flow Properties

The inlet flow properties calculated by CFturbo are shown in **Figure 5**. The main values are:

Table 9: Inlet flow properties calculated by CFturbo for the high-head Francis runner, including static and total pressure, volume and mass flow rates, and inlet swirl.

Parameter	Symbol	Value
Static pressure	p_1	28.65 bar
Total pressure	p_{t1}	49 bar
Volume flow	Q_1	$59 m^3/s$
Mass flow	$\dot{m}_1$	58894 kg/s
Swirl	s_1	$160.78\ m^2/s$

The high inlet total pressure reflects the high-head operating condition of the turbine. The presence of inlet swirl is also physically expected, because the distributor and guide vanes generate the circumferential velocity component required for energy transfer in the runner.

In a Francis turbine, the runner extracts energy mainly by reducing the angular momentum of the water. Therefore, the inlet swirl value is a key parameter for defining the inlet blade angle and the velocity triangle at the runner entrance.

3-1-8 Design Assessment of This Stage

The results shown in **Figures 2 to 5** indicate that the CFturbo model was correctly initialized as a high-head Francis turbine. The design point, turbine classification, and manually imposed runner dimensions are generally consistent with the preliminary design of the Siah Bisheh turbine.

The model represents a high-power Francis runner operating at:

$$Q = 59\ \mathrm{m^3/s}$$
$$H_n = 490\ \mathrm{m}$$
$$n = 254\ \mathrm{rpm}$$

The calculated operating point falls inside the Francis turbine region of the Cordier diagram, confirming the suitability of the selected turbine type. The manually entered runner dimensions define a large inlet diameter, a compact outlet diameter, and a relatively narrow inlet width, which are all characteristic of a high-head Francis runner.

However, one important point should be checked before finalizing the model. In the previous hand calculation, the inlet diameter was obtained as approximately:

$$D_1 = 6.868\ \mathrm{m}$$

while **Figure 4** shows:

$$dH1 = dS1 = 6.686\ \mathrm{m}$$

If the intended design value is 6.868 m, the CFturbo model should be corrected. If 6.686 m was intentionally selected after geometric refinement, this change should be mentioned in the report as a model adjustment made to improve the runner geometry.

3-1-9 Meridional Contour Fitting Using AutoCAD Data

After defining the main runner dimensions in CFturbo, the meridional contour of the runner was refined using a reference runner drawing. The purpose of this step was to obtain a more realistic hub and shroud geometry instead of relying only on the default CFturbo meridional curves. The reference drawing used for this curve-fitting process is shown in **Figure 9**.

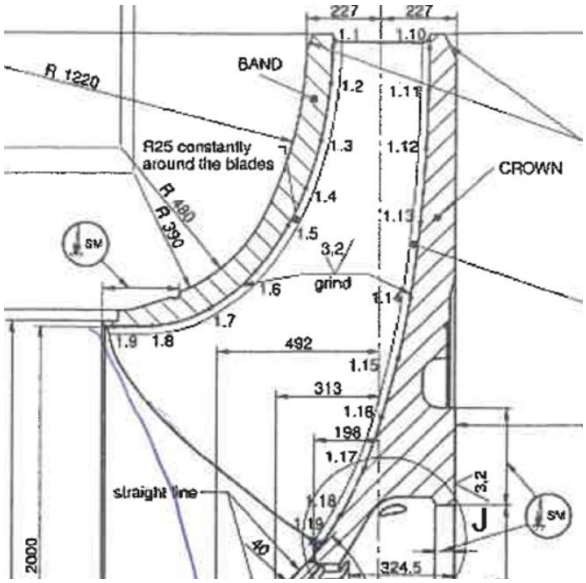


Figure 9: Reference runner drawing used for extracting the hub (crown) and shroud (band) meridional contours for curve fitting in CFturbo.

In a Francis turbine runner, the meridional contour strongly affects the flow passage area, meridional velocity distribution, blade loading, and pressure variation through the runner. Therefore, the hub and shroud contours were not treated as arbitrary curves. Instead, they were extracted from a geometrically similar runner drawing and then scaled and shifted to match the CFturbo coordinate system and the target dimensions of the Siah Bisheh turbine.

The runner drawing shown in **Figure 9** includes two main meridional boundaries: the **crown**, corresponding to the hub side of the runner, and the **band**, corresponding to the shroud side. These two curves define the primary flow passage in the meridional plane. Since the CFturbo meridional design environment uses the axial coordinate zzz and radial coordinate rrr, the extracted drawing coordinates were converted into the (z,r)(z,r)(z,r) coordinate system before being used in CFturbo.

3-1-10 Coordinate Extraction from AutoCAD

The hub and shroud curves were first traced in AutoCAD based on the reference drawing. After tracing, discrete points were extracted along each curve. The original extracted coordinates were obtained from AutoCAD as X and Y positions. Since the AutoCAD drawing data were obtained in inches, the raw coordinates could not be directly used in CFturbo.

The extracted coordinates were therefore processed in Excel. The transformation procedure consisted of three steps:

1. extracting the raw X and Y coordinates from AutoCAD;
2. converting the coordinates from inches to a consistent metric scale;
3. shifting and scaling the transformed data to match the start and end points required in CFturbo.

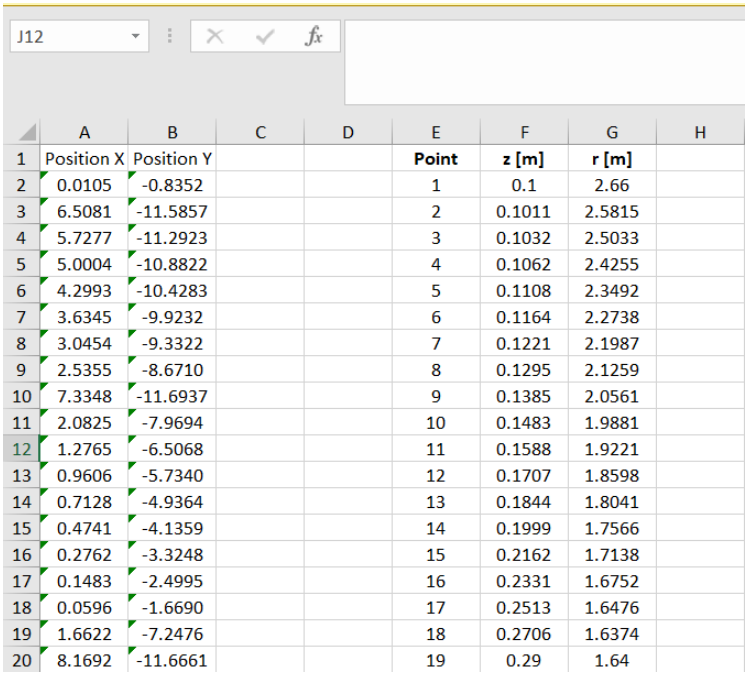
The final transformed coordinates were expressed as:

$z = z_{\mathrm{CFturbo}}$

where z is the axial coordinate and rrr is the radial coordinate in the CFturbo meridional plane.

3-1-11 Shroud / Band Curve Transformation

The transformed data for the shroud side of the runner are shown in **Figure 10**. This curve corresponds to the **band** curve in the reference drawing. In CFturbo, the shroud curve was fitted between the following start and end points:



	A	B	C	D	E	F	G	H
1	Position X	Position Y			Point	z [m]	r [m]	
2	0.0105	-0.8352			1	0.1	2.66	
3	6.5081	-11.5857			2	0.1011	2.5815	
4	5.7277	-11.2923			3	0.1032	2.5033	
5	5.0004	-10.8822			4	0.1062	2.4255	
6	4.2993	-10.4283			5	0.1108	2.3492	
7	3.6345	-9.9232			6	0.1164	2.2738	
8	3.0454	-9.3322			7	0.1221	2.1987	
9	2.5355	-8.6710			8	0.1295	2.1259	
10	7.3348	-11.6937			9	0.1385	2.0561	
11	2.0825	-7.9694			10	0.1483	1.9881	
12	1.2765	-6.5068			11	0.1588	1.9221	
13	0.9606	-5.7340			12	0.1707	1.8598	
14	0.7128	-4.9364			13	0.1844	1.8041	
15	0.4741	-4.1359			14	0.1999	1.7566	
16	0.2762	-3.3248			15	0.2162	1.7138	
17	0.1483	-2.4995			16	0.2331	1.6752	
18	0.0596	-1.6690			17	0.2513	1.6476	
19	1.6622	-7.2476			18	0.2706	1.6374	
20	8.1692	-11.6661			19	0.29	1.64	

Figure 10: Transformed shroud (band) curve data extracted from the reference drawing and fitted in CFturbo, showing the outer meridional boundary from the inlet radius (r = 2.66 m) to the outlet radius (r = 1.64 m).

$$z_{\text{start}} = 0.10 \text{ m}$$
$$r_{\text{start}} = 2.66 \text{ m}$$

and:

$$z_{\text{end}} = 0.29 \text{ m}$$
$$r_{\text{end}} = 1.64 \text{ m}$$

The transformed shroud data were arranged from the inlet region toward the outlet region. This ordering is important because CFturbo constructs the curve according to the sequence of the inserted points. If the point order is reversed or inconsistent, the resulting curve may become distorted or may create curvature discontinuities.

As shown in **Figure 10**, the shroud curve starts at the larger inlet radius and gradually moves toward the smaller outlet radius. This behavior is consistent with the meridional shape of a high-head Francis runner, where the flow enters at a larger radius and is guided inward and downward toward the runner outlet.

The final transformed shroud data used for curve fitting were approximately:

Table 10: Transformed shroud (band) curve coordinates extracted from the reference drawing and used for curve fitting in CFturbo, showing the axial (z) and radial (r) positions in meters from inlet to outlet.

Point	*z[m]*	*r[m]*	*Point*	*z[m]*	*r[m]*
1	0.1000	2.6600	**11**	0.1588	1.9221
2	0.1011	2.5815	**12**	0.1707	1.8598
3	0.1032	2.5033	**13**	0.1844	1.8041
4	0.1062	2.4255	**14**	0.1999	1.7566

5	0.1108	2.3492	**15**	0.2162	1.7138
6	0.1164	2.2738	**16**	0.2331	1.6752
7	0.1221	2.1987	**17**	0.2513	1.6476
8	0.1295	2.1259	**18**	0.2706	1.6374
9	0.1385	2.0561	**19**	0.2900	1.6400
10	0.1483	1.9881			

The small increase in radius at the final point is a result of fitting the extracted reference geometry. This local behavior was retained because it helps represent the original band curvature near the outlet region.

3-1-12 Hub / Crown Curve Transformation

The hub-side curve was extracted from the **crown** region of the reference drawing. The raw AutoCAD coordinates and the final transformed CFturbo coordinates are shown in **Figure 11**.

K8

	A	B	C	D	E	F	G
1	Position X	Position Y			**Point**	**(z) [m]**	**(r) [m]**
2	0.0168	-0.9709			1	-0.1	2.66
3	2.5515	-16.2136			2	-0.0962	2.5754
4	2.1896	-15.3127			3	-0.0939	2.4908
5	1.8936	-14.3880			4	-0.0907	2.4061
6	1.6291	-13.4536			5	-0.0862	2.3216
7	1.3774	-12.5157			6	-0.0833	2.237
8	1.1389	-11.5744			7	-0.0786	2.1525
9	0.9400	-10.6241			8	-0.0705	2.0684
10	2.9649	-17.0922			9	-0.0601	1.9848
11	0.7765	-9.6668			10	-0.0491	1.9013
12	0.4575	-7.7511			11	-0.0381	1.8178
13	0.3358	-6.7878			12	-0.0248	1.7349
14	0.2664	-5.8193			13	-0.0088	1.6527
15	0.2234	-4.8492			14	0.0081	1.5709
16	0.1557	-3.8805			15	0.0258	1.4894
17	0.1072	-2.9107			16	0.0457	1.4087
18	0.0730	-1.9403			17	0.0699	1.3301
19	0.6118	-8.7098			18	0.0976	1.2535
20	3.4479	-17.9341			19	0.13	1.18

Figure 11: Transformed hub (crown) curve data extracted from the reference drawing and fitted in CFturbo, showing the inner meridional boundary from the inlet radius (r = 2.66 m) to the outlet radius (r = 1.18 m).

For this curve, the CFturbo start and end points were defined as:

$$z_{\text{start}} = -0.10\ \text{m}$$
$$r_{\text{start}} = 2.66\ \text{m}$$

and:

$$z_{\text{end}} = 0.13\ \text{m}$$
$$r_{\text{end}} = 1.18\ \text{m}$$

This curve forms the inner boundary of the meridional flow passage. Compared with the shroud curve, the hub curve moves more strongly toward the axis, which produces the inward contraction of the runner passage.

The transformed hub data were approximately:

Table 11: Transformed hub (crown) curve coordinates extracted from the reference drawing and used for curve fitting in CFturbo, showing the axial (z) and radial (r) positions in meters from inlet to outlet.

$Point$	$z[m]$	$r[m]$			
1	−0.1000	2.6600	**11**	−0.0381	1.8178
2	−0.0962	2.5754	**12**	−0.0248	1.7349
3	−0.0939	2.4908	**13**	−0.0088	1.6527
4	−0.0907	2.4061	**14**	0.0081	1.5709
5	−0.0862	2.3216	**15**	0.0258	1.4894
6	−0.0833	2.2370	**16**	0.0457	1.4087
7	−0.0786	2.1525	**17**	0.0699	1.3301
8	−0.0705	2.0684	**18**	0.0976	1.2535
9	−0.0601	1.9848	**19**	0.1300	1.1800
10	−0.0491	1.9013			

The hub curve shown in **Figure 11** has a smooth inward trend from the runner inlet toward the outlet. This shape is important because the hub contour controls the inner wall of the passage and directly affects the outlet contraction and the local acceleration of the flow near the runner exit.

3-1-13 Importing the Fitted Curves into CFturbo

After transforming the extracted coordinates, the hub and shroud curves were manually reconstructed in the CFturbo meridional contour environment. The result of this curve-fitting process is shown in **Figure 12**.

In **Figure 12**, the blue curve represents the hub-side meridional contour, while the green curve represents the shroud-side contour. The red dashed line represents the mean meridional line between the hub and shroud boundaries. The fitted geometry shows a smooth contraction from the inlet radius toward the outlet region.

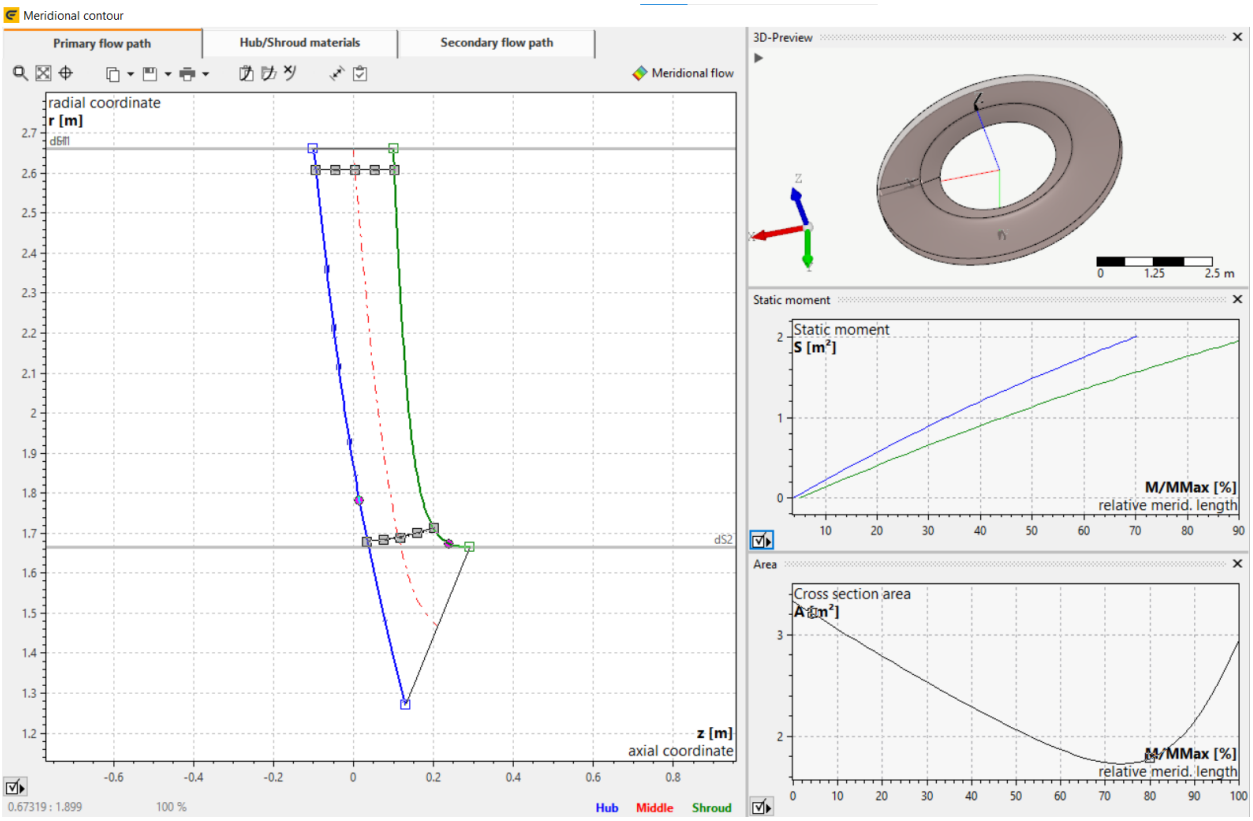


Figure 12: Final fitted meridional contour of the runner showing the hub (blue), shroud (green), and mean meridional flow line (red dashed), along with the cross-sectional area and static moment distributions along the relative meridional length.

The fitted curves were adjusted so that they remained compatible with the previously defined main dimensions. The inlet radius was kept close to:

$$r_1 = 2.66 \text{ m}$$

while the outlet side of the shroud was fitted near:

$$r_{S2} = 1.64 \text{ m}$$

and the outlet side of the hub was fitted near:

$$r_{H2} = 1.18 \text{ m}$$

This fitting approach allowed the meridional passage to follow the reference runner drawing while still satisfying the dimensional constraints of the CFturbo model.

3-1-14 Evaluation of the Fitted Meridional Passage

The right side of **Figure 12** shows the static moment and cross-sectional area distributions along the relative meridional length. These diagrams are important because they indicate whether the fitted hub and shroud curves create a smooth hydraulic passage.

The cross-sectional area plot in **Figure 12** shows that the flow area decreases from the inlet region toward the middle part of the passage. This reduction is expected because the runner geometry contracts as the flow moves inward. However, near the outlet region, the area begins to rise again. This behavior should be monitored carefully because a sudden area increase may create local diffusion, adverse pressure gradient, or non-uniform outlet flow.

The area curve does not show a sharp discontinuity, which means that the fitted contour is acceptable for the preliminary geometry stage. Nevertheless, further refinement may still be required after checking the blade angles, blade thickness, and three-dimensional blade model.

The static moment distribution in **Figure 12** also changes smoothly along the meridional length. This indicates that the fitted geometry does not introduce an abrupt change in the radial distribution of the flow passage. A smooth static moment trend is desirable because it helps maintain a stable meridional flow distribution from inlet to outlet.

3-1-15 Engineering Interpretation

The AutoCAD-based curve-fitting process improved the physical realism of the runner meridional contour. Instead of using only the default CFturbo contour, the hub and shroud boundaries were derived from a reference Francis runner drawing and adapted to the Siah Bisheh design scale.

This approach is useful for three reasons. First, it allows the runner contour to follow a geometry that is closer to a real Francis runner. Second, it helps avoid unrealistic curvature or excessive contraction in the meridional passage. Third, it provides a more reliable geometric basis for the next design steps, including blade angle definition, blade mean-line construction, and three-dimensional blade generation.

The fitted contour shown in **Figure 12** provides a reasonable preliminary meridional passage. The flow path begins at a relatively large inlet radius and contracts toward the outlet, which is consistent with a high-head Francis turbine runner. The smooth hub and shroud curves also reduce the risk of sudden local acceleration or severe curvature-induced flow separation.

However, the area increase near the final part of the passage should be considered during the next design steps. If later velocity plots or blade-passage checks show excessive diffusion or separation risk, the outlet portion of the hub or shroud curve should be slightly modified to obtain a smoother area distribution.

3-1-16 Blade Setup and Number of Blades

After completing the meridional contour fitting, the runner blade design was defined in the **Blade Properties** section of CFturbo. The blade setup used for the preliminary runner model is shown in **Figure 13**.

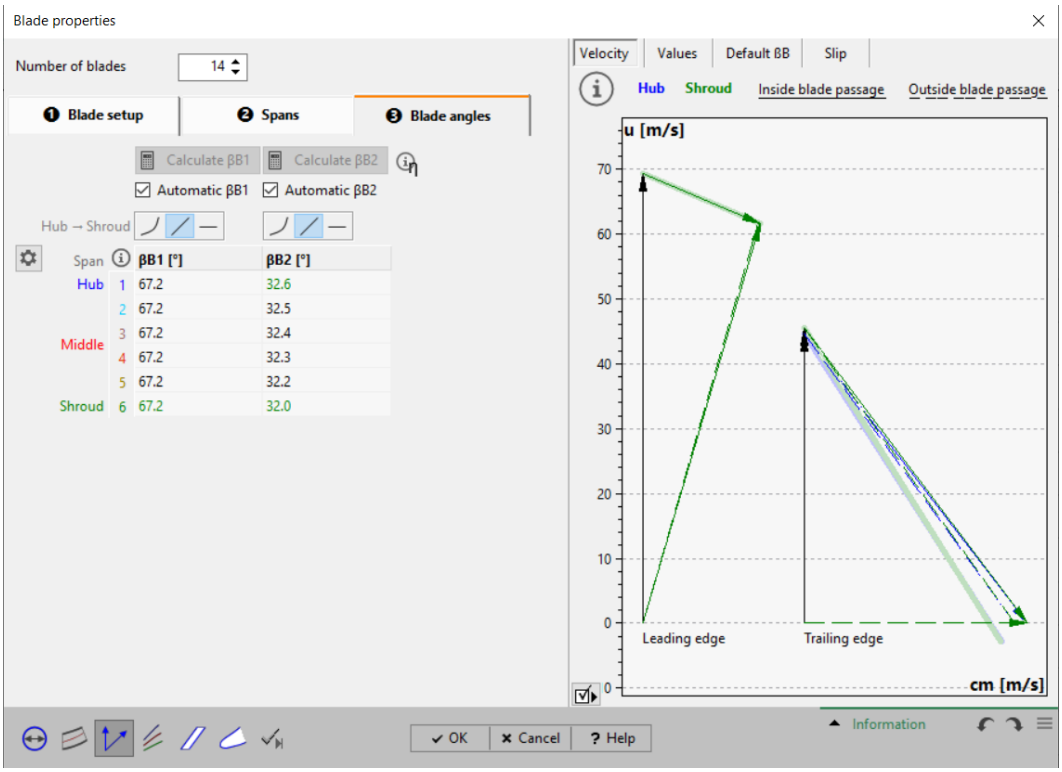


Figure 13: Blade properties setup in CFturbo, showing the number of blades (14), blade shape (Free-form 3D), leading and trailing edge thicknesses (0.0245 m), incidence angle (0°), and slip deviation (-5°) for the Francis runner.

The number of runner blades was set to:

$$Z_b = 14$$

This value was selected as a practical blade number for the high-head Francis runner geometry. A larger number of blades generally improves flow guidance and reduces flow non-uniformity inside the runner passage, but it may also increase blockage, frictional losses, and the risk of blade-to-blade interference. Therefore, the selected number of blades represents a compromise between hydraulic guidance and available blade-passage area.

The blade shape was defined as **Free-form 3D**, as shown in **Figure 13**. This option was used because the Francis runner blade is not a simple two-dimensional surface. It must follow the meridional contour, satisfy the velocity triangle requirements, and maintain a suitable blade angle distribution from hub to shroud. The free-form 3D option allows the blade to be controlled more flexibly along the span and along the meridional direction.

The blade thickness values were also defined in this step. According to **Figure 13**, the leading-edge and trailing-edge thickness values were set as:

Table 12: Leading-edge and trailing-edge thickness values for the runner blade at hub and shroud, defined in CFturbo for the preliminary geometry.

Location	Leading edge thickness [m]	Trailing edge thickness [m]
Hub	0.0245	0.0245
Shroud	0.0245	0.0245

The same thickness was used at the leading and trailing edges for both hub and shroud. This uniform initial thickness was selected to maintain a simple and stable preliminary geometry. The blockage mode was set to **Tangential**, which means that blade blockage is evaluated according to the tangential blade-passage reduction caused by the blade thickness.

The incidence angle was set to zero at both hub and shroud:

$$i_{\text{Hub}} = 0^\circ$$

$$i_{\text{Shroud}} = 0^\circ$$

This indicates that the blade inlet angle was calculated to match the incoming relative flow direction without intentional incidence deviation. This is a suitable assumption for preliminary runner design because shockless or near-shockless inflow helps reduce inlet losses and avoids excessive pressure disturbances near the leading edge.

The slip deviation was defined manually using a user-defined angular deviation model. As shown in **Figure 13**, the slip deviation was set to:

$$\delta_{\text{Hub}} = -5^\circ$$

$$\delta_{\text{Shroud}} = -5^\circ$$

This correction accounts for the deviation between the blade-congruent flow direction and the actual relative flow direction at the blade outlet. Applying a negative outlet deviation is a common preliminary correction to avoid assuming ideal blade-following flow.

3-1-17 Spanwise Blade Definition

The blade was divided into multiple spanwise sections to control the geometry between the hub and shroud. As shown in **Figure 14**, the number of spans was set to:

$$N_{\text{span}} = 6$$

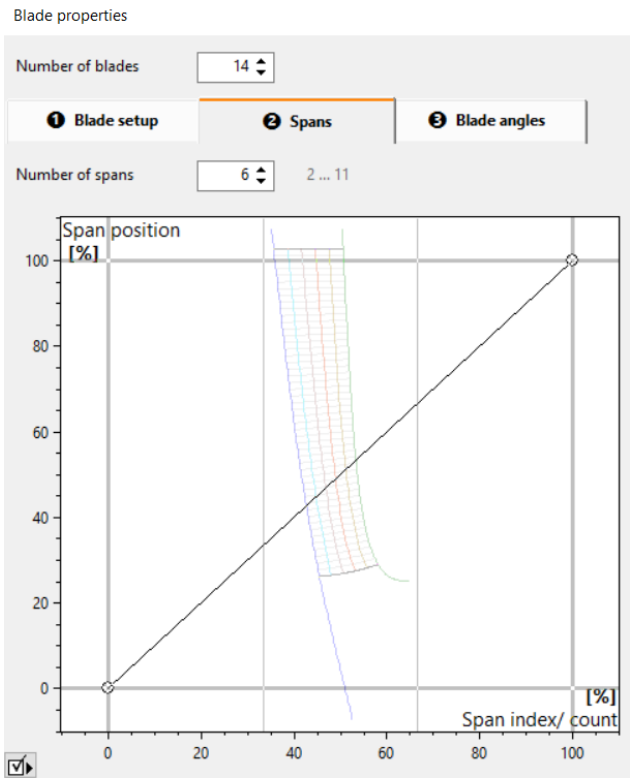


*Figure*14 *: Spanwise blade definition with six span sections from hub to shroud, and velocity triangle preview at the leading and trailing edges showing the circumferential blade speed and inlet swirl components.*

The spanwise definition allows the blade geometry to be evaluated at several radial locations between the hub and shroud. This is important because the flow conditions in a Francis runner are strongly three-dimensional. The local blade speed, meridional velocity, relative velocity, and blade angle are different at the hub, middle span, and shroud.

The spanwise curves shown in **Figure 14** indicate that the blade geometry was distributed from the hub section to the shroud section using six span locations. This provides enough resolution to capture the variation of blade angle and blade loading without making the preliminary design unnecessarily complicated.

The velocity triangle preview on the right side of **Figure 14** shows the velocity relations at the leading and trailing edges. The leading-edge triangle indicates a high circumferential blade speed and a strong inlet swirl component, which is consistent with a high-head Francis turbine. At the trailing edge, the circumferential absolute velocity is reduced, showing that the runner extracts angular momentum from the flow.

3-1-18 Blade Angle Calculation

The blade inlet and outlet angles were calculated automatically by CFturbo based on the velocity triangles. The automatic blade angle calculation is shown in **Figure 15**. The inlet blade angle was calculated as:

$$\beta_{B1} = 59.7°$$

for all span locations. The outlet blade angle varies from hub to shroud as follows:

Table 13: Spanwise distribution of inlet blade angle (β_1) and outlet blade angle (β_2) calculated automatically by CFturbo for the six span sections from hub to shroud.

Span location	β1[deg]	β2[deg]
Hub	59.7	26.2
Span 2	59.7	24.7
Middle	59.7	23.2
Span 4	59.7	21.6
Span 5	59.7	20.1
Shroud	59.7	18.6

The inlet blade angle is nearly constant along the span, while the outlet blade angle decreases from hub to shroud. This behavior is reasonable for a Francis runner because the outlet velocity triangle changes significantly along the span due to the different local blade speeds and radial positions.
The inlet angle:

$$\beta_{B1} = 59.7°$$

is inside the recommended CFturbo range, as indicated in **Figure 15**. This confirms that the selected inlet blade angle is acceptable for the preliminary design.

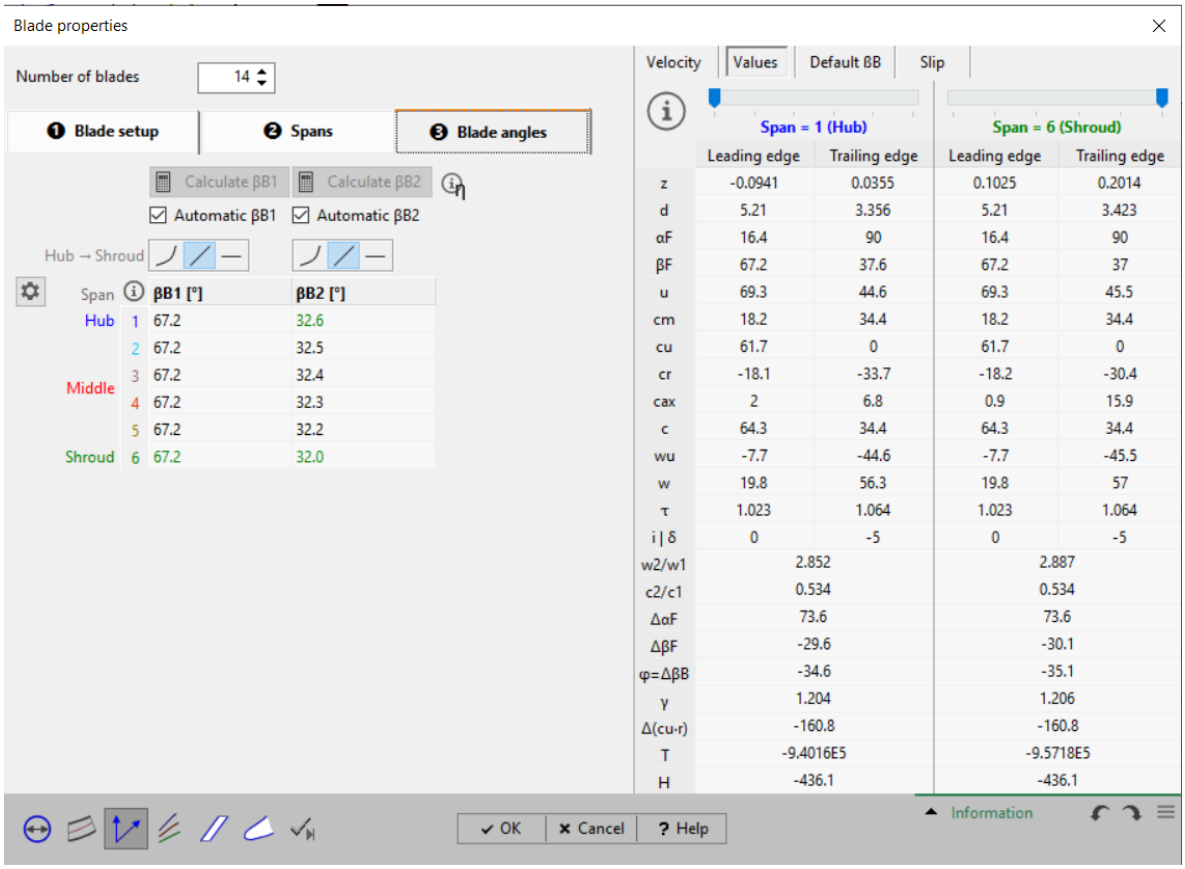


*Figure*15 *: Automatic blade angle calculation in CFturbo, showing the inlet blade angle ($\beta_{B1} = 59.7°$) and outlet blade angle variation from hub ($\beta_{B2} = 26.2°$) to shroud ($\beta_{B2} = 18.6°$), along with velocity triangle parameters at the leading and trailing edges.*

The outlet angle decreases toward the shroud:

$$\beta_{B2,\mathrm{Hub}} = 26.2°$$

$$\beta_{B2,\text{Shroud}} = 18.6^\circ$$

This indicates stronger turning of the relative flow near the shroud side. The spanwise variation of βB2\beta_{B2}βB2 is necessary because the runner blade must satisfy different local velocity conditions from hub to shroud.

3-1-19 Velocity Triangle Interpretation

The detailed values on the right side of **Figure 15** show the velocity triangle parameters at the hub and shroud. At the leading edge, the blade speed was reported as approximately:

$$u_1 = 70.8\ \text{m/s}$$

and the absolute circumferential velocity component was:

$$c_{u1} = 60.4\ \text{m/s}$$

These values show that the incoming flow has a strong tangential component. This is expected because the guide vanes and distributor system are responsible for producing the swirl required for energy transfer in the runner.

At the trailing edge, the circumferential velocity component approaches approximately zero:

$$c_{u2} \approx 0$$

This is a desirable result for a Francis turbine runner because the purpose of the runner is to extract angular momentum from the water. If the outlet swirl is too large, a significant amount of kinetic energy remains unused at the runner outlet, reducing the overall turbine efficiency.

The hub-side relative velocity ratio was reported as:

$$\frac{w_2}{w_1} = 1.799$$

while the shroud-side relative velocity ratio was:

$$\frac{w_2}{w_1} = 2.333$$

The higher value near the shroud indicates stronger acceleration of the relative flow on the shroud side. This is a point that should be monitored during later CFD analysis because excessive acceleration may increase local losses or affect pressure distribution near the blade surface.

3-1-20 Blade Mean Line Adjustment

As shown in **Figure 1**, the runner blade mean lines were adjusted using the conformal mapping design mode. The curves represent the blade passage development from the leading edge to the trailing edge at different spanwise positions between the hub and shroud. A coupled mean-line approach was used so that the blade camber variation remained consistent along the span. The figure also shows the corresponding three-dimensional preview of the runner, where the blade orientation and wrap angle distribution can be visually checked.

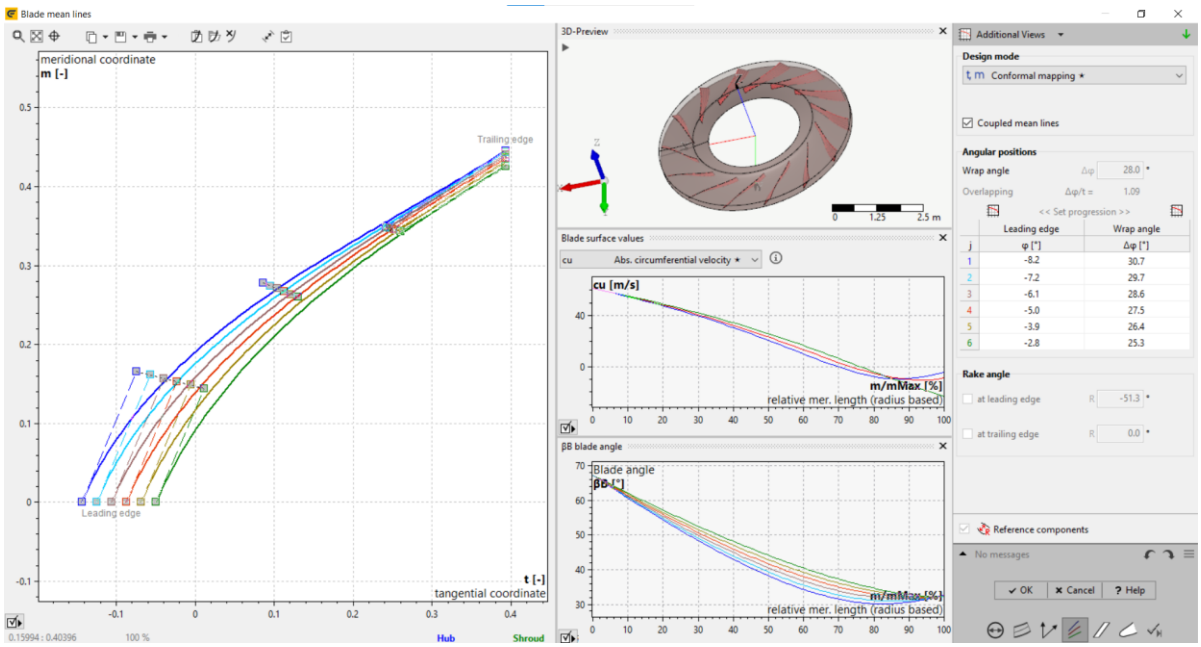


Figure 16: Blade mean line distribution in conformal mapping mode

The circumferential velocity component and blade angle plots on the right side of **Figure 16** indicate a smooth reduction of the absolute circumferential velocity along the relative meridional length. This behavior is expected for a turbine runner, since angular momentum is extracted from the flow as it passes through the blade passage. The blade angle distribution also decreases gradually from inlet to outlet, which helps to guide the flow smoothly and reduce sudden changes in flow direction.

3-1-21 Blade Loading Distribution

The blade loading distribution is presented in **Figure 17**. In this step, the blade loading mode was used to control the aerodynamic/hydraulic work distribution along the blade passage. The loading curves show how the pressure difference between the pressure side and suction side is distributed from the leading edge toward the trailing edge.

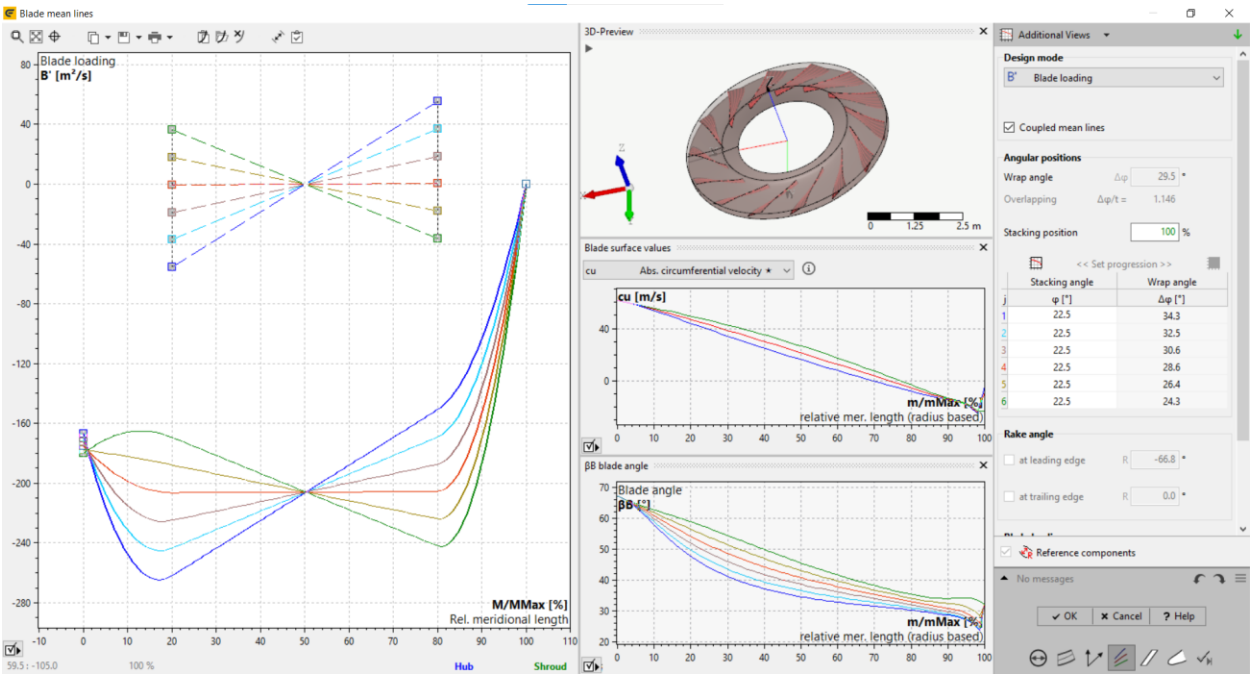


Figure 17: Blade loading distribution along the relative meridional length

The blade loading was modified in order to avoid an excessively sharp loading peak near the leading edge or trailing edge. A smoother loading distribution is desirable because sudden changes in loading may increase the possibility of flow separation, local losses, and numerical instability during CFD simulation. The final loading pattern shown in **Figure 17** provides a more continuous transition from inlet to outlet and gives a more stable blade shape for later flow analysis.

3-1-22 Blade Angle Distribution

The final blade angle distribution is shown in **Figure 18**. The blade angle decreases continuously along the relative meridional length, starting from a high inlet blade angle and reaching a lower value near the trailing edge. This gradual reduction is important in Francis turbine runners because the incoming flow enters the runner with a significant tangential component and must be turned toward the outlet direction while transferring energy to the rotating runner.

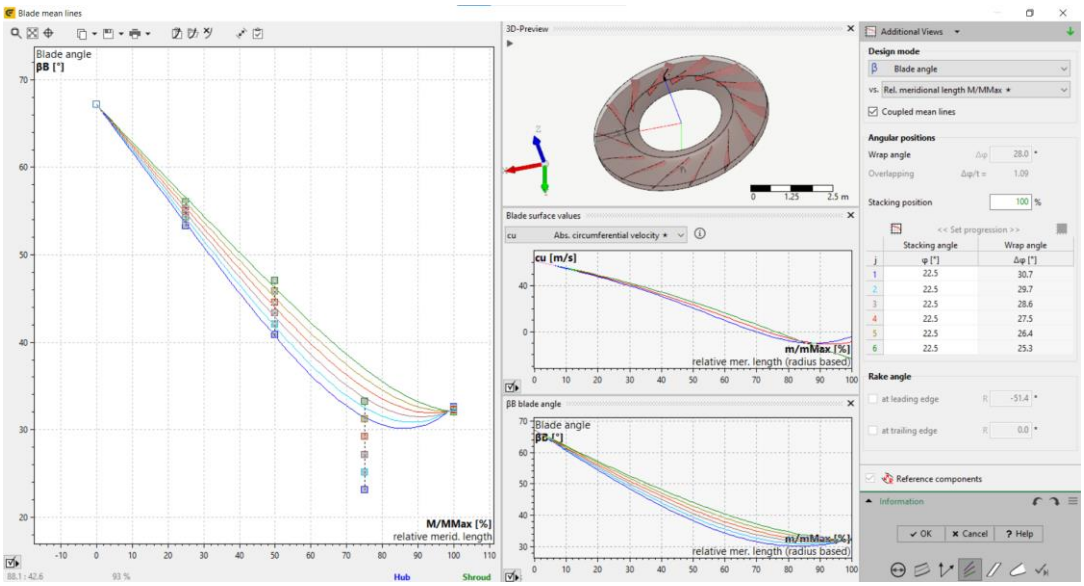


Figure 18: Final blade angle variation from leading edge to trailing edge

The spanwise curves between hub and shroud show a consistent trend, indicating that the blade angle distribution was not defined only for a single section but was adjusted across the full blade height. This helps to maintain a smooth three-dimensional flow path and reduces the risk of unfavorable incidence angles at different spanwise locations.

3-1-23 Blade Thickness Profile Based on NACA 0015

The blade thickness profile is illustrated in **Figure 19**. In this design stage, a **NACA 0015** profile was selected as the reference thickness distribution for the runner blade section. The NACA 0015 profile is a symmetric airfoil section with a thickness-to-chord ratio of 15%. This profile was selected because it provides a suitable balance between hydraulic performance, structural robustness, and meshing stability. Compared with very thin profiles, NACA 0015 provides enough thickness near the leading portion of the blade, while compared with more highly cambered or complex profiles, it reduces the possibility of geometric interference, irregular blade surfaces, and meshing problems.

Figure 19: Blade thickness distribution and three-dimensional blade profile based on NACA 0015

As shown in **Figure 19**, the blade thickness increases smoothly from the leading edge, reaches its maximum value in the front portion of the blade, and then gradually decreases toward the trailing edge. This thickness behavior is consistent with the NACA 0015 profile and is suitable for the Francis runner blade because the rounded leading edge improves the flow entrance condition, while the thinner trailing edge helps reduce wake losses at the runner outlet.

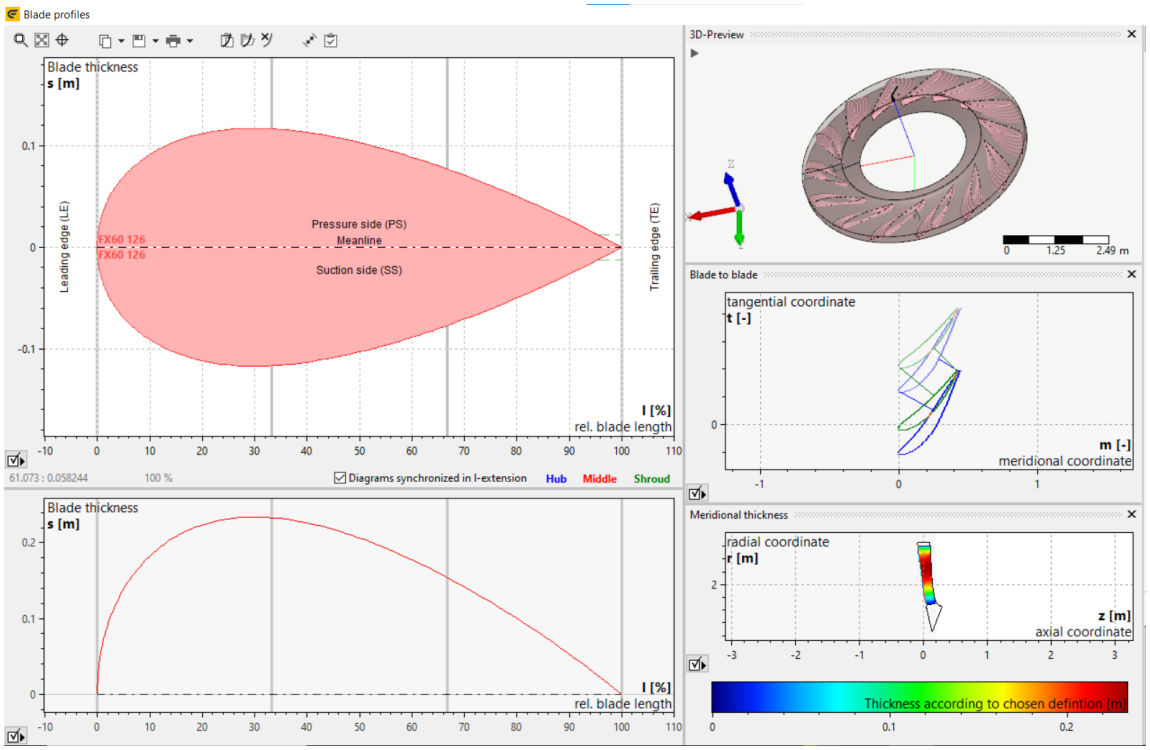
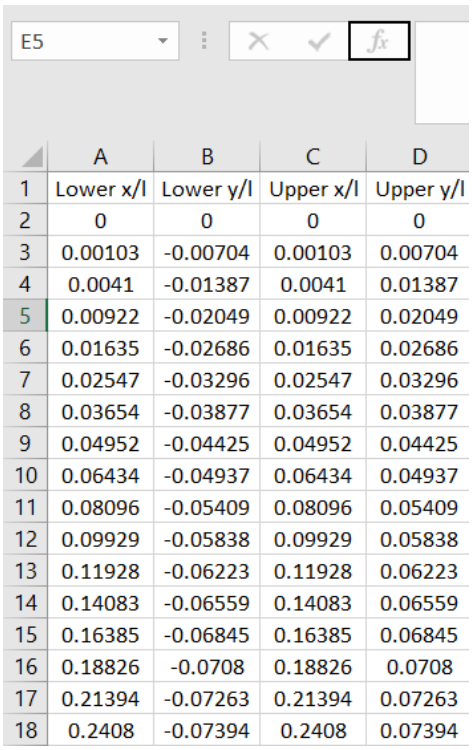


Figure 19: Blade thickness distribution and three-dimensional blade profile

The upper profile view in **Figure 19** shows the pressure side and suction side around the blade mean line. Since NACA 0015 is a symmetric profile, the upper and lower thickness distributions are arranged symmetrically about the mean line before being mapped onto the three-dimensional blade geometry. The lower plot shows the variation of blade thickness along the relative blade length. The three-dimensional preview confirms that the selected thickness profile was applied consistently along the blade span from hub to shroud. This step was important not only for hydraulic performance, but also for preparing a clean and robust blade geometry for the meshing process.

3-1-24 NACA 0015 Coordinate Data

The coordinate data used for the NACA 0015 profile are shown in **Figure 20**. The table includes the lower and upper surface coordinates of the profile, where the lower surface has negative yyy-values and the upper surface has positive yyy-values. These coordinates were used as the basis for defining the blade thickness distribution in CFturbo.



	A	B	C	D
1	Lower x/l	Lower y/l	Upper x/l	Upper y/l
2	0	0	0	0
3	0.00103	-0.00704	0.00103	0.00704
4	0.0041	-0.01387	0.0041	0.01387
5	0.00922	-0.02049	0.00922	0.02049
6	0.01635	-0.02686	0.01635	0.02686
7	0.02547	-0.03296	0.02547	0.03296
8	0.03654	-0.03877	0.03654	0.03877
9	0.04952	-0.04425	0.04952	0.04425
10	0.06434	-0.04937	0.06434	0.04937
11	0.08096	-0.05409	0.08096	0.05409
12	0.09929	-0.05838	0.09929	0.05838
13	0.11928	-0.06223	0.11928	0.06223
14	0.14083	-0.06559	0.14083	0.06559
15	0.16385	-0.06845	0.16385	0.06845
16	0.18826	-0.0708	0.18826	0.0708
17	0.21394	-0.07263	0.21394	0.07263
18	0.2408	-0.07394	0.2408	0.07394

Figure 20: Coordinate data of the NACA 0015 profile used for defining the blade thickness distribution

As shown in **Figure 20**, the profile starts from the leading edge with $x/c = 0$, where the upper and lower surfaces meet. Moving downstream along the chord, the thickness gradually increases and then decreases toward the trailing edge. This coordinate-based definition allows the blade section to preserve the smooth and symmetric characteristics of the NACA 0015 profile. Using this profile helped to create a blade section with sufficient leading-edge thickness, smooth surface curvature, and acceptable trailing-edge refinement, which are all important for achieving a stable CFD mesh and avoiding local geometric defects.

The use of NACA 0015 was mainly motivated by the need for a simple, symmetric, and numerically stable blade thickness profile. Since the runner blade already obtains its hydraulic turning from the blade mean line, wrap angle, and blade angle distribution, a symmetric thickness profile was preferred to avoid introducing unnecessary geometric complexity. In addition, the 15% thickness ratio provides enough mechanical thickness for the blade while keeping the passage blockage within a reasonable range.

Therefore, NACA 0015 was considered more suitable for this design stage than thinner profiles, which could create weak or difficult-to-mesh blade edges, or more complex cambered profiles, which could increase the risk of blade surface distortion and interference during the CFturbo-to-TurboGrid workflow

3-1-25 Leading and Trailing Edge Design

The leading and trailing edge geometry is shown in **Figure 21**. The blade edge design was checked at different spanwise locations, including hub, middle, and shroud sections. The leading edge was designed with a rounded shape to reduce incidence sensitivity and improve the flow entrance into

the blade passage. The trailing edge was shaped to provide a smooth flow exit and reduce local geometric discontinuities.

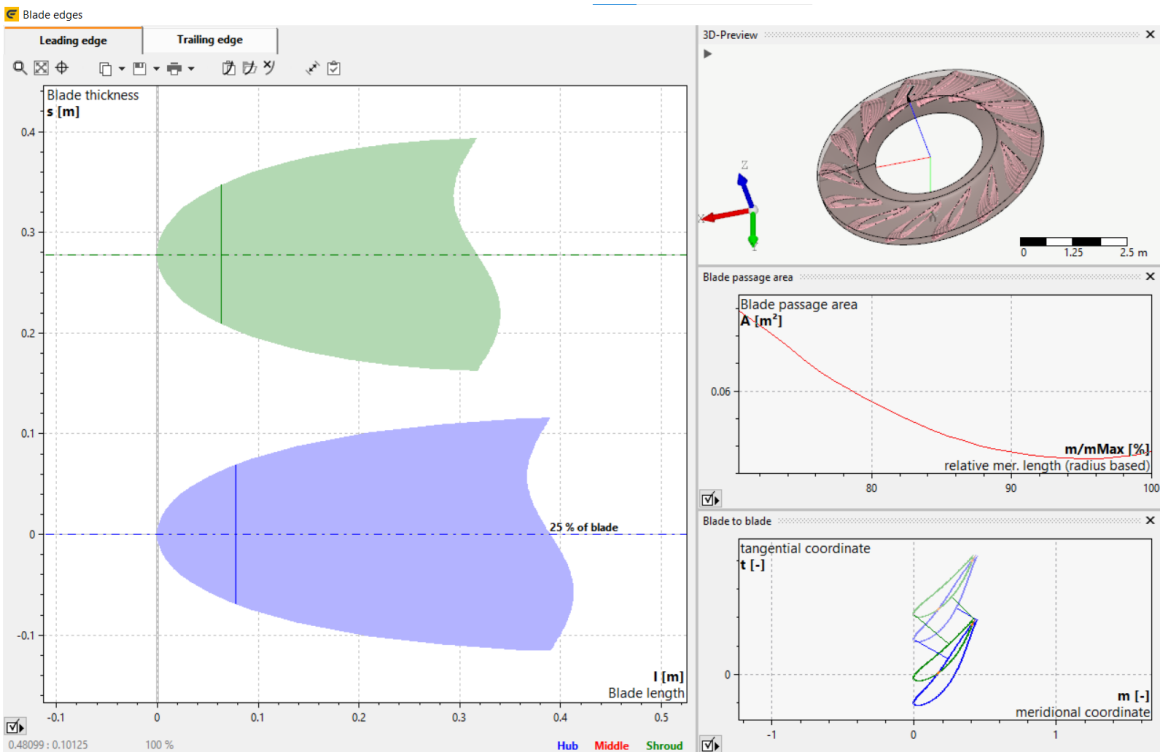


Figure 21: Leading and trailing edge shape control

The figure also includes the blade passage area variation along the relative meridional length. The passage area decreases toward the outlet region, which is consistent with the compact runner geometry and the acceleration/turning of the flow inside the passage. The blade-to-blade view confirms that the edge geometry remains continuous between neighboring blades and does not create obvious overlap or blockage.

3-1-26 Meridional Contour of the Final Runner

The final meridional contour of the runner is presented in **Figure 22**. This view shows the radial and axial development of the flow passage from the runner inlet to the outlet. The contour was modified during the design process to remove geometric inconsistencies and to obtain a smooth passage suitable for meshing and CFD simulation.

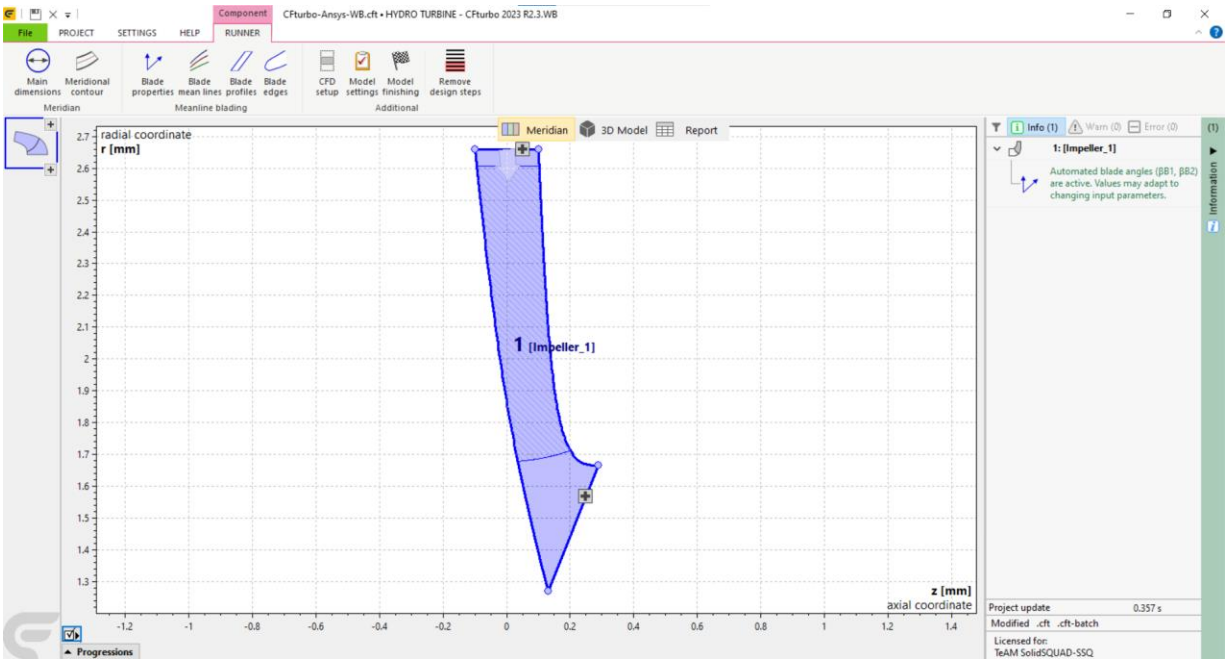


Figure 22: Final meridional contour of the runner

The inlet region is located at the larger radius, while the outlet region is directed toward the smaller radius and axial discharge direction. This configuration is consistent with a Francis turbine runner, where the flow enters the runner radially or mixed-radially and leaves in a more axial direction. The message panel indicates that the automated blade angles are active, which means that the final blade shape can still be influenced by the inlet and outlet blade-angle settings.

3-1-27 Final Three-Dimensional Runner Geometry

The completed three-dimensional runner geometry is shown in **Figure 23**. The model includes the full runner with the final blade arrangement, hub, shroud, and blade passages. The absence of error messages in the CFturbo interface indicates that the final geometry was successfully generated and is suitable for export to the next stage of the analysis.

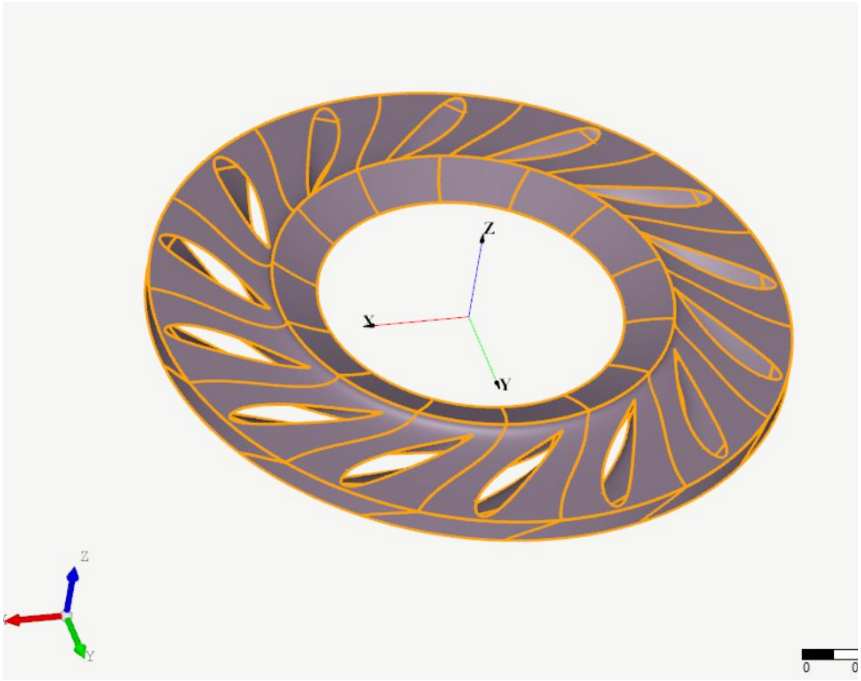


Figure 23: Final three-dimensional model of runner

The 3D view also confirms that the blade passages are regularly distributed around the runner circumference and that the blade surfaces are continuous from leading edge to trailing edge. This final inspection was necessary before exporting the geometry to TurboGrid/ANSYS CFX, since even small geometric problems at the leading edge, trailing edge, hub, or shroud can cause severe meshing difficulties.

3-1-28 Performance Prediction

The predicted performance curve of the runner is shown in **Figure 24**. The design point used in CFturbo corresponds to a volume flow rate of approximately 59 m^3/s, a head of about 516 m, and a rotational speed of 254 254 rpm. The runner diameter shown in the design point data is approximately $5.32m$.

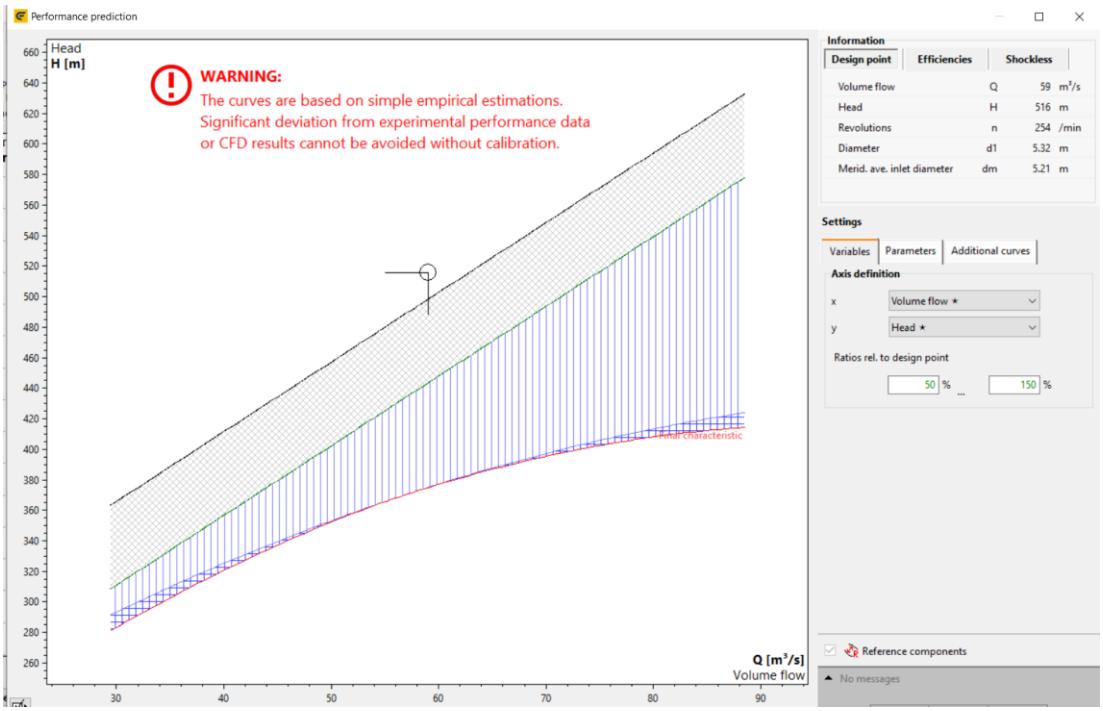


Figure 24: CFturbo performance prediction curve for the runner design

The performance prediction plot provides an initial estimate of the runner behavior over a range of flow rates. However, CFturbo also displays a warning that these curves are based on empirical estimations and may differ significantly from experimental or CFD results. Therefore, this performance curve was used only as a preliminary design reference. A more reliable evaluation of the runner performance requires CFD simulation, where pressure distribution, velocity field, torque, and hydraulic losses can be analyzed in greater detail.

3-2 Volute Design

The spiral casing (volute) was designed in CFturbo as the stationary flow-guiding component located upstream of the distributor/runner assembly. Its main function is to collect and distribute the incoming flow uniformly around the circumference before the flow enters the guide and runner passages. In the present design, the volute geometry was generated based on the outlet/inlet coupling conditions imposed by the neighboring hydraulic component, and the design was further refined by defining the spiral area development according to a velocity-based approach.

3-2-1 Volute Setup and Coupling with the Neighboring Component

The initial step in the volute design consisted of defining the coupling between the volute and the neighboring downstream component. As shown in **Figure 25**, the reverse flow direction option was activated, indicating that the volute geometry was determined consistently with the inlet conditions of the downstream component. The hub and shroud reference positions of the neighboring component were both set at a radial coordinate of **r = 2.66 m**, while the axial coordinates were defined as **z = -0.1 m** for the hub side and **z = 0.1 m** for the shroud side. These values established the basic connection interface between the spiral casing and the hydraulic passage located after it.

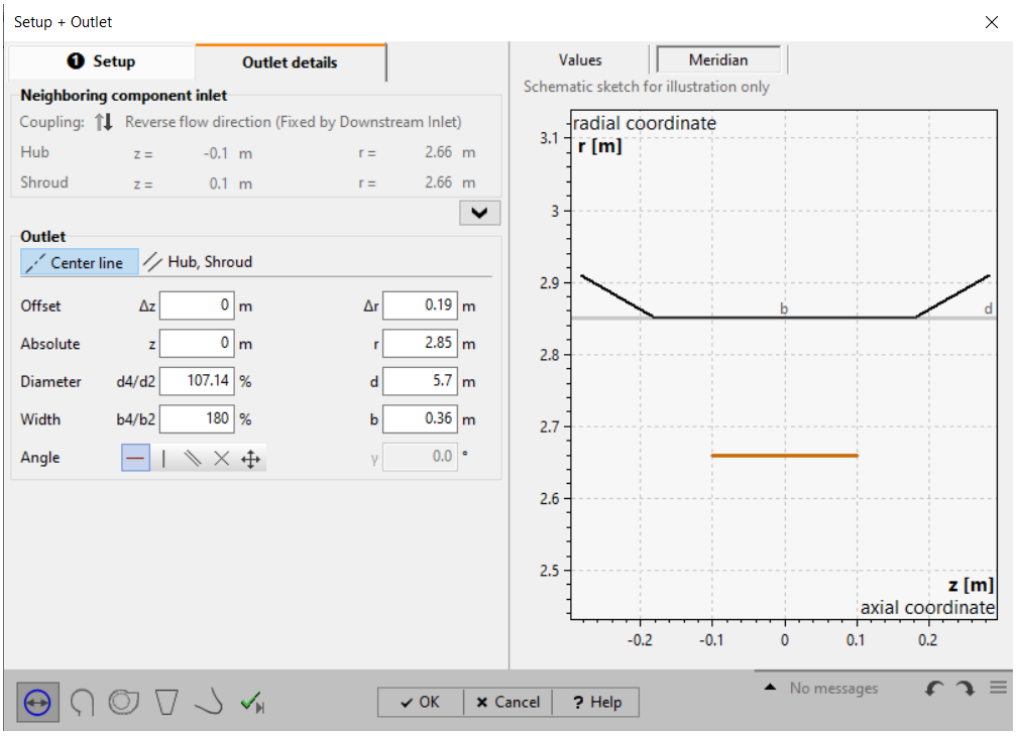


Figure 25: Volute setup and outlet definition in CFturbo, showing the coupling with the downstream component, radial and axial offsets, outlet radius (r = 2.85 m), outlet width (b = 0.36 m), and outlet angle (γ = 0°).

In the same setup window, the outlet centerline parameters of the volute were specified. The radial offset was selected as **Δr = 0.19 m**, while the axial offset was taken as **Δz = 0 m**, meaning that the outlet centerline was shifted radially but not axially. The absolute outlet radius was set to **r = 2.85 m**, corresponding to an outlet diameter of **d = 5.7 m**, which is equal to **107.14%** of the reference diameter d2d_2d2. The outlet width was selected as **b = 0.36 m**, corresponding to **180%**

of b2b_2b2. In addition, the outlet angle was defined as $\boldsymbol{\gamma = 0°}$, which indicates a non-inclined outlet centerline in the meridional representation. These values are illustrated schematically in **Figure 25** and define the main boundary dimensions used for the spiral casing design. Volute setup and outlet definition in CFturbo, showing the coupling with the neighboring component and the main outlet dimensions.

3-2-2 Meridional Outlet Definition

The meridional sketch shown in **Figure 25** provides a simplified but useful geometric interpretation of the volute outlet section. The sketch indicates that the volute cross-section was centered around a mean radius of approximately **2.85 m**, with a width of **0.36 m**. The top and bottom sidewalls transition smoothly from the outlet section toward the neighboring component interface. This kind of outlet definition is important because it directly affects the compatibility between the spiral casing and the adjacent hydraulic element. If the outlet section is poorly matched, non-uniform flow distribution and secondary losses may appear before the fluid even enters the runner zone.

From a hydraulic point of view, the chosen outlet dimensions appear reasonable for the design discharge of **59 m³/s**, since they provide a sufficiently large passage area for the incoming flow while preserving a compact casing geometry. The absence of axial offset further simplifies the connection in the meridional plane and reduces the likelihood of unnecessary flow distortion at the junction.

3-2-3 Spiral Area Development

After defining the outlet interface, the spiral casing area distribution was generated in the **Spiral areas** design module, as shown in **Figure 26**.

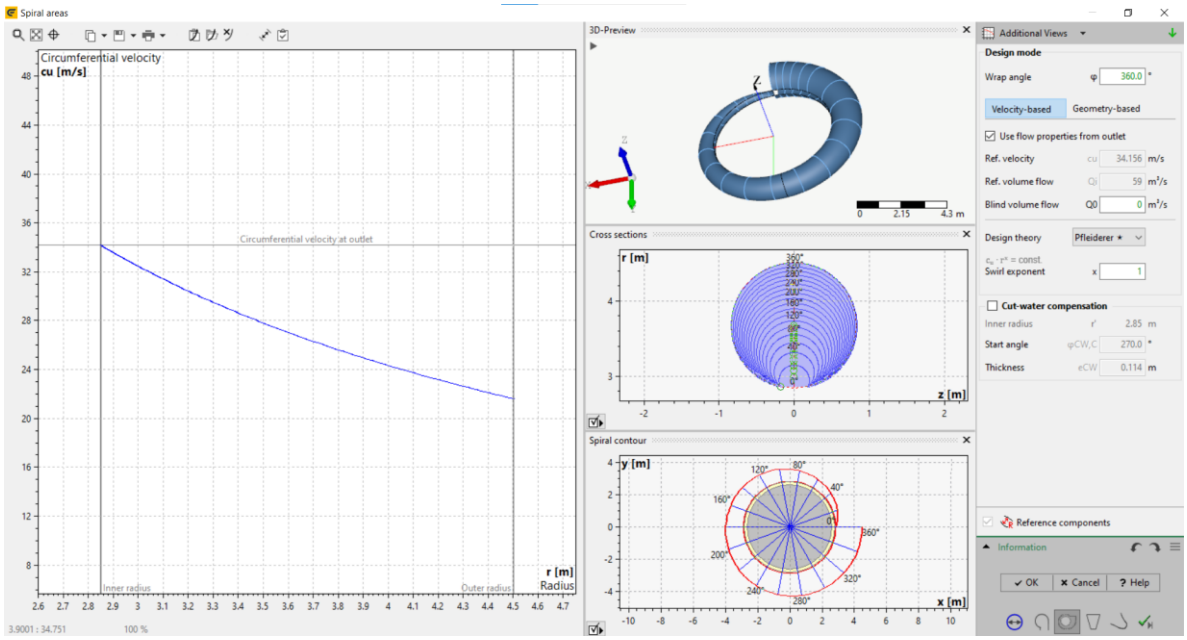


Figure 26: Volute spiral area design in CFturbo, showing the velocity-based design mode, Pfleiderer theory with $c_u \cdot r = const$, circumferential velocity distribution, cross-sectional area variation along the 360° wrap angle, and the spiral contour plan view.

In this step, the total wrap angle of the volute was set to $\boldsymbol{\varphi = 360^{\circ}}$, meaning that the casing fully surrounds the machine. This is a standard and hydraulically suitable choice for a Francis turbine, since a full spiral casing promotes a more uniform circumferential distribution of flow toward the stay vane and guide vane region.

The design mode was selected as **velocity-based**, and the option **Use flow properties from outlet** was enabled. This means that the spiral casing was not defined purely through geometric assumptions; instead, the flow distribution was derived from the hydraulic conditions at the outlet section. The reference velocity was approximately **34.156 m/s**, and the reference volume flow was **59 m³/s**, while the blind volume flow was taken as **0 m³/s**. This indicates that the entire design discharge was considered available for the spiral distribution, without accounting for any bypass or dead-flow region.

The design theory was selected as **Pfleiderer**, with the formulation $\boldsymbol{c_u \cdot r = const}$ and a **swirl exponent of 1**. This is a classical and well-established assumption for spiral casing design, since it implies conservation of angular momentum in an idealized form and produces a gradual reduction in circumferential velocity as the radius increases. This behavior can be observed directly in the velocity plot of **Figure 26**, where the circumferential velocity decreases continuously from the inner radius toward the outer radius. Spiral area design of the volute in CFturbo, including the velocity distribution, cross-sections, and spiral contour.

3-2-4 Velocity Distribution in the Volute

The circumferential velocity plot shown on the left side of **Figure 26** is particularly important for evaluating the hydraulic consistency of the spiral casing. The curve starts near the inner radius at a circumferential velocity of approximately **34 m/s** and gradually decreases to nearly **22 m/s** as the radius increases toward the outer region. This trend is expected and physically meaningful.

The reason for this decrease is that the flow progressively leaves the spiral casing along the circumference and enters the downstream stationary passages. Therefore, as the remaining flow rate within the casing decreases, the required local transport velocity also decreases. A smooth reduction of circumferential velocity is desirable because abrupt changes could produce local separation, excessive pressure gradients, or non-uniform flow feeding around the periphery.

The plotted trend suggests that the designed spiral casing provides a controlled and continuous redistribution of momentum, which is favorable for obtaining a more even inflow to the downstream blade passages. Although the performance predicted in CFturbo at this stage remains preliminary, the shape of the circumferential velocity curve indicates that the casing was designed in a hydraulically coherent manner.

3-2-5 Cross-Sectional Variation Along the Wrap Angle

The cross-section view in **Figure 26** shows a family of sections distributed along the full wrap angle of the volute. These sections appear to maintain a nearly circular or smoothly rounded form, which is beneficial from both hydraulic and structural viewpoints. A smooth cross-sectional

evolution helps avoid sudden changes in local area, thereby reducing losses and promoting stable flow guidance.

In addition, the lower-right panel of **Figure 26** presents the spiral contour in plan view. This plot confirms that the casing expands gradually along the circumferential direction, in agreement with the area increase required by the decreasing through-flow remaining inside the casing. The arrangement of the radial divisions and angular markers further indicates that the area distribution was applied consistently around the full 360-degree casing.

Such a geometric progression is one of the defining characteristics of a well-designed turbine volute: at the beginning of the spiral, the local area is larger because it carries the full discharge, whereas near the end of the spiral the remaining flow is much smaller and the cross-sectional area can therefore be reduced. The plot shown in **Figure 26** confirms that this principle was captured in the present design.

3-2-6 Volute Geometry and Wall Thickness

The design settings in **Figure 26** also show that the spiral casing inner radius was taken as **2.85 m**, the **start angle** was set to **270°**, and the casing **thickness** was selected as **0.114 m**. These parameters define the geometric envelope of the volute. The inner radius controls the proximity of the volute to the runner/stator region, while the thickness affects both structural integrity and the available hydraulic passage. The 270-degree starting angle defines the angular position from which the spiral growth begins.

The **cut-water compensation** option was not activated in the shown design. This means that the current geometry was generated without additional correction for the tongue or cut-water region. At the present conceptual design stage, this is acceptable, although in a more advanced design iteration this area may later require refinement because the tongue region often has a strong influence on local pressure fluctuations, flow non-uniformity, and casing losses.

3-2-7 General Assessment of the Volute Design

Based on the setup and spiral area results shown in **Figures 25 and 26**, the designed volute appears to be consistent with the hydraulic requirements of the turbine. The use of the full design discharge $Q = 59\,\mathrm{m}^3/\mathrm{s}$, together with a complete 360-degree wrap angle and Pfleiderer-based velocity distribution, produced a spiral casing with a realistic and physically meaningful variation of circumferential velocity and cross-sectional area.

The chosen outlet dimensions $(\mathrm{d} = 5.7\mathrm{m}, \mathrm{b} = 0.36\,\mathrm{m}, \mathrm{r} = 2.85\mathrm{m})$ also appear compatible with the scale of the machine and with the interface required by the downstream hydraulic component. The meridional sketch confirms that the casing outlet is properly positioned with respect to the neighboring component, while the spiral area module demonstrates that the area progression is smooth and hydraulically reasonable.

Therefore, this stage of the design can be considered a satisfactory preliminary volute configuration in CFturbo. However, as is common in turbine design practice, the final quality of

the casing should later be verified through more detailed hydraulic analysis, especially near the tongue region and at the interface with the distributor or stay vane assembly.

3-2-8 Diffuser Geometry Definition

After defining the basic spiral casing layout and its area development, the next stage of the volute design involved specifying the diffuser geometry that connects the circular inlet conduit to the spiral casing. This step is shown in **Figure 27**, where the diffuser was constructed in the x-y plane and aligned with the circumferential direction of the volute. The graphical window on the left shows the diffuser footprint, while the panels on the right display the corresponding 3D preview, cross-sections, and area progression along the diffuser length.

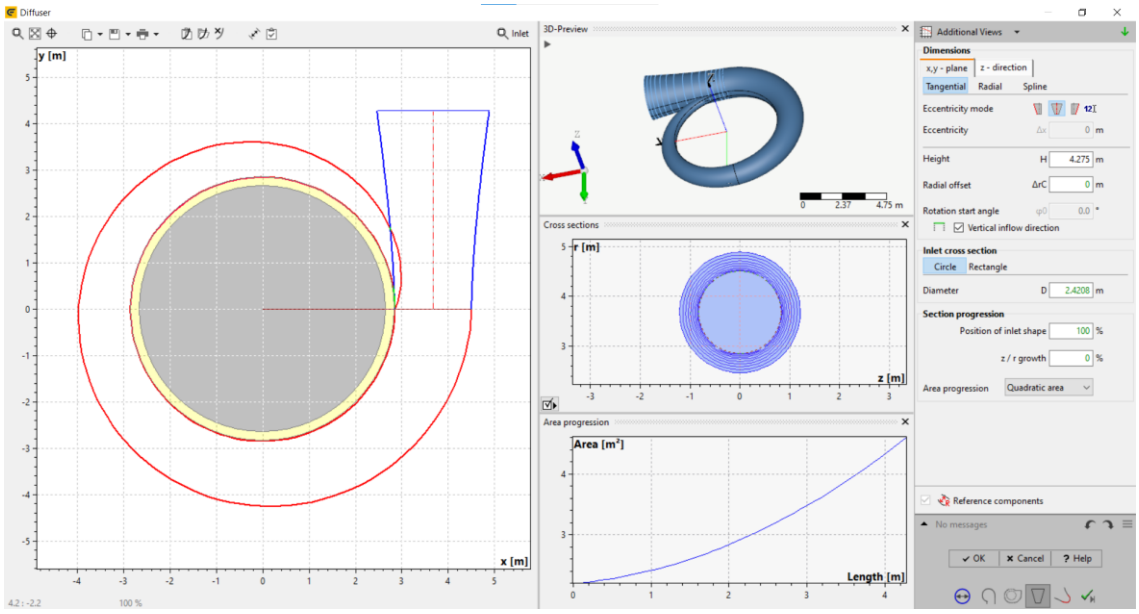


Figure 27: Volute diffuser design in CFturbo, showing the plan-view geometry, 3D preview, cross-section evolution, and quadratic area progression along the diffuser length. The diffuser has a circular inlet section (diameter = 2.4200 m) and a total height of 4.275 m.

As seen in **Figure 27**, the diffuser cross-section was defined as a **circular inlet section** with a diameter of **2.4200 m**. The total diffuser height was selected as **4.275 m**, while the radial offset was set to **0 m**, indicating that the diffuser was introduced without additional radial displacement relative to the reference path. The eccentricity was also defined as **0 m**, meaning that the section centerline remains geometrically centered and no offset eccentric deformation was introduced. The rotation start angle was kept at **0°**, which implies that the diffuser section orientation was not intentionally rotated at the start of the passage.

A particularly important design choice in this stage was the selection of **quadratic area progression**, as shown in the lower plot of **Figure 27** This means that the flow area increases gradually and nonlinearly along the diffuser length. Such a progression is preferred over an abrupt or purely linear expansion because it provides a smoother reduction in mean flow velocity and helps limit local pressure losses. In turbine casing design, a controlled area growth in the diffuser is desirable because it improves flow adaptation from the feeding pipe to the spiral casing and reduces the likelihood of separation or strong local recirculation zones.

The cross-section distribution shown in **Figure 27** confirms that the diffuser was discretized into a series of smoothly varying circular sections. This representation indicates that the transition from the inlet conduit to the volute body was built as a continuous geometric connection rather than a sharp junction. From both hydraulic and geometric points of view, this is an appropriate design strategy for a high-head Francis turbine, where losses at the casing entry should be minimized as much as possible. Diffuser design of the volute in CFturbo, showing the plan-view geometry, 3D preview, cross-section evolution, and quadratic area progression.

3-2-9 Cutwater (Tongue) Design

Once the diffuser geometry had been established, the cutwater region of the volute was designed. The cutwater, also referred to as the tongue region, is one of the most sensitive parts of the spiral casing because it strongly affects the local redistribution of flow and the uniformity of the inflow delivered to the downstream stationary and rotating components. The cutwater design used in the present work is shown in **Figure 28**.

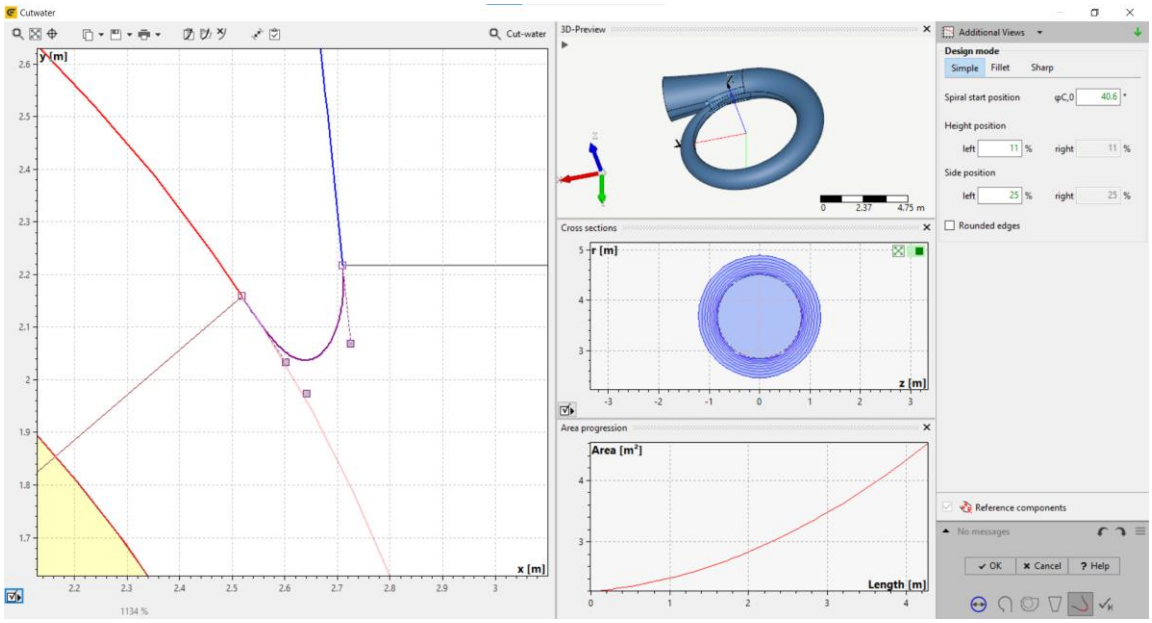


Figure 28: Cutwater (tongue) design of the volute in CFturbo, showing the spiral start position (φC,0 = 40.6°), height and side positions (11% and 25%), rounded-edge definition, and the associated area progression plot.

As can be seen in **Figure 28**, the cutwater was designed in **simple mode**. The spiral start position was defined as $\boldsymbol{\varphi_{C,0} = 40.6^{\circ}}$. This parameter determines the angular location at which the cutwater begins relative to the spiral wrap and therefore controls the spatial relationship between the inlet conduit and the initial spiral development. The cutwater location is critical because it determines how the incoming flow is introduced into the spiral casing and how the remaining circumferential passage is distributed around the runner.

The right-side design settings in **Figure 28** show that the **height position** of the cutwater was set to **11%** on both the left and right sides, while the **side position** was set to **25%** on both sides. In addition, the option **Rounded edges** was activated. This is a hydraulically beneficial choice, since rounded cutwater edges reduce sharp local gradients, decrease the risk of local separation, and moderate the strong pressure non-uniformities that can otherwise arise near the tongue region.

The geometry shown in the left panel of **Figure 28** confirms that the cutwater was shaped as a smooth transition curve rather than a sharp geometric corner. This is important because the cutwater region represents the point where the spiral casing effectively terminates and where the smallest remaining circumferential flow region exists. In practical hydraulic design, the tongue region often governs pressure pulsation behavior and can have a considerable effect on rotor-stator interaction. Although the CFturbo geometry at this stage represents a preliminary design, the use of a rounded and smooth tongue profile is fully consistent with good engineering practice.

The lower-right area progression plot in **Figure 28** further confirms that the area variation continues smoothly through the casing development and does not exhibit abrupt local discontinuities. This is another indication that the cutwater region was integrated properly into the global casing geometry. Cutwater design of the volute, showing the tongue geometry, cutwater position parameters, rounded-edge definition, and associated area progression.

3-2-10 Three-Dimensional Assembly of the Volute and Runner

After completing the diffuser and cutwater definitions, the three-dimensional geometry of the spiral casing was generated and assembled together with the runner. This full configuration is shown in **Figure 29**. The blue solid represents the volute, while the pink component inside it corresponds to the runner previously designed in CFturbo.

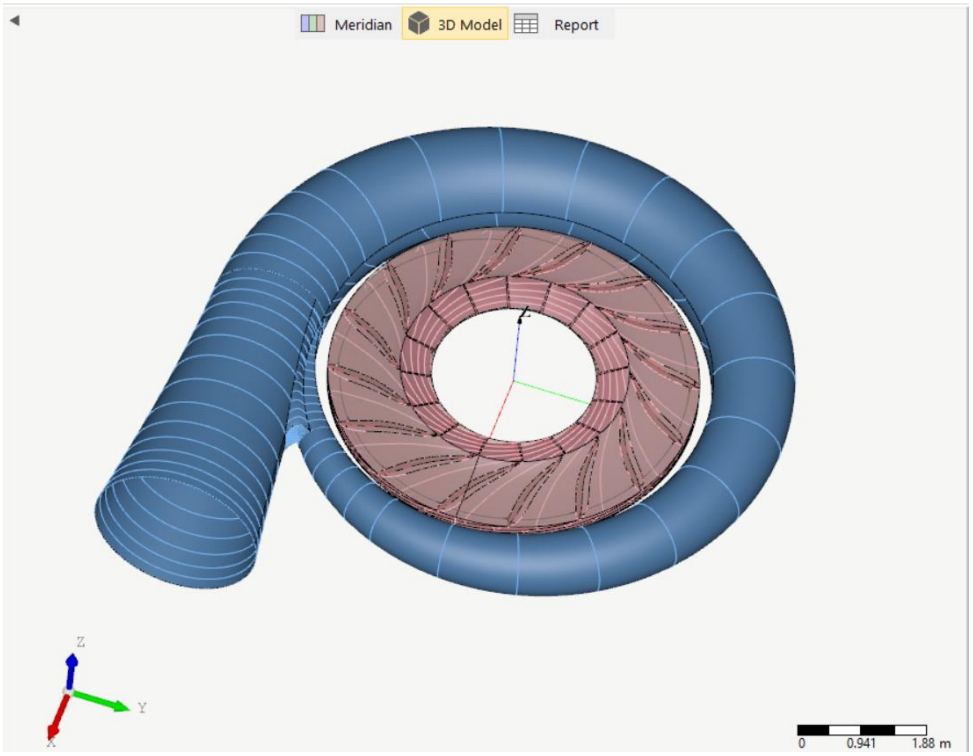


Figure 29: Three-dimensional assembly of the volute (blue) and runner (pink) components, illustrating the overall geometric integration and spatial compatibility of the casing and rotating component.

The 3D representation in **Figure 29** clearly shows that the spiral casing fully surrounds the runner region and that the inlet conduit is smoothly connected to the casing body. This confirms that the diffuser and spiral sections were successfully integrated into a unified casing geometry. The assembled view also makes it possible to assess the overall packaging and spatial

compatibility of the machine. In particular, the geometry indicates that the casing diameter and wrap angle are appropriate relative to the runner outer envelope.

From a design standpoint, this figure is highly useful because it provides a direct visual confirmation that the stationary and rotating components occupy consistent spatial regions. A proper relative placement between volute and runner is essential, since the casing must deliver the flow uniformly into the stay vane/guide vane zone and subsequently toward the runner inlet. Although guide vane and stay vane components are not shown in this particular image, the general relationship between the casing and runner can already be evaluated.

Another important observation from **Figure 29** is the continuity of the casing wall and the smooth curvature of the spiral development. No abrupt contraction or nonphysical geometric distortion is visible in the casing body. This is desirable because highly curved or discontinuous casing surfaces tend to introduce unnecessary hydraulic losses. Therefore, the resulting 3D model can be considered geometrically sound at the preliminary design level. Three-dimensional model of the volute assembled around the runner, illustrating the overall geometric integration of the casing and rotating component.

3-2-11 Meridional Integration of the Volute with the Runner

A further confirmation of the compatibility between the spiral casing and the runner is provided by the meridional view shown in **Figure 30**. In this figure, the volute is displayed as the upstream stationary component labeled **Volute_1**, while the runner is shown below as **Impeller_1**. The figure provides a sectional representation of the relative position between the two components in the meridional plane.

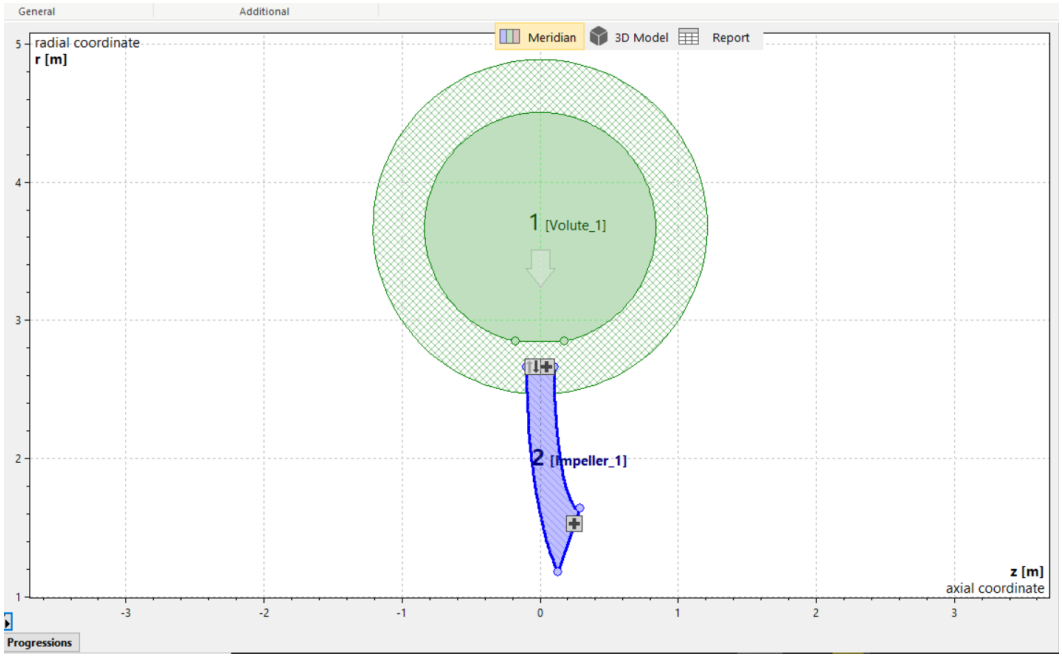


Figure 30: Meridional view of the volute and runner assembly, showing the geometric compatibility between the upstream spiral casing (Volute_1) and the downstream runner (Impeller_1), with the flow direction indicated by the downward arrow.

The cross-hatched green annular region in **Figure 30** corresponds to the volute passage, while the blue meridional runner contour represents the previously designed runner inlet and outlet

region. This view is very important because it reveals whether the casing outlet aligns properly with the runner inlet zone. Based on the figure, the volute outlet is positioned directly above the runner entry region, which is consistent with the intended hydraulic flow path. The axial alignment appears reasonable, and the radial position of the volute also corresponds well to the runner inlet radius.

The central downward arrow shown inside the volute in **Figure 30** indicates the general direction of flow transfer from the casing toward the downstream hydraulic system. This graphical representation supports the physical interpretation that the spiral casing is feeding the flow toward the internal turbine passage. The meridional compatibility observed here is a necessary condition for achieving an acceptable final hydraulic design.

It should be emphasized that in a full Francis turbine system, the spiral casing does not deliver the flow directly into the runner without intermediate control elements. Rather, the flow typically passes through stay vanes and guide vanes before reaching the runner. However, even at this stage, the meridional alignment shown in **Figure 30** is useful because it ensures that the casing outlet is located in the correct general region with respect to the runner entrance. Therefore, the figure confirms that the volute geometry is appropriately positioned for further integration with the remaining turbine components. Meridional view of the volute and runner assembly, showing the geometric compatibility between the upstream spiral casing and the downstream runner region.

3-2-12 Hydraulic and Geometric Interpretation of the Volute Design

Taken together, **Figures 27-30** provide a coherent picture of the volute design process. The procedure began with the definition of the diffuser section and its area progression, continued through the shaping of the cutwater region, and culminated in the complete 3D and meridional integration of the casing with the runner.

From the hydraulic point of view, the use of a circular diffuser inlet with quadratic area progression is a reasonable choice because it promotes a smooth transfer of water from the supply conduit into the spiral casing. The cutwater design, with rounded edges and carefully selected positional parameters, further improves the flow guidance at the most critical circumferential region of the casing. Finally, the assembled 3D and meridional views indicate that the volute can be integrated into the overall turbine layout without apparent geometric inconsistency.

3-3 Stator Design

3-3-1 Stator Component Definition

After completing the runner and volute design stages, the next component added to the CFturbo model was the stator. The stator was introduced to provide a controlled stationary passage downstream of the runner region and to improve the geometric continuity of the hydraulic path. The initial stator setup and main meridional dimensions are shown in **Figure 31**.

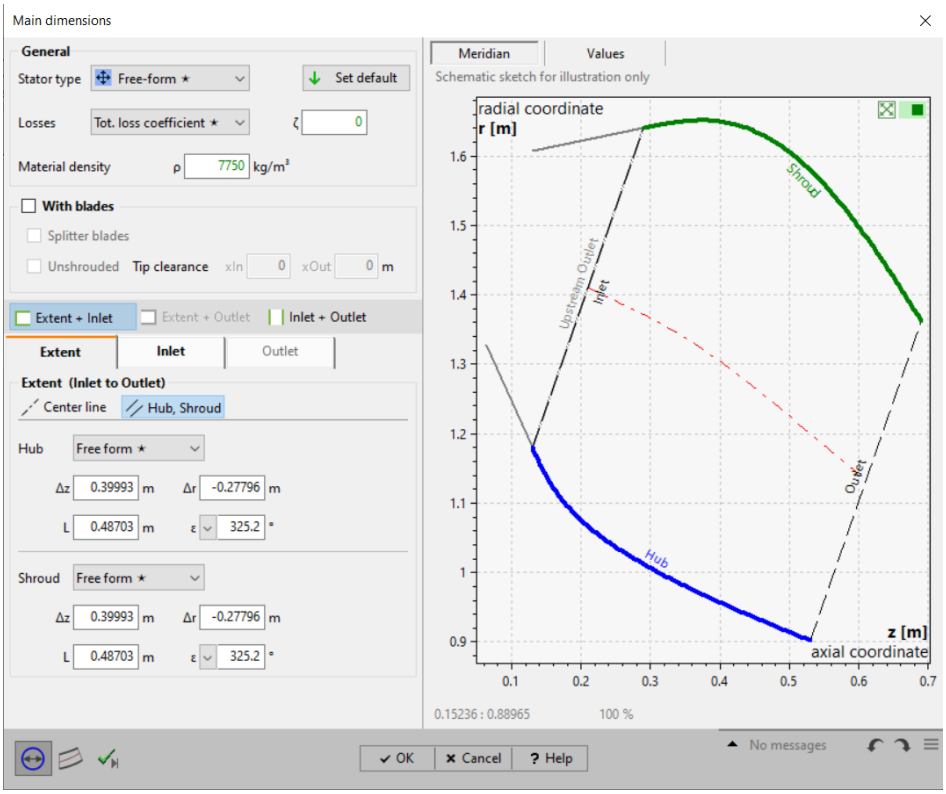


Figure 31: Initial stator component setup in CFturbo, showing the Free-form type, loss coefficient ($\zeta = 0$), material density ($\rho = 7750$ kg/m³), and main meridional dimensions including hub and shroud extents ($\Delta z = 0.39993$ m, $\Delta r = -0.27796$ m) without blade activation.

In this design step, the stator type was selected as **Free-form**, allowing direct control of the hub and shroud contours. This approach was appropriate because the stator passage had to be adjusted according to the previously generated runner outlet geometry rather than being generated from a purely automatic default shape.

The loss model was defined using the total loss coefficient option, with the loss coefficient initially set to:

$$\zeta = 0$$

This means that at this stage the stator was primarily treated as a geometric component, and detailed hydraulic loss estimation was not yet imposed. The material density was set to:

$$\rho = 7750\ \mathrm{kg/m^3}$$

which is consistent with the steel-based material assumption already used for the runner and other turbine components.

A key observation from **Figure 31** is that the **With blades** option is not activated. Therefore, this stage should be interpreted as the design of a **bladeless stator/diffuser-type stationary passage**, not as the final

detailed design of stator vanes or guide vanes. If a fully bladed stator is required later, the blade option must be enabled and a separate blade design procedure must be completed.

3-3-2 Meridional Extent of the Stator

The stator geometry was defined using the **Extent + Inlet** setting, as shown in **Figure 31**. Both the hub and shroud contours were defined using the free-form option. The same axial and radial extent parameters were used for both hub and shroud:

$$\Delta z = 0.39993\ m, \Delta r = -0.27796\ m$$

The meridional length was:

$$L = 0.48703\ \text{m}$$

and the corresponding contour angle was approximately:

$$\varepsilon = 325.2°$$

The negative radial offset indicates that the stator passage contracts inward in the radial direction as it progresses along the meridional path. This is visible in **Figure 31**, where both the hub and shroud contours move toward smaller radial coordinates.

The hub contour is shown in blue and the shroud contour is shown in green. The red dashed line represents the mean meridional flow path. The purpose of this geometry is to create a smooth stationary passage connected to the runner outlet, while avoiding sudden geometric discontinuity between components.

3-3-3 Meridional Contour Refinement

After defining the initial stator extent, the hub and shroud contours were refined in the meridional contour environment. The refined contour is shown in **Figure 32**.

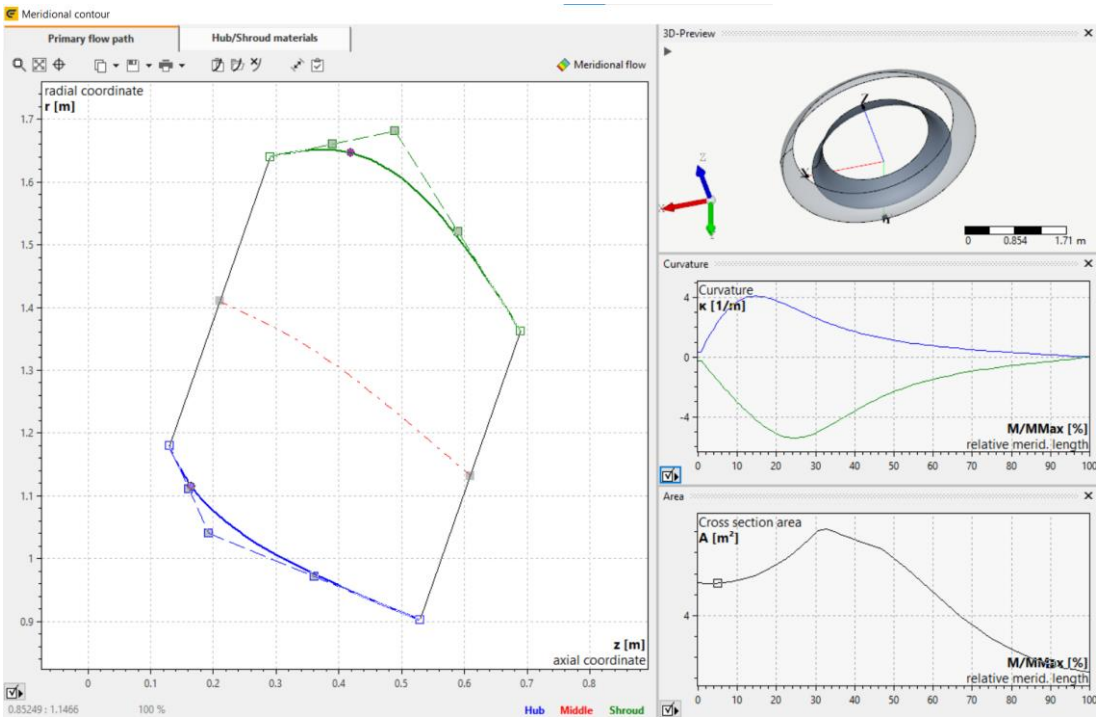


Figure 32: Refined meridional contour of the stator, showing the hub (blue), shroud (green), and mean meridional flow path (red dashed), along with the curvature and cross-sectional area distributions along the relative meridional length.

The hub contour follows a smooth inward-curving path, while the shroud contour defines the outer boundary of the stationary passage. The stator passage begins near the runner outlet region and then

develops into a curved annular passage. This geometry is important because the flow leaving the runner may still contain residual swirl and non-uniform velocity distribution. A properly shaped stationary passage can help guide this flow more gradually and reduce local geometric losses.

The right side of **Figure 32** presents the curvature and cross-sectional area evaluations. These plots were used to check the quality of the free-form geometry. The curvature plot shows that both hub and shroud contours vary continuously along the meridional length. Although the curvature is not perfectly uniform, no extreme discontinuity is visible. This indicates that the contour is acceptable for the preliminary model.

The cross-sectional area plot in **Figure 32** shows an initial slight increase followed by a decrease along the relative meridional length. This behavior should be interpreted carefully. A moderate change in area is acceptable, but a strong contraction could increase local velocity and hydraulic losses. Therefore, this region should be checked again during CFD validation, especially if the stator is intended to act as a diffuser or flow-recovery passage.

3-3-4 5.4 Integration with Runner and Volute in Meridional View

The complete meridional assembly of the volute, runner, and stator is shown in **Figure 33**. In this figure, the volute is labeled as **Volute_1**, the runner is labeled as **Impeller_1**, and the stator is labeled as **Stator_1**.

The stator is positioned downstream of the runner outlet region. This arrangement is geometrically reasonable because the stator component must receive the flow after it leaves the runner and guide it toward the next downstream hydraulic passage. The meridional position shown in **Figure 33** confirms that the stator is connected to the runner region and occupies the correct general location in the turbine assembly.

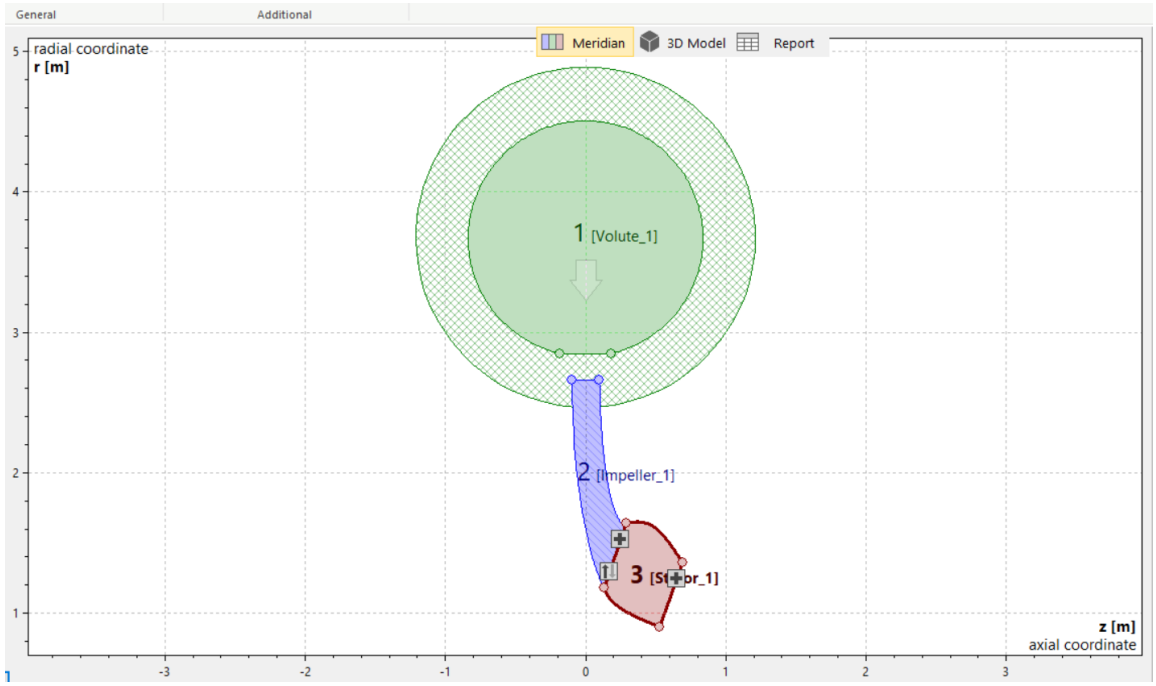


Figure 33: Complete meridional assembly of the volute (Volute_1), runner (Impeller_1), and stator (Stator_1), showing the component order (Volute → Runner → Stator) and the geometric continuity of the hydraulic path.

The figure also shows that the volute remains upstream and surrounds the runner region, while the stator is located after the runner. This confirms the intended component order:

$$Volute \rightarrow Runner \rightarrow Stator$$

This component arrangement is consistent with the CFturbo model structure used in the current design.

However, based on **Figure 33**, the stator geometry is still relatively compact and should be treated as a preliminary stationary outlet passage. If the final turbine configuration requires a full distributor or guide-vane system upstream of the runner, that component must be designed separately.

3-3-5 Three-Dimensional Assembly Check

The three-dimensional assembly of the turbine components is shown in **Figure 34**. The blue component represents the volute, while the central rotating runner region and the added stator passage are visible inside the casing assembly.

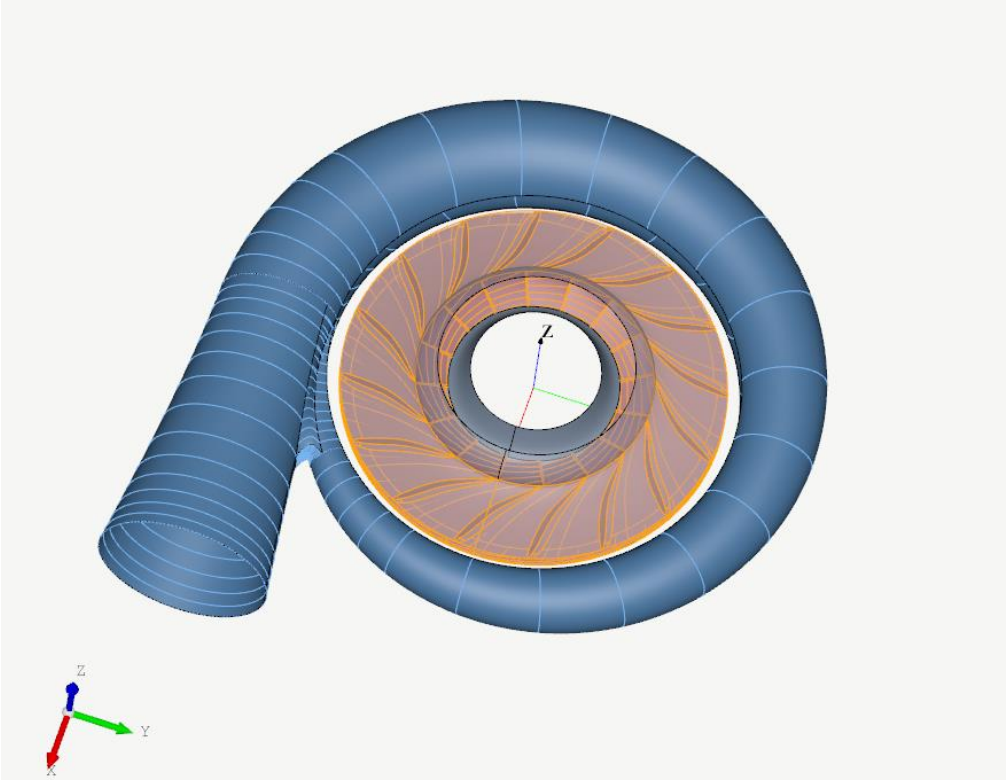


Figure 34: Three-dimensional assembly of the complete turbine model including the volute (blue), runner (central rotating region), and stator (downstream passage), showing the successful integration and spatial compatibility of all hydraulic components.

The 3D view confirms that the stator component was successfully integrated into the turbine model. The geometry does not show an obvious interference with the volute or runner in the displayed view. The annular arrangement also appears consistent with the axisymmetric nature of the turbine flow path.

This step is important because a component can appear acceptable in the meridional view but still create geometric problems in the full three-dimensional model. In **Figure 34**, the stator is shown as part of the assembled hydraulic passage, and the overall geometry remains continuous.

The 3D view also confirms that the turbine model has now progressed from isolated runner and volute design to a more complete multi-component assembly. This geometry can be used as the basis for later export, meshing, and CFD analysis.

Meshing with Turbo Mesh

The meshing process was carried out using **ANSYS TurboGrid** inside the ANSYS Workbench environment. As shown in **Figure 35**, the final runner geometry designed in CFturbo was transferred to TurboGrid for structured blade-passage meshing. The generated TurboGrid mesh was then connected to the CFX setup for flow simulation.

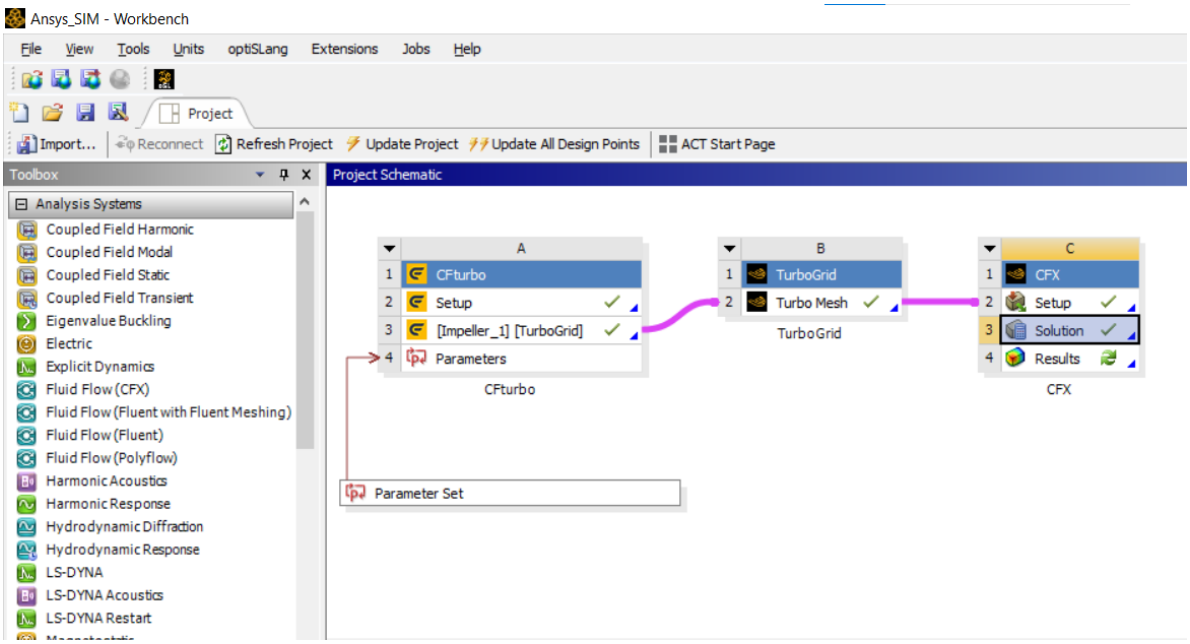


Figure 35: ANSYS Workbench workflow for transferring the CFturbo runner geometry to TurboGrid and CFX

The use of TurboGrid was selected because the geometry belongs to a turbomachinery blade passage. In the earlier stage of the project, attempts were made to mesh the runner domain using a general-purpose meshing approach. However, due to the high curvature of the Francis runner blade, the narrow blade-to-blade passage, and the sensitivity of the leading and trailing edge regions, generating a reliable mesh directly in a general meshing environment was difficult. Therefore, TurboGrid was used because it is specifically developed for turbomachinery passages and can generate high-quality structured meshes around blade, hub, shroud, inlet, outlet, and periodic surfaces.

The Workbench schematic in **Figure 35** shows the complete data transfer path. The CFturbo component contains the finalized impeller design, which was sent to TurboGrid as the blade-row geometry. After the mesh generation was completed, the TurboGrid mesh was passed directly to CFX. This workflow reduced the chance of geometry translation errors and allowed the same periodic passage definition to be preserved during the CFD setup.

4-1 Single-Passage Mesh Strategy

The computational domain was meshed as a **single periodic runner passage**, as shown in **Figure 36**. Since the runner consists of 14 blades, the full annular runner can be represented by one periodic sector. This approach significantly reduces the computational cost while preserving the physical behavior of the complete runner through periodic boundary conditions.

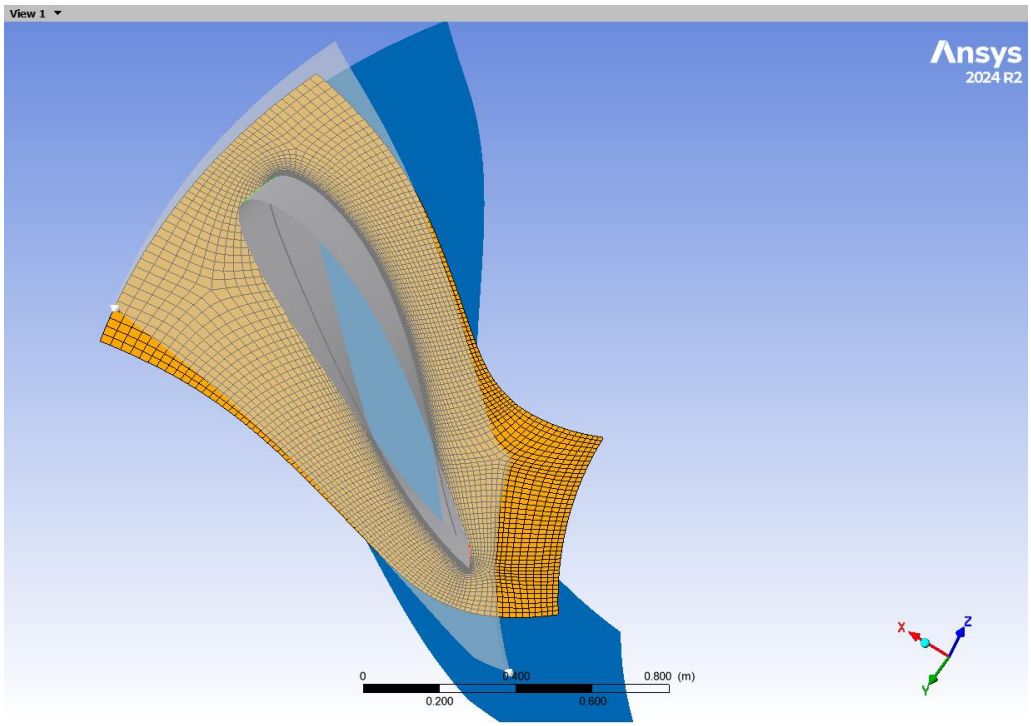


Figure 36: Structured mesh of the single periodic runner passage generated in TurboGrid

The single-passage method is commonly used in turbomachinery CFD when the geometry and flow field are assumed to be circumferentially periodic. Instead of solving the entire 360-degree runner, only one blade-to-blade passage is modeled. The flow quantities on the two side surfaces of the passage are linked using periodic interfaces. Therefore, the solution in this sector represents the repeating flow pattern of the full runner.

As shown in **Figure 36**, the mesh follows the curved geometry of the runner blade passage. The cells are distributed from the inlet region to the outlet region and are aligned with the hub, shroud, and blade surfaces. The structured arrangement of elements around the blade helps to capture the strong flow curvature and pressure variation inside the passage. The mesh also extends sufficiently across the inlet and outlet regions, allowing the boundary conditions to be applied away from the most curved parts of the blade.

The use of a structured blade-passage mesh was important because the Francis runner has a complex three-dimensional shape. The blade twists from inlet to outlet, and the passage area changes along the meridional direction. A structured TurboGrid mesh provides better control over element alignment compared with an unstructured mesh, especially near the blade leading edge, trailing edge, hub, and shroud.

4-2 Topology Selection

The selected topology is shown in **Figure 37**. The final topology used for the runner was **Single Round Round Symmetric**. This topology is suitable for a single-bladed periodic passage with rounded leading and trailing edges. It was selected because the runner geometry did not include splitter blades, and therefore a single-blade topology was more appropriate than splitter-based topologies.

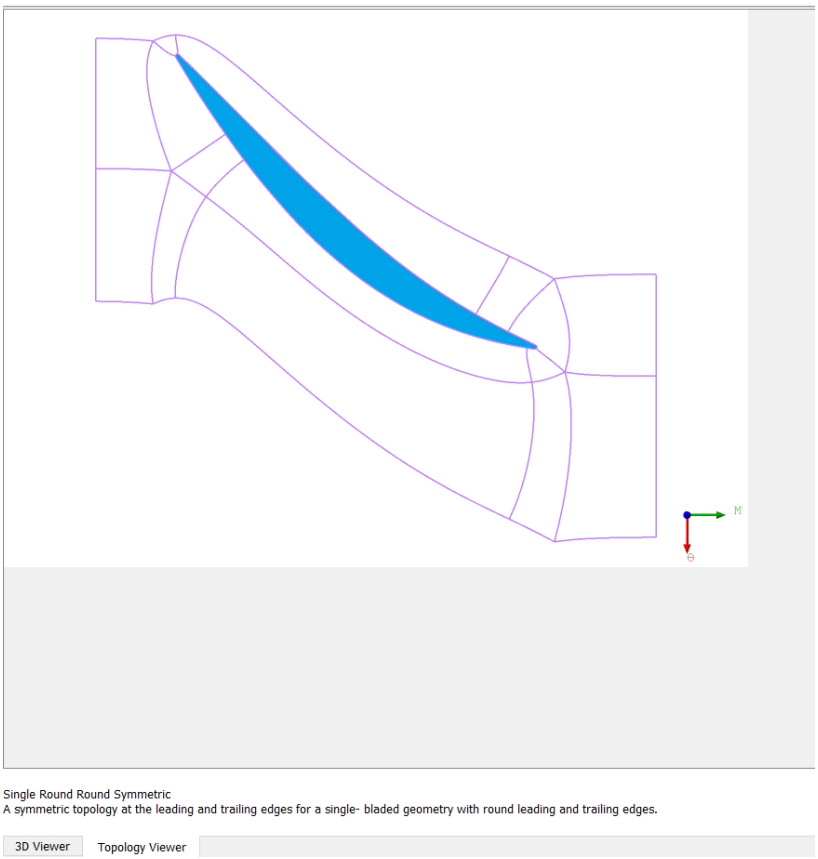


Figure 37: Single Round Round Symmetric topology used for the runner blade passage

During the meshing process, the topology selection was an important step. Some automatically suggested topology options were intended for splitter-blade configurations, which were not suitable for this runner. Therefore, a single-blade topology was selected to match the actual blade arrangement. The **Single Round Round Symmetric** topology creates a symmetric mesh structure around both the leading and trailing edge regions. This is important for avoiding highly distorted cells in the regions where the blade thickness becomes small and where the flow direction changes rapidly.

As shown in **Figure 37**, the blade is located inside a periodic passage bounded by the hub and shroud surfaces. The topology lines define how the computational grid is distributed around the blade surface and through the passage. The rounded leading edge and rounded trailing edge treatment help to improve mesh smoothness and reduce the possibility of negative volumes or poor-quality cells in these sensitive regions.

This topology was therefore selected as the final meshing topology because it matched the geometry of the runner blade and provided a stable basis for the CFD mesh generation.

4-3 Mesh Quality Statistics

The final mesh quality was evaluated using the TurboGrid mesh statistics shown in **Figure 38**.

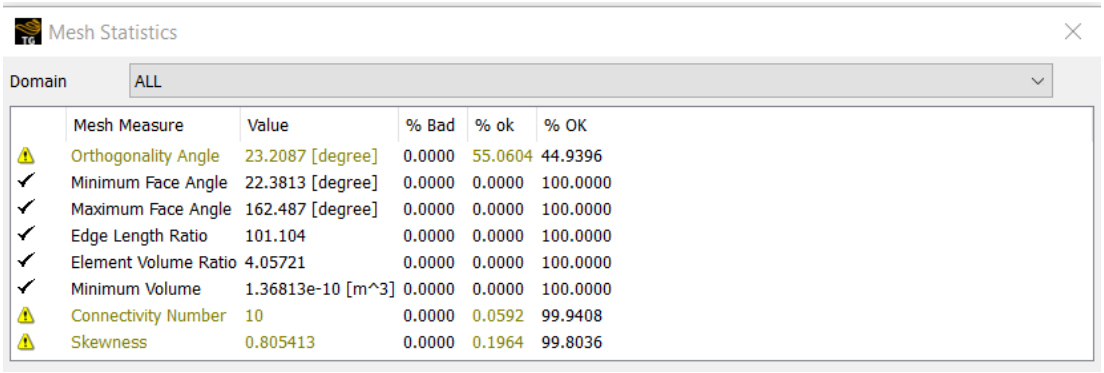


Figure 38: Mesh quality statistics of the TurboGrid runner passage mesh

The generated mesh contained approximately 169047 *nodes* and 156910 *elements*.

The mesh quality statistics are summarized in Table 14.

Table 14: Mesh quality parameters of the final TurboGrid mesh

Mesh Measure	Value	% Bad	Interpretation
OrthogonalityAngle	23.2087 ∘ 23.2087°23.2087 ∘	0.0000	*Acceptable; nobadcells*
MinimumFaceAngle	22.3813 ∘ 22.3813°22.3813 ∘	0.0000	*Good*
MaximumFaceAngle	162.487 ∘ 162.487°162.487 ∘	0.0000	*Good*
Edge Length Ratio	101.104	0.0000	Acceptable
Element Volume Ratio	4.05721	0.0000	Good
MinimumVolume	$1.36813 \times 10 - 10\,m31.36813 \times 10^{-10}\,\mathrm{m}^3 1.36813 \times 10 - 10m3$	0.0000	*Positivecellvolume*
Connectivity Number	10	0.0000	Acceptable
Skewness	0.805413	0.0000	Acceptable for CFD run

The most important point in **Figure 38** is that all mesh-quality parameters show:

$$\%Bad = 0$$

This means that no elements were classified as bad according to the TurboGrid quality criteria. In addition, the minimum element volume is positive, which confirms that the mesh does not contain negative-volume cells. This is a critical requirement before transferring the mesh to CFX, because negative or highly distorted elements can lead to solver divergence.

The orthogonality angle was 23.2087°. Although this value was marked with a warning symbol, the percentage of bad elements was zero. Therefore, the orthogonality was considered acceptable for the CFD simulation. The mesh also showed a maximum skewness value of 0.805413. This value is not ideal, but it remained within an acceptable range for a first structured turbomachinery simulation, especially considering the high curvature and compact blade-passage shape of the Francis runner.

The edge length ratio was 101.104, which indicates that some cells are stretched. This behavior is expected in turbomachinery blade-passage meshes because finer cells are needed near blade surfaces, leading and trailing edges, hub, and shroud, while larger cells can be used farther from the walls. Since the percentage of bad cells was zero, the stretching was not considered harmful for the solver.

The element volume ratio was 4.05721, and the minimum volume was $1.36813 \times 10^{-10}\,\mathrm{m}^3$. These values indicate that the element size variation was controlled and that the mesh did not contain collapsed

or inverted cells. The minimum and maximum face angles were also within acceptable limits, with no bad elements reported.

4-4 Periodic Passage Representation

Because only one runner passage was modeled, periodicity was a necessary part of the mesh and simulation setup. The side surfaces of the mesh correspond to the lower and upper periodic boundaries of the blade passage. These periodic surfaces were later connected in CFX using a periodic interface. Since the runner contains 14 blades, the pitch angle of one passage is:

$$\theta = \frac{360^\circ}{14} = 25.714^\circ$$

Therefore, the single passage represents one 25.714° sector of the complete runner. This periodic representation allowed the full runner behavior to be modeled with much lower computational cost.

The periodic approach was especially useful because the final CFD model required several trial runs to check rotation direction, inlet flow condition, and convergence stability. Solving the full runner would have required significantly more computational resources and longer simulation time. The single-passage approach therefore provided a practical and physically reasonable solution for the present design study.

4-5 Mesh Preparation for CFX

After checking the mesh quality, the TurboGrid mesh was transferred to ANSYS CFX. The wall group included the blade, hub, and shroud surfaces. The inlet and outlet surfaces were used to define the mass-flow inlet and pressure outlet boundary conditions. The two periodic sides were used to define the circumferential periodicity of the runner passage.

This region definition was important because the CFD setup depended directly on correct boundary assignment. In particular, the periodic sides had to remain periodic interfaces and not be included in the wall boundary. Similarly, the blade, hub, and shroud surfaces had to remain no-slip rotating walls. The correct separation of these regions made it possible to calculate the runner torque and pressure drop from the final CFD solution.

4-6 Final Evaluation of Mesh Suitability

Based on the mesh visualization and mesh-quality statistics, the final TurboGrid mesh was considered suitable for the CFD simulation. The mesh was structured, aligned with the blade passage, and free of bad cells. The leading and trailing edge regions were handled using a suitable rounded-edge topology, and the mesh maintained positive cell volumes throughout the domain.

The final mesh was therefore accepted for the CFX analysis. The subsequent CFD simulations confirmed the suitability of the mesh because the solver was able to converge successfully after applying the corrected swirl inlet condition and high-resolution numerical schemes

Simulation with Ansys CFX

5-1 CFX Workflow and Computational Setup

After generating the structured mesh in ANSYS TurboGrid, the mesh was transferred to ANSYS CFX for the numerical simulation of the Francis runner passage. As shown in **Figure 39**, the computational domain represents a single periodic runner passage. The simulation was performed using a rotating domain, periodic interfaces, a mass-flow inlet, and a pressure outlet.

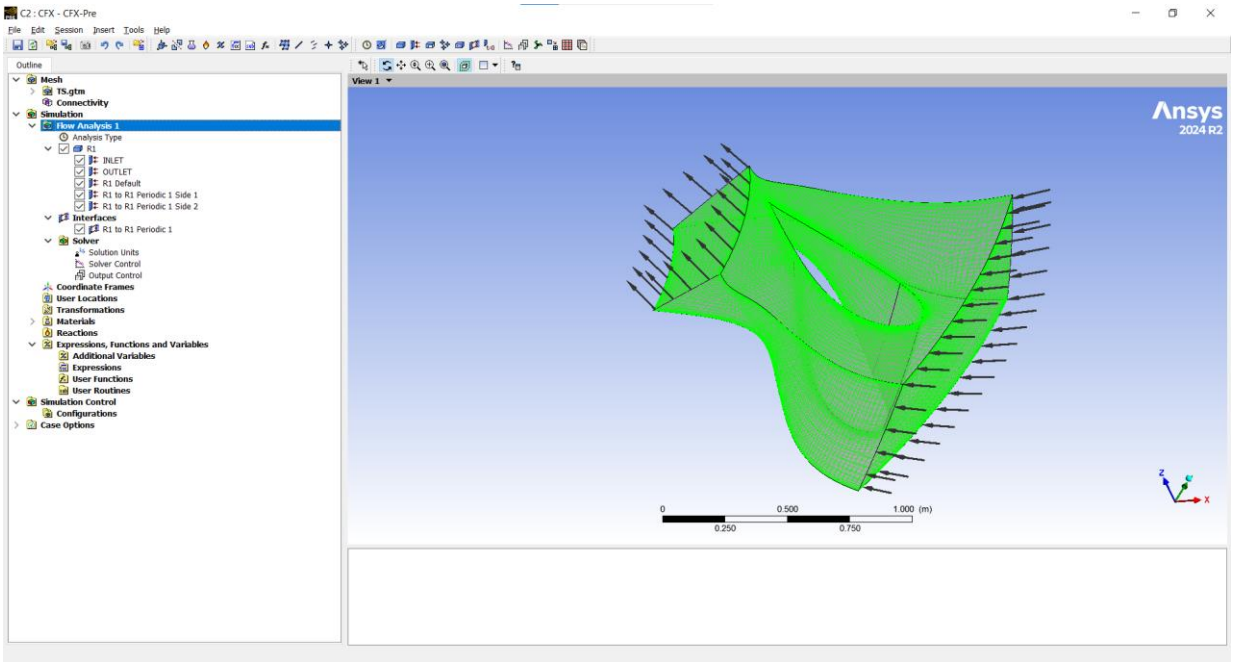


Figure 39: CFX setup environment for the runner-passage simulation

The runner passage was modeled as a rotating fluid domain because the objective of the CFD analysis was to calculate the energy exchange between the water flow and the rotating runner blade. The rotational speed was defined according to the design condition of the turbine. Since the complete runner has 14 blades, the numerical model represents one periodic sector of the complete runner. The remaining blade passages are represented through the periodic boundary condition.

In the initial CFX setup, different operating conditions were tested. A normal inlet flow direction was first applied, but the pressure and torque results were not consistent with the expected design-point performance. Therefore, the inlet condition was improved by applying a swirl inlet condition based on the velocity components at the runner inlet. This modification was necessary because the flow entering a Francis turbine runner is not purely normal to the inlet boundary; it has a significant circumferential velocity component generated by the guide vanes.

5-2 Swirl Inlet Boundary Condition

The inlet boundary condition used for the final CFD simulation is shown in **Figure 40**. The inlet was defined as a subsonic mass-flow inlet. Since the runner model contains only one periodic passage, the total mass flow rate of the turbine was divided by the number of runner blades.

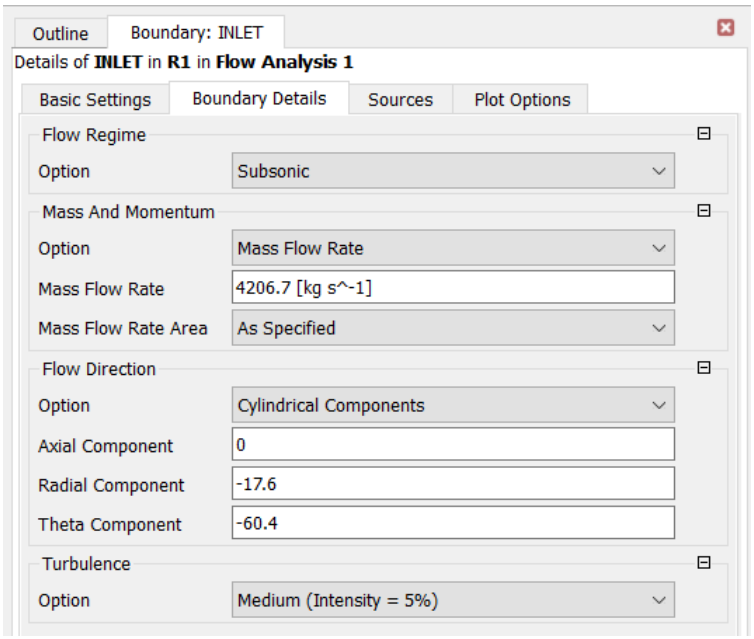


Figure 40: Swirl inlet boundary condition applied at the runner passage inlet

For a total runner flow rate of $59\,\mathrm{m}^3/\mathrm{s}$ and 14 blades, the mass flow rate per passage was calculated and applied as approximately:

$$\dot{m}_{passage} = 4206.7\,\mathrm{kg/s}$$

The flow direction was defined using **cylindrical components** instead of a simple normal direction. The applied inlet velocity direction components were:

$$V_z = 0\,, V_r = -17.6\,, V_\theta = -60.4$$

The negative radial component represents the inward movement of the flow toward the smaller radius of the Francis runner. The negative tangential component was applied to match the corrected runner rotation direction and to represent the swirl entering the runner from the guide-vane region.

This inlet definition was one of the most important corrections in the CFD setup. When the inlet flow was assumed to be normal to the boundary, the predicted head and torque were not close to the design values. After applying the swirl inlet condition, the total pressure drop, shaft power, and runner head became much closer to the expected design-point performance.

The turbulence condition at the inlet was set to medium intensity, corresponding to a turbulence intensity of 5%. This value was considered suitable for the present runner-passage simulation because the inlet flow is expected to be turbulent due to the upstream guide-vane region and high flow velocity.

5-3 Initial Stabilization Run

Before performing the final high-resolution simulation, an initial stabilization run was carried out. The purpose of this run was to obtain a stable flow field after applying the swirl inlet condition. As shown in **Figure 41**, the initial run used a more conservative numerical setup in order to prevent divergence and reduce residual oscillations.

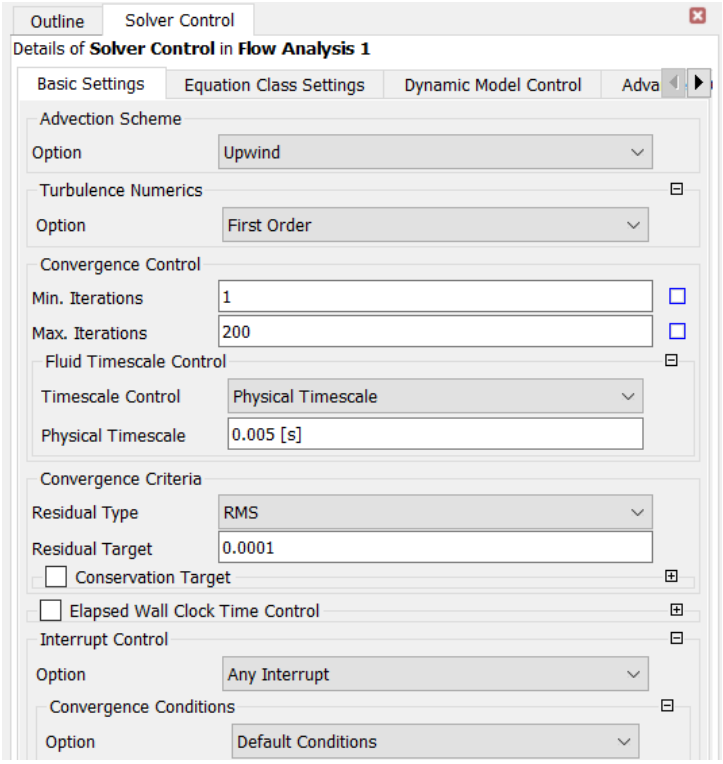


Figure 41: Residual convergence history of the initial stabilization run

For the initial stabilization run, the advection scheme was set to **Upwind**, and the turbulence numerics were set to **First Order**. These lower-order schemes are more numerically diffusive than high-resolution schemes, but they provide better stability during the early stages of the solution. The maximum number of iterations was set to 200, and a physical timescale was used for the steady-state solution.

The residual history in **Figure 41** shows that the solution became significantly more stable after applying the swirl inlet condition. The mass residual decreased to the order of 10^{-6}, while the momentum residuals decreased to approximately 10^{-4}. Unlike the earlier simulations with a normal inlet direction, the residuals decreased smoothly and did not show strong oscillatory behavior. Therefore, this run was used as an intermediate stabilized solution for the final CFD simulation.

This first run was not used as the final CFD result because the numerical schemes were intentionally conservative. Instead, the solution field obtained from this run was used as the initial condition for the final high-resolution run.

5-4 Final High-Resolution Run

After the stabilization run was completed, the simulation was continued using higher-order numerical schemes. In the final run, the advection scheme and turbulence numerics were changed to **High Resolution**. This improved the accuracy of the pressure field, velocity distribution, and torque prediction.

The final run was started from the converged solution of the initial stabilization run using the **Current Solution Data** option. This approach allowed the solver to continue from an already stable flow field instead of starting from a poor initial condition. As a result, the high-resolution simulation was able to converge successfully.

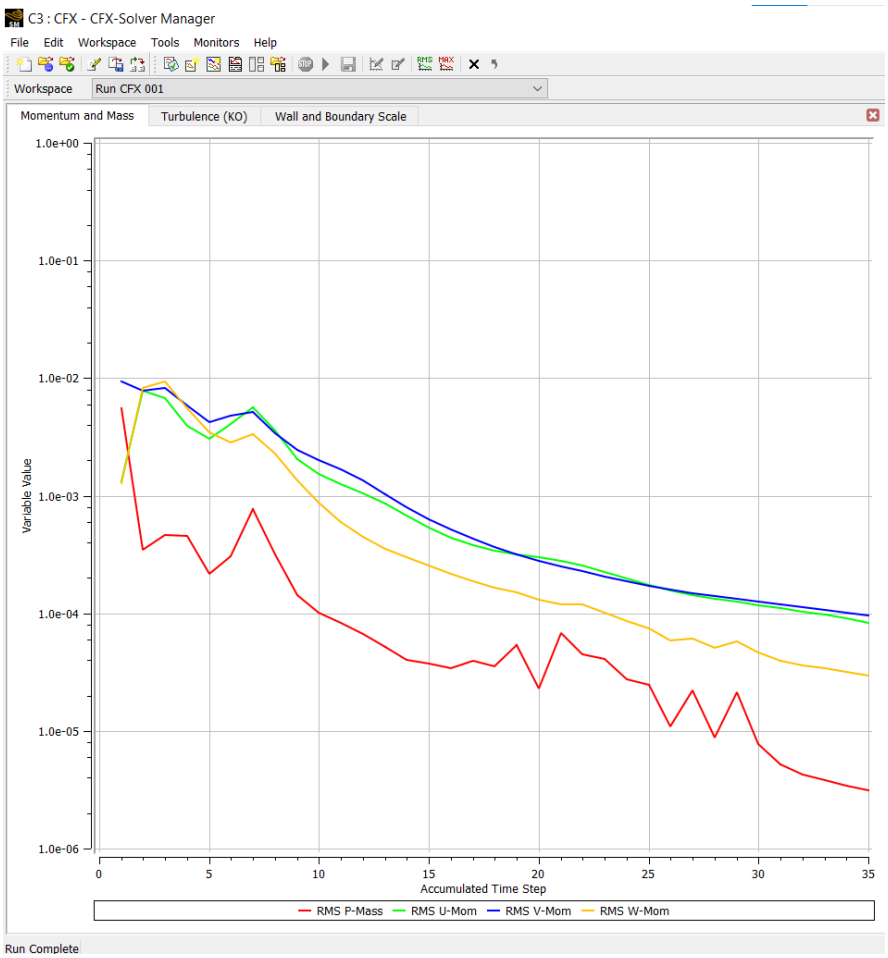


Figure 42: Residual convergence history of the final high-resolution CFX simulation

As shown in **Figure 42**, the residuals decreased to the specified convergence target of 1×10−51 \times 10^{-5}1×10−5, and the solver stopped automatically after reaching the convergence criterion. The residual curves show a small increase during the transition from the stabilization solution to the high-resolution solution, but after that point the residuals continued to decrease smoothly. This behavior is expected when switching from lower-order to higher-order numerical schemes.

The final convergence behavior confirms that the CFD setup was numerically stable and suitable for post-processing. Therefore, the final result file was used to calculate the mass balance, total pressure drop, runner head, torque, shaft power, and runner efficiency.

5-5 Boundary Conditions Used in the Final CFD Simulation

The final CFD setup can be summarized as follows:

Table 15: Summary of the final CFD setup and boundary conditions used for the runner passage simulation in ANSYS CFX.

Boundary / Setting	Applied Condition
Domain type	Rotating fluid domain
Rotational speed	$-254\,rpm - 254\,rpm - 254rpm$
Inlet type	Mass-flow inlet
Inlet mass flow rate	$4206.7\,kg/s$4206.7 kg/s$4206.7kg/s$
Flow direction	Cylindrical components
Axial component	000

Radial component	−17.6-17.6−17.6
Tangential component	−60.4-60.4−60.4
Inlet turbulence	Medium intensity, 5%
Outlet type	Static pressure outlet
Outlet pressure	0 Pa0 Pa0$Pagauge$
Periodic boundaries	Rotational periodic interface
Wall condition	No-slip rotating wall
Turbulence model	$SSTk-\omega k-\backslash omegak-\omega$
Initial stabilization advection scheme	Upwind
Initial stabilization turbulence numerics	First Order
Final advection scheme	High Resolution
Final turbulence numerics	High Resolution
Final residual target	$1\times10-51\times10^{-5}1\times10-5$

5-6 Final CFD Performance Parameters

The final converged CFD results were extracted from the high-resolution simulation. The mass flow rate at the inlet was:

$$\dot{m_{in}} = 4206.51\,\text{kg/s}$$

and the outlet mass flow rate was:

$$\dot{m_{out}} = -4209.58\,\text{kg/s}$$

The resulting mass imbalance was approximately 0.073% which confirms that the continuity requirement was satisfied.

The mass-flow-averaged total pressure in the stationary frame was:

$$P_{t,in} = 5.50467\times10^{6}\,\text{Pa}$$

at the inlet and:

$$P_{t,out} = 5.32743\times10^{5}\,\text{Pa}$$

at the outlet. Therefore, the total pressure drop across the runner passage was:

$$\Delta P_t = 4.9719\times10^{6}\,\text{Pa}$$

This corresponds to a runner head of approximately:

$$H = 507.8\,m$$

The torque extracted from one periodic runner passage was:

$$T_{passage} = -760708\,\text{N.m}$$

Since the runner contains 14 blades, the total runner torque was estimated as:

$$T_{total} = 14T_{passage} = -10.65\times10^{6}\,\text{N.m}$$

Using the angular velocity of:

$$\omega = -26.60\,\text{rad/s}$$

the shaft power was calculated as:

$$P_{shaft} = T_{total}\omega = 283.3MW$$

The hydraulic power based on the total pressure drop and total flow rate was:

$$P_{hyd} = Q\Delta P_t = 293.34\,\text{MW}$$

Therefore, the runner efficiency was estimated as:

$$\eta = \frac{P_{shaft}}{P_{hyd}} = 96.6\%$$

This efficiency should be interpreted as the **runner-passage efficiency**, not the overall turbine efficiency, because the present CFD model includes only the runner passage. Losses related to the spiral casing, guide vanes, stay vanes, draft tube, and full-stage interactions were not included in this simplified numerical model.

5-7 Final CFD Results Summary

Table 16: Summary of the final CFD results for the Francis runner passage, including mass flow balance, total pressure drop, runner head, torque, shaft power, hydraulic power, and runner efficiency.

Parameter	Value
Number of blades	14
Mass flow rate per passage	4206.51 kg/s4206.51 kg/s4206.51kg/s
Outlet mass flow rate	−4209.58 kg/s − 4209.58 kg/s − 4209.58kg/s
Mass imbalance	0.073
Total pressure at inlet	5.50467 MPa5.50467 MPa5.50467MPa
Total pressure at outlet	0.532743 MPa0.532743 MPa0.532743MPa
Total pressure drop	4.972 MPa4.972 MPa4.972MPa
CFD runner head	507.8 m507.8 m507.8m
Torque per passage	−760708 $N.m$ − 760708 N. m − 760708$N.m$
Total runner torque	−10.65 $MN.m$ − 10.65 MN. m − 10.65$MN.m$
Angular velocity	−26.60 rad/s − 26.60 rad/s − 26.60rad/s
Shaft power	283.3 MW283.3 MW283.3MW
Hydraulic power	293.34 MW293.34 MW293.34MW
Runner efficiency	96.6

5-8 Final Solver Control Settings

After obtaining a stable initial solution using the first-order stabilization run, the simulation was continued using a more accurate high-resolution numerical setup. The solver control settings used for the final CFD run are shown in **Figure 43**.

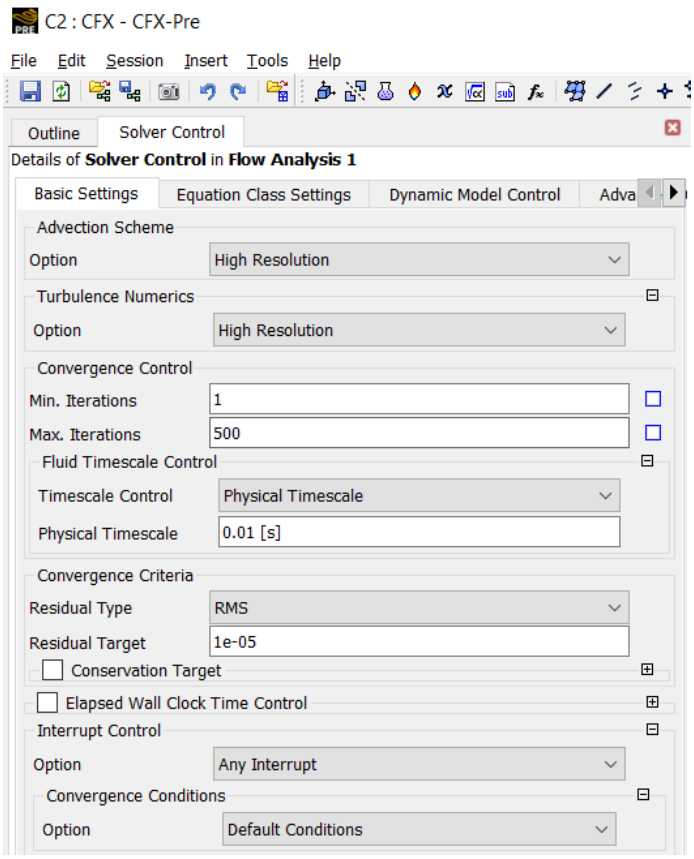


Figure 43: Solver control settings for the final high-resolution CFX simulation

In this stage, both the advection scheme and the turbulence numerics were set to **High Resolution** in order to improve the accuracy of the pressure, velocity, and torque predictions.
The final solver settings were defined as follows:

Table 17: Final solver control settings for the high-resolution CFD simulation in ANSYS CFX, including advection scheme, turbulence numerics, timescale control, and convergence criteria.

Solver Parameter	Final Setting
Advection scheme	High Resolution
Turbulence numerics	High Resolution
Minimum iterations	1
Maximum iterations	500
Timescale control	Physical Timescale
Physical timescale	$0.01\,s 0.01\,s 0.01 s$
Residual type	RMS
Residual target	$1 \times 10 - 51 \times 10^{-5} 1 \times 10 - 5$
Interrupt control	Any Interrupt

The use of **High Resolution** schemes was important because the final solution was intended for performance evaluation, including total pressure drop, runner torque, shaft power, and efficiency. Although the first-order run provided better numerical stability, it was more diffusive and therefore not ideal for final quantitative analysis. For this reason, the high-resolution run was performed after the flow field had already been stabilized.

A physical timescale of $0.01s$ was selected instead of the automatic timescale option. This was done to improve the stability of the steady-state solution, especially because the runner passage includes strong curvature, swirl inlet flow, blade loading, and rotating-frame effects. The maximum iteration number was set to 500, but the solver was allowed to stop earlier if the specified residual target was reached.

5-9 Final High-Resolution Run

The convergence history of the final CFX simulation is shown in **Figure 44**. This run was started from the stabilized solution obtained in the initial run. Therefore, the solver did not start from an undeveloped flow field, but from a physically improved initial condition that already included the swirl inlet effect.

At the beginning of the final run, the residuals showed a temporary increase due to the change from first-order schemes to high-resolution numerical schemes. This behavior is expected because the higher-order discretization reduces numerical diffusion and modifies the predicted pressure and velocity fields. After this transition, all residuals decreased continuously.

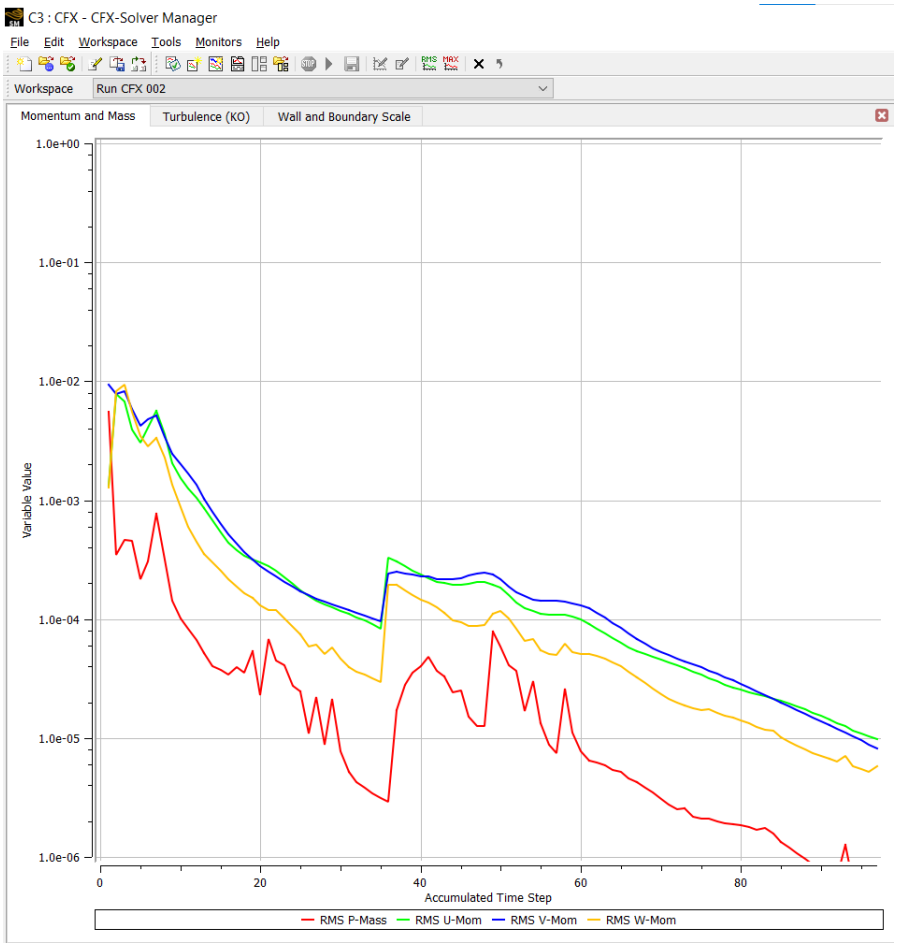


Figure 44: Residual convergence history of the final high-resolution CFX simulation

The solver stopped automatically after reaching the specified RMS residual target of 10^{-5} .The final residual level indicates that the numerical solution was sufficiently converged for post-processing. The pressure-mass residual decreased to the order of 10^{-6}, while the momentum residuals also dropped to approximately 10^{-5}. Compared with the earlier simulations, the final convergence behavior was significantly more stable and physically meaningful.

The convergence improvement was mainly achieved through two modifications:

1. applying a swirl inlet condition instead of a purely normal inlet flow direction;

2. using a two-step solution strategy, including a first-order stabilization run followed by a high-resolution final run.

This approach allowed the CFD solution to converge without strong residual oscillations and produced final results that were much closer to the design-point performance of the Francis runner.

5-9-1 Two-Step CFX Solution Strategy

The CFD solution was completed using a two-step procedure. First, an initial stabilization run was performed using conservative numerical schemes. Then, the final run was continued from the stabilized solution using high-resolution schemes.

5-9-2 Initial Stabilization Run

The first run was used only to stabilize the flow field after applying the swirl inlet condition. The main settings of this run were:

Table 18: Numerical settings for the initial stabilization run using first-order schemes and upwind advection to stabilize the swirl-inlet solution before the final high-resolution simulation.

Parameter	Initial Stabilization Run
Advection scheme	Upwind
Turbulence numerics	First Order
Maximum iterations	200
Residualtarget	$1 \times 10 - 41 \times 10^{-4} 1 \times 10 - 4$
Physicaltimescale	$0.005\ s 0.005\ \mathrm{s} 0.005 s$
Purpose	Stabilizing the swirl − inlet solution

This run was not considered the final CFD result because the numerical schemes were intentionally diffusive. However, it provided a stable initial condition for the final high-resolution simulation.

5-9-3 Final High-Resolution Run

The second run was used as the final CFD solution. The main settings were:

Table 19: Numerical settings for the final high-resolution run using second-order schemes for accurate performance prediction, started from the converged solution of the stabilization run.

Parameter	Final Run
Advection scheme	High Resolution
Turbulence numerics	High Resolution
Maximum iterations	500
Residualtarget	$1 \times 10 - 51 \times 10^{-5} 1 \times 10 - 5$
Physicaltimescale	$0.01\ s 0.01\ \mathrm{s} 0.01 s$
Purpose	Final performance prediction

The final run was solved using the **Current Solution Data** from the stabilization run. This reduced convergence difficulty and allowed the high-resolution simulation to reach the residual target successfully.

Post processing with CFX Post Processing

After completing the final high-resolution CFX simulation, the converged result file was opened in **ANSYS CFD-Post** for extracting the main hydraulic and mechanical performance parameters of the runner. The post-processing stage included checking the mass balance, calculating the total pressure drop, estimating the runner head, extracting the torque, calculating the shaft power, and evaluating the runner-passage efficiency.

The final CFD result was obtained from the swirl-inlet simulation, which used the corrected runner rotational direction and high-resolution numerical schemes. Since the numerical model represents only one periodic passage of the runner, all passage-based quantities such as torque were scaled according to the total number of runner blades.

6-1 CFD-Post Workflow

The complete simulation workflow is shown in **Figure 45**. The final runner geometry was first designed in CFturbo and then transferred to ANSYS TurboGrid for structured meshing. The generated TurboGrid mesh was connected to the CFX setup, where the runner-passage CFD simulation was performed. After the solver reached convergence, the result file was opened in CFD-Post for the extraction of final performance parameters.

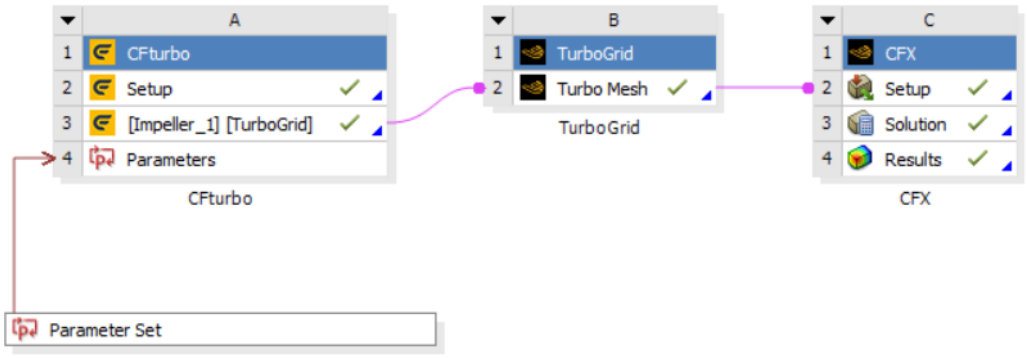


Figure 45: ANSYS Workbench workflow after completing the CFX simulation and opening the result file in CFD-Post

This workflow confirms that the CFD analysis was performed through a consistent turbomachinery simulation chain:

$$CFturbo \rightarrow TurboGrid \rightarrow CFX \rightarrow CFD - Post$$

Using this connected workflow helped preserve the blade-passage definition, periodic surfaces, inlet and outlet regions, and rotating domain setup from the meshing stage to the final post-processing stage.

6-2 Torque Extraction

The runner torque was extracted using the **Function Calculator** in CFD-Post, as shown in **Figure 46**.

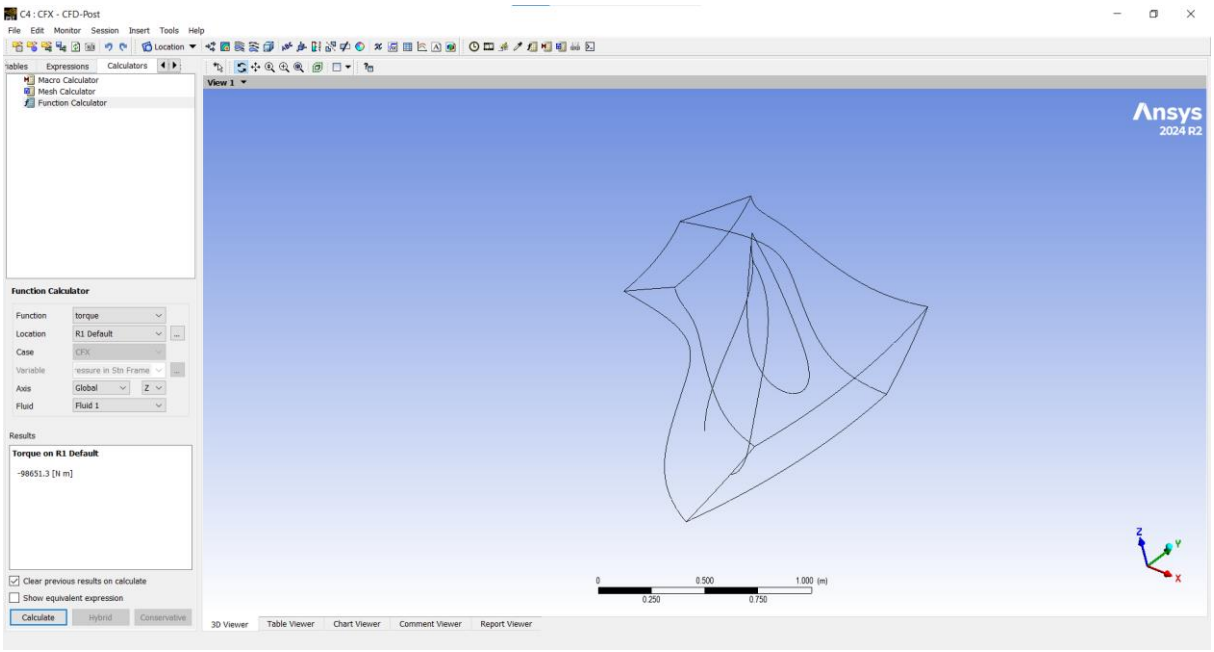


Figure 46: Torque extraction from the runner wall group using CFD-Post Function Calculator

The selected function was **torque**, and the selected location was **R1 Default**, which represents the main wall group of the runner passage, including the runner blade, hub, and shroud wall surfaces. The torque was calculated about the **Global Z-axis**, which corresponds to the rotational axis of the runner.

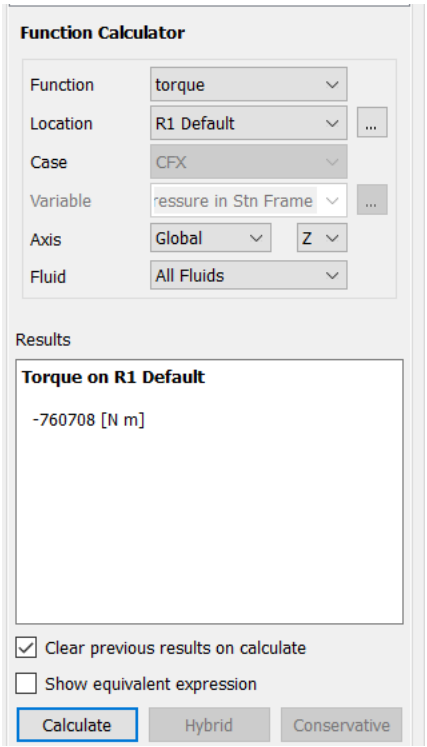


Figure 47: Calculated torque for one periodic runner passage

The torque obtained from one periodic runner passage was $T_{passage} = -760708$ N.m .

The negative sign is related to the selected rotational direction and coordinate-system convention. In the final CFX setup, the runner angular velocity was also negative $\omega = -26.60\ rad/s$.

Therefore, the product of torque and angular velocity gives a positive shaft power, which indicates turbine power extraction.

Since the runner contains 14 blades, the total runner torque was obtained by multiplying the passage torque by 14:

$$T_{total} = 14T_{passage} = 14 \times (-760708) = -10{,}649{,}912\ N.m$$

Therefore, the magnitude of the total runner torque is:

$$|T_{total}| = 10.65 \times 10^6 \text{ N.m}$$

6-3 Total Pressure Extraction

The total pressure at the runner inlet was extracted using the mass-flow-averaged value of **Total Pressure in Stn Frame**, as shown in **Figure 48**.

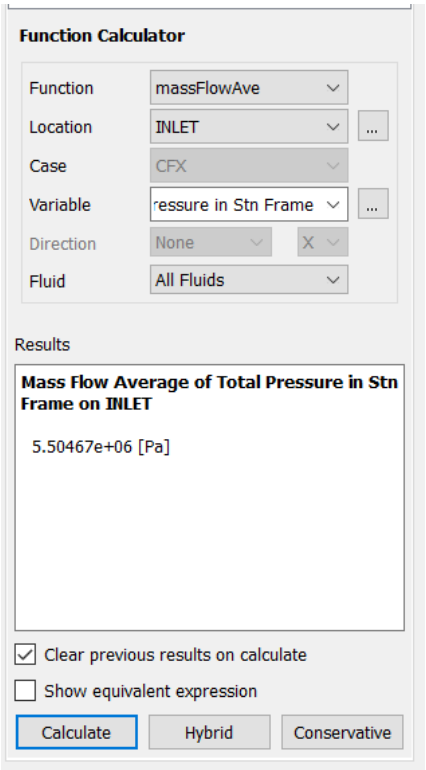


Figure 48: Mass-flow-averaged total pressure in the stationary frame at the runner inlet

The use of the stationary-frame total pressure is important because the energy drop across the runner must be evaluated in a consistent reference frame.

The extracted inlet total pressure was:

$$P_{t,in} = 5.50467 \times 10^6 \text{ Pa}$$

The corresponding mass-flow-averaged total pressure at the runner outlet is shown in **Figure 49**.

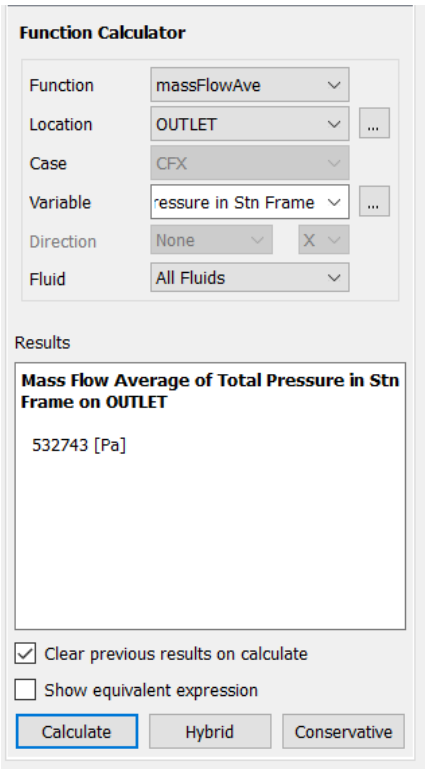


Figure 49: Mass-flow-averaged total pressure in the stationary frame at the runner outlet

The extracted outlet value was:

$$P_{t,out} = 532743 \text{ Pa}$$

Therefore, the total pressure drop across the runner passage was calculated as:

$$\Delta P_t = P_{t,in} - P_{t,out} = 5.50467 \times 10^6 - 532743 = 4{,}971{,}927\ Pa$$

or:

$$\Delta P_t = 4.972 \text{ MPa}$$

The positive pressure drop indicates that the fluid loses total pressure while passing through the runner, which is consistent with turbine operation.

6-4 Mass Flow Balance

The inlet mass flow rate was extracted using the Function Calculator, as shown in **Figure 50**.

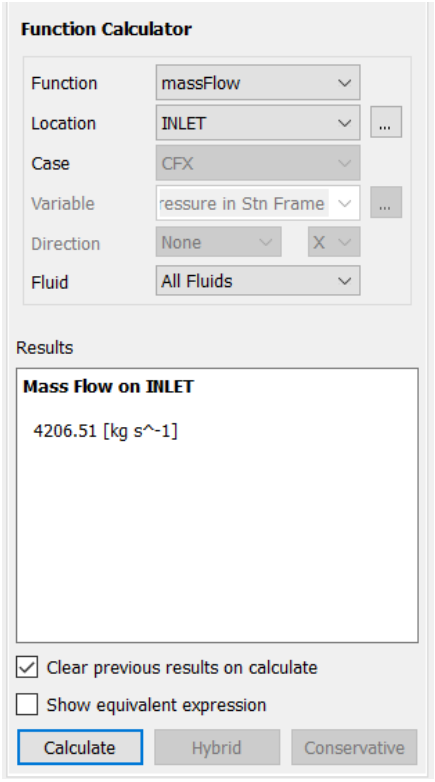


Figure 50: Mass flow rate at the runner inlet

The calculated mass flow rate at the inlet was:

$$\dot{m_{in}} = 4206.51 \text{ kg/s}$$

This value corresponds to one periodic runner passage. Since the runner has 14 blades, the total mass flow rate represented by the complete runner is approximately:

$$\dot{m_{total}} = 14 \times 4206.51 = 58{,}891.14\ kg/s$$

which is consistent with the design flow rate of approximately:

$$Q = 59 \text{ m}^3/\text{s}$$

for water.

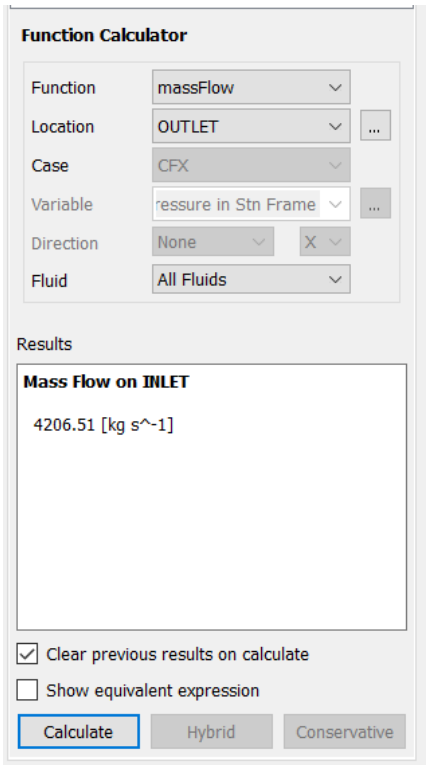


Figure 51: Mass flow rate at the runner outlet

The outlet mass flow rate obtained from the final CFD result was:

$$\dot{m_{out}} = -4209.58\ \text{kg/s}$$

The negative sign is expected because the outlet surface normal direction is opposite to the flow direction. For mass-balance evaluation, the absolute value of the outlet mass flow rate is used.

The mass imbalance was calculated as:

[2.10]
$$\text{Mass imbalance} = \frac{|\dot{m_{out}}| - |\dot{m_{in}}|}{|\dot{m_{in}}|}$$
$$= \frac{|4209.58 - 4206.51|}{4206.51} \times 100 = 0.073\%$$

This value is lower than 0.1%, indicating that the continuity requirement was satisfied with very good accuracy.

6-5 Runner Head Calculation

The runner head was calculated from the total pressure drop using:

[2.11]
$$H = \frac{\Delta P_t}{\rho g}$$

where:

$$\Delta P_t = 4{,}971{,}927\ \text{Pa}$$
$$\rho = 998\ \text{kg/m}^3$$
$$g = 9.81\ \text{m/s}^2$$

Therefore:

$$H = \frac{4{,}971{,}927}{998 \times 9.81} 507.8\ m$$

The calculated CFD runner head is therefore:

$$H_{CFD} = 507.8 \text{ m}$$

This value is close to the design head used in the CFturbo design stage. The result shows that applying the swirl inlet condition improved the agreement between the CFD simulation and the expected operating point.

6-6 Shaft Power Calculation

The shaft power was calculated using:

[2.12] $$P_{shaft} = T_{total}\omega$$

For the final simulation:

$$T_{total} = -10{,}649{,}912 \text{ N.m}$$

and:

$$\omega = -26.60\, rad/s$$

Thus:

$$P_{shaft} = (-10{,}649{,}912)(-26.60) = 283.3 \times 10^6 = 283.3 \text{ MW}$$

The positive power value confirms that the runner extracts energy from the fluid.

6-7 Hydraulic Power and Runner Efficiency

The hydraulic power based on the total pressure drop was calculated as:

[2.13] $$P_{hyd} = Q\Delta P_t$$

where:

$$Q = 59 \text{ m}^3/\text{s}$$

and:

$$\Delta P_t = 4{,}971{,}927 \text{ Pa}$$

Therefore:

$$P_{hyd} = 59 \times 4{,}971{,}927 = 293.34 \times 10^6 \text{ W} = 293.34 \text{ MW}$$

The runner-passage efficiency was then calculated from:

$$\eta = \frac{P_{shaft}}{P_{hyd}} = \frac{283.3}{293.34} = 0.966$$

Therefore:

$$\eta = 96.6\%$$

This value represents the **runner-passage efficiency**, not the overall turbine efficiency. The present CFD model only includes the runner passage and does not include losses in the spiral casing, stay vanes, guide vanes, draft tube, bearings, or generator. Therefore, the obtained efficiency is expected to be higher than the complete turbine efficiency.

6-8 Final Post-Processing Results

Table X: Final CFD post-processing results for the Francis runner passage

Table 20: Final CFD post-processing results for the Francis runner passage, extracted from CFD-Post, including mass flow balance, total pressure drop, runner head, torque, shaft power, hydraulic power, and runner-passage efficiency.

Parameter	Value
Number of runner blades	14
Massflowrateatinlet	4206.51 kg/s4206.51 kg/s4206.51 kg/s
Massflowrateatoutlet	$-4209.58\ kg/s$ − 4209.58 kg/s − 4209.58 kg/s
Massimbalance	0.073

$\boldsymbol{Totalpressureatinlet}$	5.50467 MPa5.50467 MPa5.50467 MPa
$\boldsymbol{Totalpressureatoutlet}$	0.532743 MPa0.532743 MPa0.532743 MPa
$\boldsymbol{Totalpressuredrop}$	4.972 MPa4.972 MPa4.972 MPa
$\boldsymbol{CFDrunnerhead}$	507.8 m507.8 m507.8 m
$\boldsymbol{Torqueperpassage}$	$-760708\ N.m - 760708$ N. m $- 760708\ N.m$
$\boldsymbol{Totalrunnertorque}$	$-10.65\ MN.m - 10.65$ MN. m $- 10.65\ MN.m$
$\boldsymbol{Angularvelocity}$	$-26.60\ rad/s - 26.60$ rad/s $- 26.60\ rad/s$
$\boldsymbol{Shaftpower}$	283.3 MW283.3 MW283.3 MW
$\boldsymbol{Hydraulicpower}$	293.34 MW293.34 MW293.34 MW
$\boldsymbol{Runner-passageefficiency}$	96.6

6-9 Flow Field Visualization

After extracting the numerical performance parameters from CFD-Post, several contour plots were generated to evaluate the flow behavior inside the runner passage. These contours were used to examine the static pressure field, total pressure variation, and velocity magnitude distribution. The visual results help confirm whether the final CFD solution is physically consistent with turbine operation.

6-9-1 Static Pressure Distribution on Runner Blade

The static pressure distribution on the runner blade and surrounding wall surfaces is shown in **Figure 36**. The contour was plotted on the R1 Default wall group, which includes the blade, hub, and shroud surfaces of the single periodic runner passage.

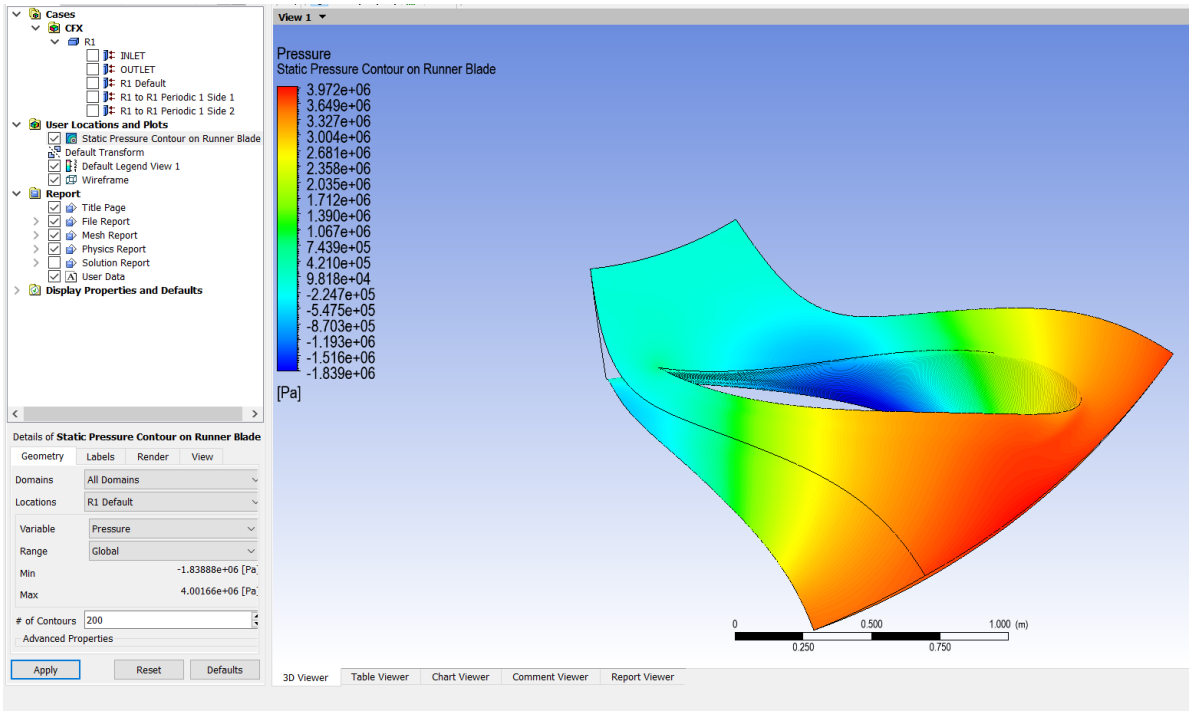


Figure 52: Static pressure distribution on the runner blade and wall surfaces

As shown in **Figure 52**, a clear pressure variation exists across the blade passage. The high-pressure region is mainly located on one side of the blade passage, while a lower-pressure region appears near the

opposite side of the blade surface. This pressure difference between the pressure side and suction side is the main reason for torque generation in the runner.

The contour also shows that the static pressure changes smoothly along the blade passage, without abrupt discontinuities. This indicates that the final mesh and boundary conditions were able to produce a physically reasonable pressure field. The presence of a low-pressure zone near the blade surface is expected in a Francis runner because the blade curvature and flow turning create a pressure difference across the blade. This pressure difference transfers angular momentum from the water to the rotating runner.

The maximum pressure in the contour is approximately:

$$P_{max} \approx 4.0 \times 10^6 \text{ Pa}$$

while the minimum pressure reaches approximately:

$$P_{min} \approx -1.84 \times 10^6 \text{ Pa}$$

These values are gauge pressure values in the CFD domain. Therefore, the relative variation of pressure is more important than the absolute value when interpreting the contour.

6-9-2 Total Pressure Distribution in the Stationary Frame

The total pressure distribution in the stationary frame is presented in **Figure 53**. This variable was used because the energy drop through the runner should be evaluated in a consistent reference frame. In a rotating turbomachinery domain, using the stationary-frame total pressure is more meaningful for evaluating the hydraulic energy change of the flow.

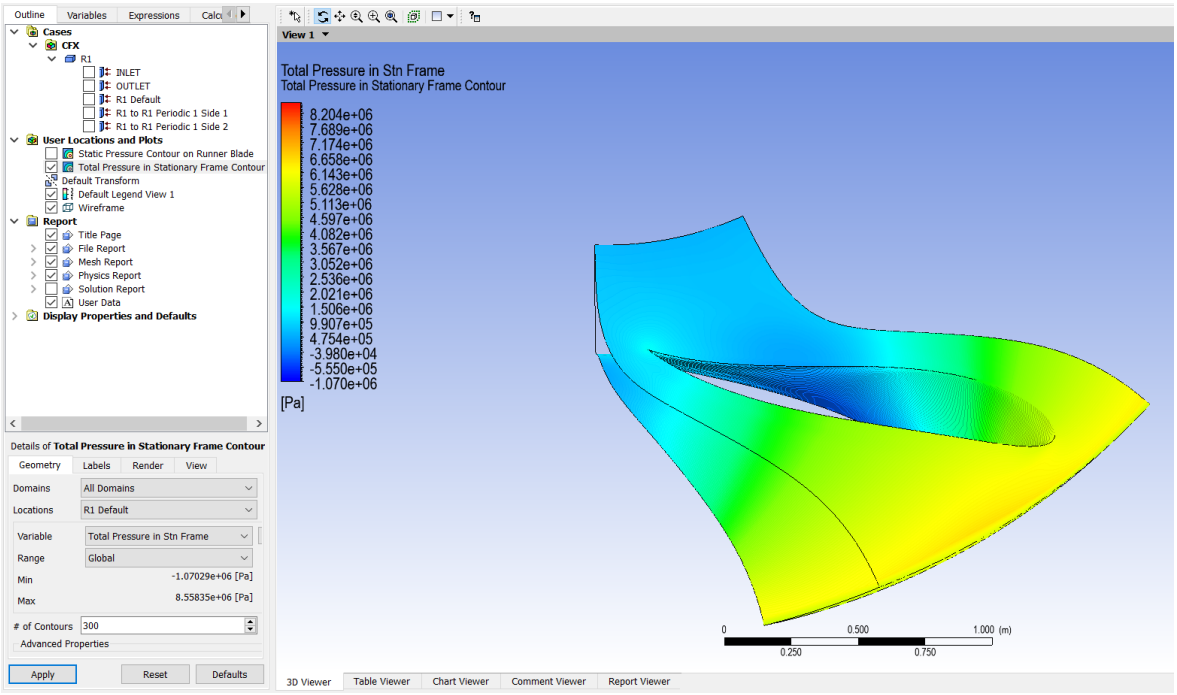


Figure 53: Total pressure distribution in the stationary frame on the runner passage wall surfaces

The contour shows a significant total pressure variation across the runner passage. The total pressure decreases as the flow passes through the runner, which confirms that hydraulic energy is extracted from the fluid. This behavior is consistent with turbine operation. The total pressure field also shows that the energy extraction is not uniform over the whole passage, which is expected because of the three-dimensional blade curvature, non-uniform loading, and strong turning of the flow.

The total pressure contour in **Figure 53** supports the numerical values extracted earlier from CFD-Post. The mass-flow-averaged total pressure decreased from:

$$P_{t,in} = 5.50467 \times 10^6 \text{ Pa}$$

at the inlet to:

$$P_{t,out} = 5.32743 \times 10^5 \text{ Pa}$$

at the outlet. Therefore, the pressure-based post-processing and the contour visualization both confirm that the runner operates in turbine mode.

6-9-3 Total Pressure Contour on a Passage Plane

The total pressure distribution on an internal passage plane is shown in **Figure 54**. Unlike the previous contour, which was plotted on the wall surfaces, this figure provides a sectional view of the pressure field inside the flow passage. This type of plot is useful for understanding how the flow energy varies within the blade-to-blade channel.

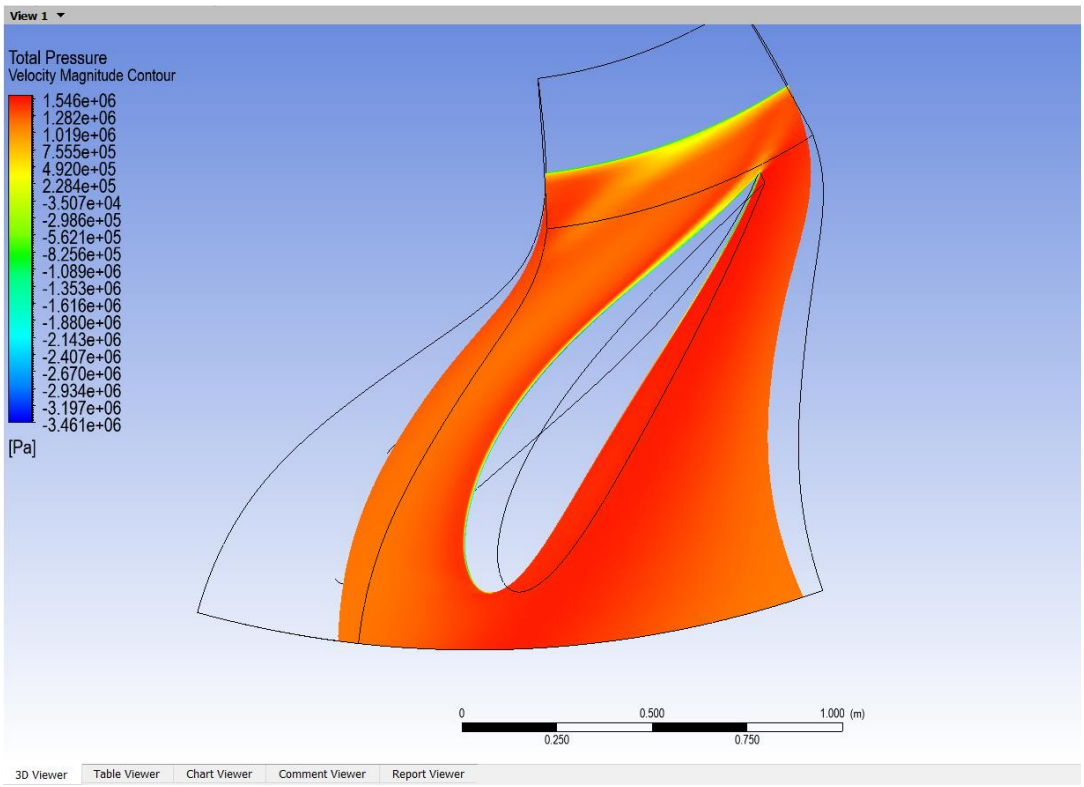


Figure 54: Total pressure distribution on an internal passage plane

As shown in **Figure 54**, the total pressure remains relatively high over most of the passage plane, while local variations are observed near the blade surface and curved regions of the flow path. These local variations are caused by the blade loading, curvature of the passage, and interaction between the flow and the runner blade. The smooth distribution of the contour indicates that there are no severe numerical discontinuities in the internal flow field.

This figure is also useful for showing that the energy transfer is not limited to the wall surfaces. The pressure field changes throughout the passage, and the runner extracts energy from the flow as it is guided from the inlet region toward the outlet.

6-9-4 Velocity Magnitude Distribution

The velocity magnitude distribution inside the runner passage is shown in **Figure 55**. The contour was plotted on an internal plane to visualize the flow acceleration and turning within the blade-to-blade passage.

The velocity contour shows that the flow velocity is not uniform across the passage. Higher velocity regions appear along the curved blade channel, especially near the narrow section of the passage where the flow is accelerated. Lower velocity regions are observed near wider parts of the passage and close to some wall regions. This behavior is physically reasonable because the blade passage area and flow direction change continuously from inlet to outlet.

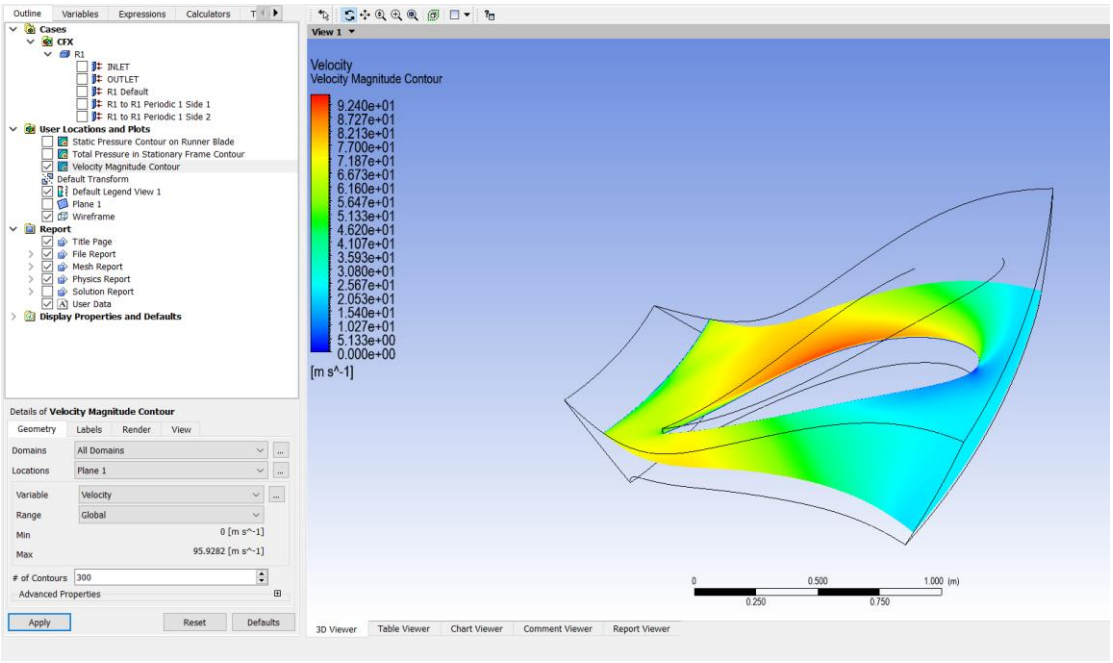


Figure 55: Velocity magnitude distribution inside the runner passage

The maximum velocity shown in the contour is approximately:

$$V_{max} \approx 95.9\,\mathrm{m/s}$$

This relatively high velocity is expected for a high-head Francis turbine runner. The contour also shows that the velocity field follows the blade passage smoothly, which indicates that the swirl inlet condition improved the alignment of the incoming flow with the runner blade geometry.

Compared with the earlier simulations using a purely normal inlet direction, the final velocity field is more consistent with the expected flow behavior in a Francis runner. The flow enters the passage with a tangential component, follows the curved blade channel, and exits toward the outlet region. This confirms that the inlet swirl definition was necessary for obtaining a realistic runner-passage simulation.

6-9-5 Interpretation of the Flow Field

The contour results confirm that the final CFD simulation produced a physically consistent flow field. The static pressure contour shows a clear pressure difference across the runner blade, which explains the generation of runner torque. The total pressure contours show that the flow loses hydraulic energy as it passes through the runner. The velocity contour shows smooth acceleration and turning of the flow inside the blade passage.

The final visual results support the numerical post-processing values obtained from the Function Calculator. The total pressure drop, calculated runner head, torque, and shaft power are consistent with the observed pressure and velocity distributions. Therefore, the final CFD solution can be considered suitable for evaluating the hydraulic behavior of the runner passage.

However, these contours should be interpreted as results of a **single periodic runner-passage simulation**. The complete turbine system was not modeled. Therefore, the effects of the spiral casing, stay

vanes, guide vanes, draft tube, leakage, and full-stage interactions are not included in these flow visualizations

6-9-6 Additional Velocity Field Visualization

The velocity magnitude distribution on an internal passage plane is shown in **Figure 56**.

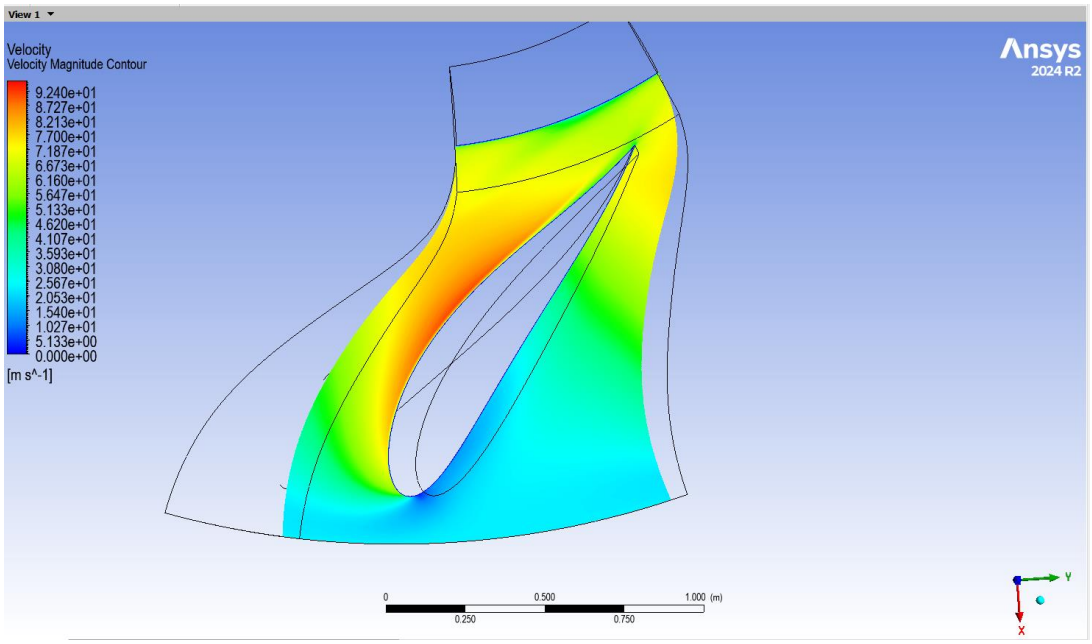


Figure 56: Velocity magnitude distribution on an internal runner-passage plane

This figure provides a clearer view of the flow acceleration and velocity variation inside the blade-to-blade channel. The contour was plotted using the velocity magnitude variable, and the velocity range extends from nearly zero to approximately:

$$V_{max} \approx 92.4 \text{ m/s}$$

The contour shows that the velocity is not uniformly distributed across the runner passage. Higher velocity regions are observed along the curved blade channel, especially near the narrow passage between the blade surface and the surrounding wall. This is physically expected because the flow is guided through a curved and contracting passage, which results in local acceleration.

A lower velocity region can also be observed near the inner curved part of the passage. This may be associated with local flow deceleration caused by the strong curvature of the blade passage and the turning of the incoming swirl flow. However, the velocity field remains continuous and does not show severe discontinuities or unrealistic numerical artifacts. Therefore, the contour supports the conclusion that the final swirl-inlet simulation produced a physically reasonable velocity field.

The velocity distribution also confirms the importance of using the cylindrical inlet components. In the earlier simulations with a normal inlet direction, the flow was not properly aligned with the blade passage. In the final simulation, the inlet swirl helped the flow follow the runner blade channel more naturally, leading to better agreement between the CFD-predicted head and the design head.

6-10 Streamline Visualization

The streamline settings used in CFD-Post are shown in **Figure 57**.

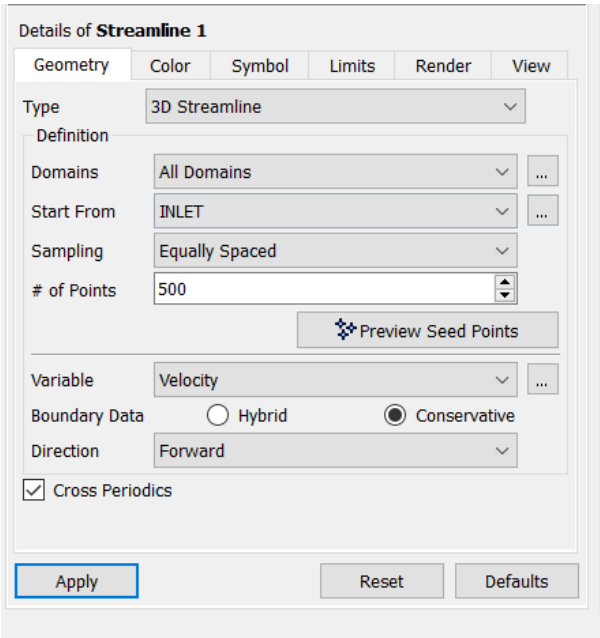


Figure 57: Streamline definition settings in CFD-Post

A three-dimensional streamline object was created to visualize the flow path inside the runner passage. The streamlines were started from the **INLET** boundary and were distributed using the **Equally Spaced** sampling method. The number of seed points was set to:

$$N = 500$$

The streamline variable was defined as **Velocity**, and the streamline direction was set to **Forward**. The option **Cross Periodics** was also activated. This option is important in a periodic turbomachinery passage because it allows the streamline visualization to pass through periodic boundaries and represent the repeated flow path more naturally.

These settings were selected to provide a sufficiently dense streamline pattern while still keeping the figure readable. Using too few streamlines would not clearly show the three-dimensional flow direction, while too many streamlines could make the figure visually crowded.

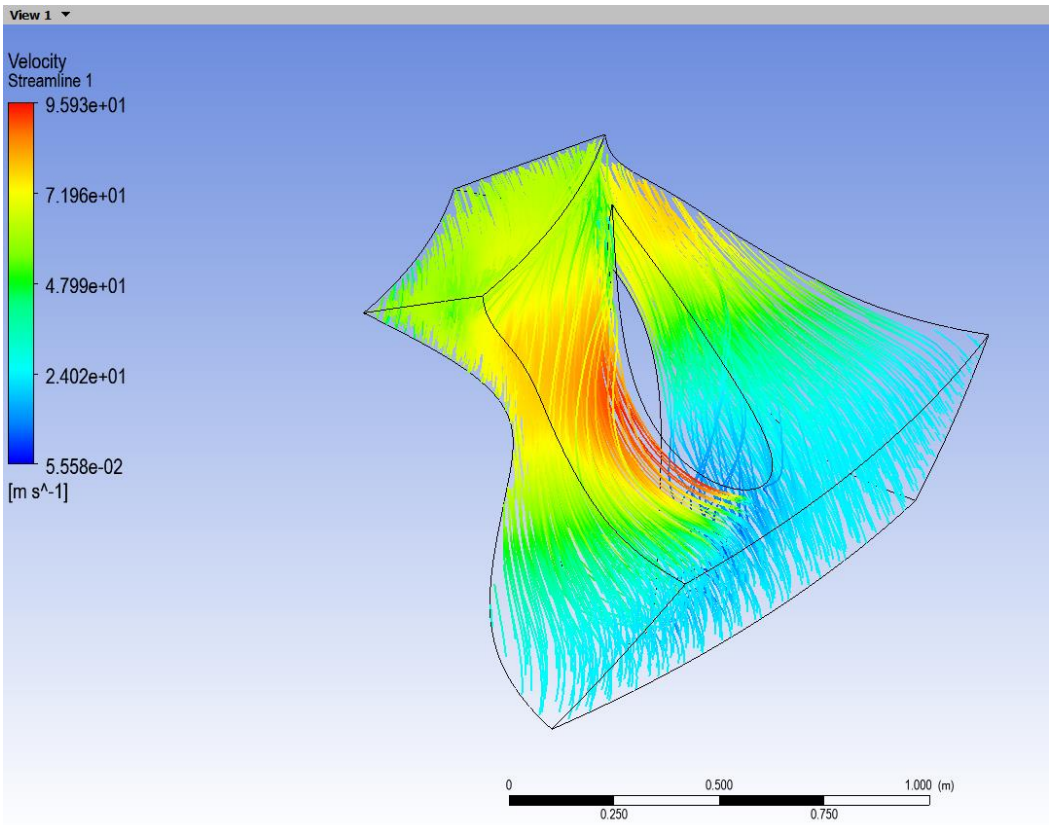


Figure 58: Three-dimensional streamlines through the runner passage colored by velocity magnitude

The three-dimensional streamlines through the runner passage are shown in **Figure 58**. The streamlines are colored by velocity magnitude, allowing both the flow direction and the local velocity variation to be visualized in a single plot.

The streamline pattern shows that the flow enters the runner passage from the inlet region and follows the curved blade-to-blade channel toward the outlet. The streamlines bend smoothly around the blade, which indicates that the inlet swirl condition improved the alignment between the incoming flow and the blade geometry. This behavior is consistent with the expected flow structure in a Francis turbine runner, where the flow enters with a tangential component and is turned by the rotating blade passage.

The highest velocity regions appear along the main flow path near the blade passage throat, where the flow is accelerated. Lower velocity regions are observed farther from the main passage and near some wall regions. The streamline pattern does not show a large-scale separated or reversed flow structure, which supports the quality of the final CFD result.

Overall, **Figure 58** confirms that the final CFX solution produces a smooth and physically meaningful three-dimensional flow path inside the runner passage. This is an important qualitative validation of the final simulation, especially because the final performance results were strongly affected by the inlet flow direction.

6-11 Wall y^+ Evaluation

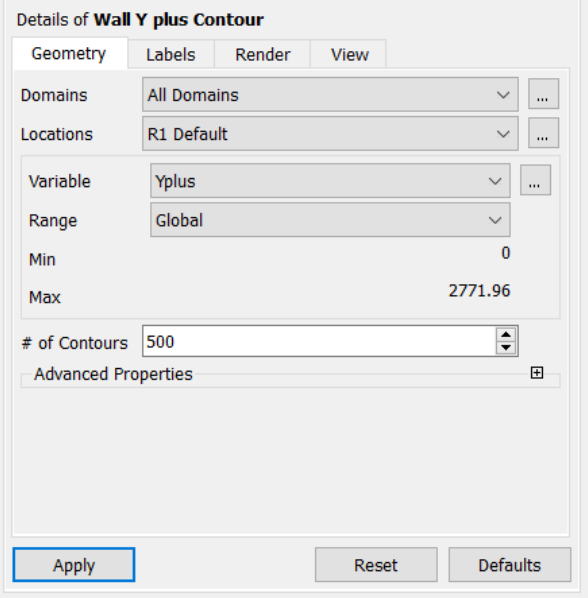


Figure 59: Wall y^+ contour setup in CFD-Post

The wall y^+ contour setup is shown in **Figure 59**. The contour was created on the R1 Default wall group, which includes the runner blade, hub, and shroud wall surfaces. The selected variable was **Yplus**, and the contour range was defined globally.

The wall y^+value is an important near-wall mesh parameter in turbulent flow simulations. It indicates the nondimensional distance of the first near-wall cell from the wall surface. Therefore, it is useful for evaluating whether the near-wall mesh resolution is suitable for the selected turbulence model and wall treatment.

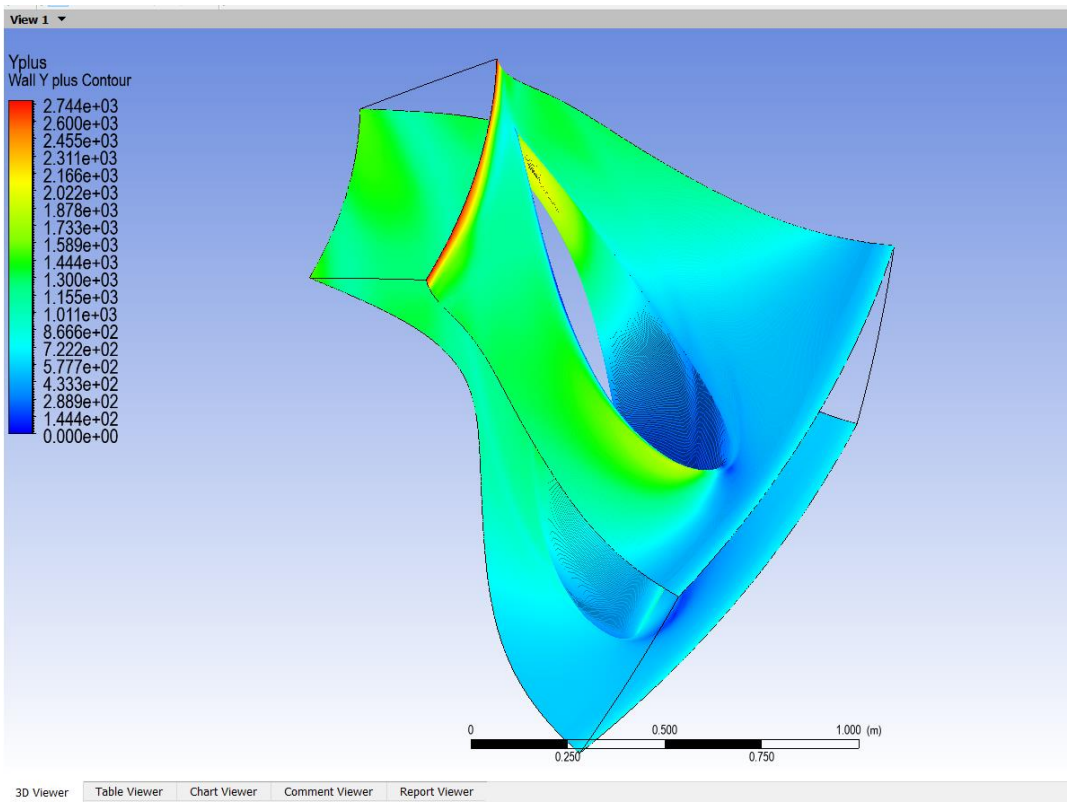


Figure 60: Wall y^+ distribution on the runner wall surfaces

The wall y^+distribution on the runner wall surfaces is shown in **Figure 60**. The contour indicates that the y^+ value varies significantly over the blade, hub, and shroud surfaces. The maximum value reported in the plot is approximately:

$$y^+_{max} \approx 2772$$

Higher y^+values are mainly observed near regions with high velocity gradients and strong flow acceleration, especially around the blade passage and near the high-curvature regions. Lower y^+values appear in regions where the local wall shear and velocity gradients are smaller.

The relatively high maximum y^+value indicates that the near-wall mesh is not fine enough to fully resolve the viscous sublayer. Therefore, the final simulation should be interpreted as a wall-function-based runner-passage CFD analysis rather than a fully wall-resolved boundary-layer simulation. This is acceptable for the present project because the main objective was to evaluate the overall runner-passage performance, including pressure drop, torque, shaft power, and flow pattern, rather than to perform a detailed wall-shear or cavitation inception study.

However, if a more advanced CFD study is required in the future, the mesh should be refined near the blade, hub, and shroud surfaces to reduce the y^+values and improve the prediction of near-wall flow behavior. This would be especially important for detailed loss analysis, cavitation assessment, or accurate wall shear stress prediction.

Despite the high local y^+values, the final CFD solution showed good mass balance, stable residual convergence, and physically reasonable pressure and velocity fields. Therefore, the mesh and wall treatment were considered acceptable for the scope of this runner-passage performance evaluation.

6-12 Final Interpretation of Flow Visualization Results

The final post-processing results provide both quantitative and qualitative confirmation of the runner performance. The pressure contours showed a clear pressure difference across the runner blade, which explains the generation of torque. The total pressure contours confirmed that the flow loses hydraulic energy while passing through the runner. The velocity contours showed smooth acceleration and turning of the flow inside the blade passage.

The streamline visualization further confirmed that the flow follows the blade passage in a physically reasonable manner. This result supports the decision to use a swirl inlet condition instead of a purely normal inlet flow direction. The wall y^+contour showed that the near-wall mesh is more suitable for a wall-function-based simulation than for a fully wall-resolved boundary-layer analysis.

Overall, the contour and streamline results support the numerical performance values obtained from CFD-Post. The final CFD model was therefore considered suitable for estimating the runner-passage head, torque, shaft power, and efficiency.

References

1 https://www.farab.com/en/portfolio/siah-bisheh-pumped-storage-power-plant/

2 https://cdn2.jahannews.com/images/docs/000266/n00266489-r-b-002.jpg

3 https://www.researchgate.net/figure/Francis-turbine-for-conduit-hydropower-generation-79_fig6_333997954

Appendix A: CFturbo Final Design Check Table

			Symbol	Unit	Value
PROJECT					
	Actual power output		PD	[kW]	2.52E+05
	Efficiency		η	[%]	84.5
	Information				
		Project type			Hydro Turbine
		Title			
		Classification			
		Author			
		Comment			
		Creation date			24 Jul 2026, 17:53
		Last modification			24 Jul 2026, 20:50
	Global setup				
		Design Point			
			Volume flow	Q	[m³/s]	59
			Revolutions	n	[/min]	254
			Addt'l. Hydraulic efficiency	ηh+	[%]	95
			Rotation direction			Right (clockwise)
			Specific speed (EU)	nq	[-]	18
			Head	H	[m]	516
			Net head	Hn	[m]	490
		Fluid			
			Name			Water (20°C)
			Description			
			Symbol			
			Heat capacity	cp	[J/(kg·K)]	4182
			Density	ρ	[kg/m³]	998
			Vapor pressure	pv	[bar]	0.0234
			Kinematic viscosity	ν	[mm²/s]	1
			Thermal conductivity	λ	[W/(m·K)]	0.556
		Inlet conditions			
			Pressure (total)	pt	[bar]	51.5
			Temperature	T	[°C]	20
	1: [Impeller_1]				
		Active			✓

Inlet					
	Center point		z, r	[m; m]	[0; 2.66]
	Center line diameter		d	[m]	5.32
	Area		A	[m^2]	3.343
	Width		b	[m]	0.2
	Angle		γ	[°]	0
	Angle mode				User-defined
Outlet					
	Center point		z, r	[m; m]	[0.21; 1.41]
	Center line diameter		d	[m]	2.82
	Area		A	[m^2]	4.315
	Width		b	[m]	0.487
	Angle		γ	[°]	70.8
	Angle mode				User-defined
Component flow properties					
	Inlet - Flow properties				
		Static pressure	p1	[bar]	31.46
		Total pressure	pt1	[bar]	51.5
		Inlet - Velocity triangle			
			Peripheral speed	u1	[m/s]	70.8
			Meridional velocity	cm1	[m/s]	19
			Meridional velocity (internal)	cm*1	[m/s]	19.5
			Abs. circumferential velocity	cu1	[m/s]	60.4
			Absolute velocity	c1	[m/s]	63.3
			Rel. circumferential velocity	wu1	[m/s]	-10.3
			Relative velocity	w1	[m/s]	21.6
			Absolute flow angle	α1	[°]	17.4
			Relative flow angle	β1	[°]	61.5
	Outlet - Flow properties				
		Static pressure	p2	[bar]	-1.776

Total pressure		pt2	[bar]	1
Outlet - Velocity triangle				
Peripheral speed		u2	[m/s]	21.8
Meridional velocity		cm2	[m/s]	23.6
Meridional velocity (internal)		cm*2	[m/s]	24.3
Abs. circumferential velocity		cu2	[m/s]	0.9
Absolute velocity		c2	[m/s]	23.6
Rel. circumferential velocity		wu2	[m/s]	-20.9
Relative velocity		w2	[m/s]	31.5
Absolute flow angle		α2	[°]	87.9
Relative flow angle		β2	[°]	48.5
Main dimensions				
Version 1.				
Inflow swirl				According to outl. swirl
Setup				
Unshrouded				-
Splitter blades				-
Material density		ρ	[kg/m³]	7750
Manual dimensioning				-
Outlet swirl velocity		cu2	[m/s]	0
Stage info				
Impeller type				Standard
Parameter types				

	Used for runner diameter dS1			Work coefficient
	Used for runner diameter dH1			Diameter ratio dH/dS
	Used for runner diameter dS2			Diameter ratio d2/d1
	Used for inlet width b1z			Horiz. inlet width ratio
Parameters				
	Work coefficient	ψ	[-]	2.022
	Diameter ratio dH1/dS1	dH1/dS1	[-]	1
	Diameter ratio dS2/dS1	dS2/dS1	[-]	0.615
	Inlet width / diameter	b1z/dS1	[-]	0.035
Power				
	Torque	T	[Nm]	9.47E+06
	Actual power output	PD	[kW]	2.52E+05
	Power loss	PL	[kW]	46025
Efficiency				
	Hydraulic efficiency	ηh	[%]	89
	Volumetric efficiency	ηv	[%]	97
	Mechanical efficiency	ηm	[%]	99
Stage efficiency				
	Internal efficiency	ηI	[%]	82
	Stage efficiency incl. mech. losses	ηSt	[%]	81.2
Global values				

	Work coefficient	ψ	[-]	2.021
	Specific diameter	δ	[-]	6.16
	Total flow coefficient	φt	[-]	0.038
	Meridional flow coefficient	φm	[-]	0.268
	Meridional velocity ratio	cm2/cm1	[-]	0.37
	Relative velocity ratio	w2/w1	[-]	1.059
	Inlet area	A1	[m^2]	3.112
	Outlet area	A2	[m^2]	8.41
	Area ratio	A2/A1	[-]	2.702
	Outlet width ratio	b2/d2	[-]	1
Reynolds numbers				
	Reynolds number (d1)	Re(d1)	[-]	3.76E+08
	Reynolds number (b1)	Re(b1)	[-]	1.32E+07
	Reynolds number (d2)	Re(d2)	[-]	1.42E+08
	Reynolds number (b2)	Re(b2)	[-]	7.12E+07
Dimensions				
	Automatic			✓
	Hub diameter inlet	dH1	[m]	5.32
	Shroud diameter inlet	dS1	[m]	5.32
	Hub diameter outlet	dH2	[m]	0
	Shroud diameter outlet	dS2	[m]	3.272

		Horizontal inlet width	b1z	[m]	0.1862
		Work coefficient	ψ	[-]	2.021
		Specific diameter	δ	[-]	6.16
		Total flow coefficient	φt	[-]	0.038
		Meridional flow coefficient	φm	[-]	0.268
	Shaft				
		Allowable stress	τ	[MPa]	15
		Factor of safety	SF	[-]	1.15
		Min. shaft diameter	d	[m]	1.546
Meridional contour					
	Leading edge fixed on inlet				✓
	Trailing edge fixed on outlet				✓
	Axial extension max.		ΔzMax	[m]	0.39
	Radial extension max.		ΔrMax	[m]	1.48
	Hub Contour Controls		z, r	[m; m]	[[-0.1; 2.66], [-0.0248; 1.992], [-0.002; 1.769], [0.08; 1.412], [0.13; 1.18]]
	Shroud Contour Controls		z, r	[m; m]	[[0.1; 2.66], [0.1244; 1.892], [0.1372; 1.636], [0.2289; 1.636], [0.29; 1.64]]
	Inlet angle			[°]	-180
	Outlet angle			[°]	109.2
	Axial start position		z	[m]	0
	Axial extension		Δz	[m]	0.21
	Leading Edge				
		Width circle diameter	b	[m]	0.1999

	Width hub to shroud	b*	[m]	0.2
	Diameter at mid span	d	[m]	5.32
	Diameter average	d*	[m]	5.32
Trailing Edge				
	Width circle diameter	b	[m]	0.373
	Width hub to shroud	b*	[m]	0.487
	Diameter at mid span	d	[m]	2.82
	Diameter average	d*	[m]	2.82
Blade properties				
Blade shape				Free-form 3D
Number of blades				16
Number of spans				6
Blockage mode				Tangential
Beta interpolation mode LE				Linear
Beta interpolation mode TE				Linear
Automatic βB1				✓
Automatic βB2				✓
Rel. deceleration hub		w2/w1	[-]	1.799
Rel. deceleration shroud		w2/w1	[-]	2.333
Abs. deflection angle hub		ΔαF	[°]	73.8
Abs. deflection angle shroud		ΔαF	[°]	73.8
Rel. deflection angle hub		ΔβF	[°]	-28.4
Rel. deflection angle shroud		ΔβF	[°]	-36.1
Main blade				
	Thickness LE@hub	sLEH	[m]	0.0245
	Thickness LE@shroud	sLES	[m]	0.0245
	Thickness TE@hub	sTEH	[m]	0.0245
	Thickness TE@shroud	sTES	[m]	0.0245
	Blade angle leading edge	βB1	[°]	[59.7, 59.7]
	Blade angle trailing edge	βB2	[°]	[26.2, 18.6]
Span distribution				
	CurveMode			Straight line
Incidence				
	Definition			Angle direct
	Angle hub	iHub	[°]	0
	Angle shroud	iShr	[°]	0
Slip				
	Slip model			User defined

	Angle hub	δHub	[°]	-5
	Angle shroud	δShr	[°]	-5
Suction side [hub, shroud]				
	Circumferential velocity	uS	[m/s]	[70.8, 70.8]
	Angle of absolute flow	αFS	[°]	[15.8, 15.8]
	Angle of relative flow	βFS	[°]	[59, 59]
	Circumferential component of relative velocity	wuS	[m/s]	[-10.3, -10.3]
	Relative velocity	wS	[m/s]	[20, 20]
	Meridional component of absolute velocity	cmS	[m/s]	[17.1, 17.1]
	Radial component of absolute velocity	crS	[m/s]	[-17.1, -17.1]
	Axial component of absolute velocity	caxS	[m/s]	[0.8, 0.2]
	Circumferential component of absolute velocity	cuS	[m/s]	[60.4, 60.4]
	Absolute velocity	cS	[m/s]	[62.8, 62.8]
Leading edge (outside blade passage) [hub, shroud]				
	Angle of absolute flow	αF0	[°]	[15.8, 15.8]
	Angle of relative flow	βF0	[°]	[59, 59]
	Circumferential component of relative velocity	wu0	[m/s]	[-10.3, -10.3]
	Relative velocity	w0	[m/s]	[20, 20]
	Meridional component of absolute velocity	cm0	[m/s]	[17.1, 17.1]
	Radial component of absolute velocity	cr0	[m/s]	[-17.1, -17.1]
	Axial component of absolute velocity	cax0	[m/s]	[0.8, 0.2]
	Circumferential component of absolute velocity	cu0	[m/s]	[60.4, 60.4]
	Absolute velocity	c0	[m/s]	[62.8, 62.8]
Leading edge (inside blade passage) [hub, shroud]				
	Circumferential velocity	u1	[m/s]	[70.8, 70.8]
	Blade blockage	τ1	[-]	[1.028, 1.028]
	Incidence	i	[°]	[0, 0]
	Angle of absolute flow	αF1	[°]	[16.2, 16.2]
	Angle of relative flow	βF1	[°]	[59.7, 59.7]

	Circumferential component of relative velocity	wu1	[m/s]	[-10.3, -10.3]
	Relative velocity	w1	[m/s]	[20.4, 20.4]
	Meridional component of absolute velocity	cm1	[m/s]	[17.6, 17.6]
	Radial component of absolute velocity	cr1	[m/s]	[-17.6, -17.6]
	Axial component of absolute velocity	cax1	[m/s]	[0.8, 0.2]
	Circumferential component of absolute velocity	cu1	[m/s]	[60.4, 60.4]
	Absolute velocity	c1	[m/s]	[63, 63]
Trailing edge (inside blade passage) [hub, shroud]				
	Circumferential velocity	u2	[m/s]	[31.4, 43.6]
	Blade blockage	τ2	[-]	[1.136, 1.136]
	Deviation	δ	[°]	[-5, -5]
	Angle of absolute flow	αF2	[°]	[90, 90]
	Angle of relative flow	βF2	[°]	[31.2, 23.6]
	Circumferential component of relative velocity	wu2	[m/s]	[-31.4, -43.6]
	Relative velocity	w2	[m/s]	[36.7, 47.6]
	Meridional component of absolute velocity	cm2	[m/s]	[19, 19]
	Radial component of absolute velocity	cr2	[m/s]	[-17.4, 1.8]
	Axial component of absolute velocity	cax2	[m/s]	[7.8, 19]
	Circumferential component of absolute velocity	cu2	[m/s]	[0, 0]
	Absolute velocity	c2	[m/s]	[19, 19]
Trailing edge (outside blade passage) [hub, shroud]				
	Angle of absolute flow	αF3	[°]	[90, 90]
	Angle of relative flow	βF3	[°]	[28.1, 21]
	Circumferential component of relative velocity	wu3	[m/s]	[-31.4, -43.6]
	Relative velocity	w3	[m/s]	[35.6, 46.7]
	Meridional component of absolute velocity	cm3	[m/s]	[16.8, 16.8]
	Radial component of absolute velocity	cr3	[m/s]	[-15.3, 1.6]

		Axial component of absolute velocity	cax3	[m/s]	[6.9, 16.7]
		Circumferential component of absolute velocity	cu3	[m/s]	[0, 0]
		Absolute velocity	c3	[m/s]	[16.8, 16.8]
	Blade mean lines				
	Design mode				Blade loading
	Blade Passage				
		Max. sphere radius		[m]	0
	Blade				
		Coupled mean lines			✓
		Stacking position		[%]	100
		Spec. work blade loading	Y	[m^2/s^2]	-5058
		Head	H	[m]	516
		Automatic			✓
		Overlapping	Δφ/t	[-]	2.552
		Meanline 1 (Hub)			
		Leading edge position	φLE	[°]	-72.8
		Trailing edge position	φTE	[°]	22.5
		Wrap angle	Δφ	[°]	95.3
		Blade angle leading edge	βB1	[°]	59.7
		Blade angle trailing edge	βB2	[°]	26.2
		Front position	relLE	[%]	20
		Aft position	relTE	[%]	80
		Blade loading @ LE	B'LE	[-]	-214.8
		Slope	s	[-]	75.8
		Meanline 2			
		Leading edge position	φLE	[°]	-53.6
		Trailing edge position	φTE	[°]	22.5
		Wrap angle	Δφ	[°]	76.1

	Blade angle leading edge	βB1	[°]	59.7
	Blade angle trailing edge	βB2	[°]	24.7
	Front position	relLE	[%]	20
	Aft position	relTE	[%]	80
	Blade loading @ LE	B'LE	[-]	-216.4
	Slope	s	[-]	-2.249
Meanline 3				
	Leading edge position	φLE	[°]	-37.7
	Trailing edge position	φTE	[°]	22.5
	Wrap angle	Δφ	[°]	60.2
	Blade angle leading edge	βB1	[°]	59.7
	Blade angle trailing edge	βB2	[°]	23.2
	Front position	relLE	[%]	20
	Aft position	relTE	[%]	80
	Blade loading @ LE	B'LE	[-]	-218
	Slope	s	[-]	-80.3
Meanline 4				
	Leading edge position	φLE	[°]	-24.8
	Trailing edge position	φTE	[°]	22.5
	Wrap angle	Δφ	[°]	47.3

	Blade angle leading edge	βB1	[°]	59.7
	Blade angle trailing edge	βB2	[°]	21.6
	Front position	relLE	[%]	20
	Aft position	relTE	[%]	80
	Blade loading @ LE	B'LE	[-]	-219.6
	Slope	s	[-]	-158.4
Meanline 5				
	Leading edge position	φLE	[°]	-14.3
	Trailing edge position	φTE	[°]	22.5
	Wrap angle	Δφ	[°]	36.8
	Blade angle leading edge	βB1	[°]	59.7
	Blade angle trailing edge	βB2	[°]	20.1
	Front position	relLE	[%]	20
	Aft position	relTE	[%]	80
	Blade loading @ LE	B'LE	[-]	-221.2
	Slope	s	[-]	-236.5
Meanline 6 (Shroud)				
	Leading edge position	φLE	[°]	-6.2
	Trailing edge position	φTE	[°]	22.5
	Wrap angle	Δφ	[°]	28.7

			Blade angle leading edge	βB1	[°]	59.7
			Blade angle trailing edge	βB2	[°]	18.6
			Front position	relLE	[%]	20
			Aft position	relTE	[%]	80
			Blade loading @ LE	B'LE	[-]	-222.8
			Slope	s	[-]	-314.6
		Blade profiles				
		Thickness definition				Perpendicular to mean line
		Blade Passage				
			Max. sphere radius		[m]	0.0259
			Leading Edge			
				Area	[m^2]	0.0424
				Distance blade-to-blade	[m]	0.0518
				Distance hub-to-shroud	[m]	0.604
			Trailing Edge			
				Area	[m^2]	0.0424
				Distance blade-to-blade	[m]	0.0518
				Distance hub-to-shroud	[m]	0.604
			Throat			
				Area	[m^2]	0.0424
				Distance blade-to-blade	[m]	0.0518
				Distance hub-to-shroud	[m]	0.604
		Blade				
			Design mode			Freeform

	SS-PS-Coupling			Symmetric
	Thickness Hub (LE → TE)		[m]	[0, 0.185, 0.1232, 0.0039]
	Thickness Shroud (LE → TE)		[m]	[0, 0.185, 0.1232, 0.0039]
	PS hub control point thickness	s	[m]	[0.0122, 0.0122]
Blade edges				
Blade Passage				
	Max. sphere radius		[m]	0
Blade				
	Leading edge mode			None
	Edge position		[%]	5
	Trailing edge mode			Ellipse
	Hub-to-shroud			Uniform
	Hub axis ratio		[-]	3
Model finishing				
Create extended blade				-
Performance prediction Euler				
Flow factor			[-]	0
Hydraulic resistance		ζh	[-]	0
Shock losses		ζs	[-]	0
Slip coefficient		γ	[-]	1.264